

Dimensioning Standards

Rules and Practices

- Accurate dimensioning is one of the most demanding undertakings when designing parts.
- *Use the checklist to insure you have followed the basic dimensioning rules.*
- *Keep in mind there may be a case where the need to break a standard could occur to give clarity to the part and manufacturer.*

Standards

- In order for the drawings to be dimensioned so that all people can understand them, we need to follow standards that every company in the world must follow. Standards are created by these organizations:

-ANSI	-MIL
-ISO	-DOD
-DIN	-CEN
-JIS	

Standards Institutions

- ANSI - American National Standards Institute - This institute creates the engineering standards for North America.
- ISO - International Organization for Standardization - This is a world wide organization that creates engineering standards with approximately 100 participating countries.

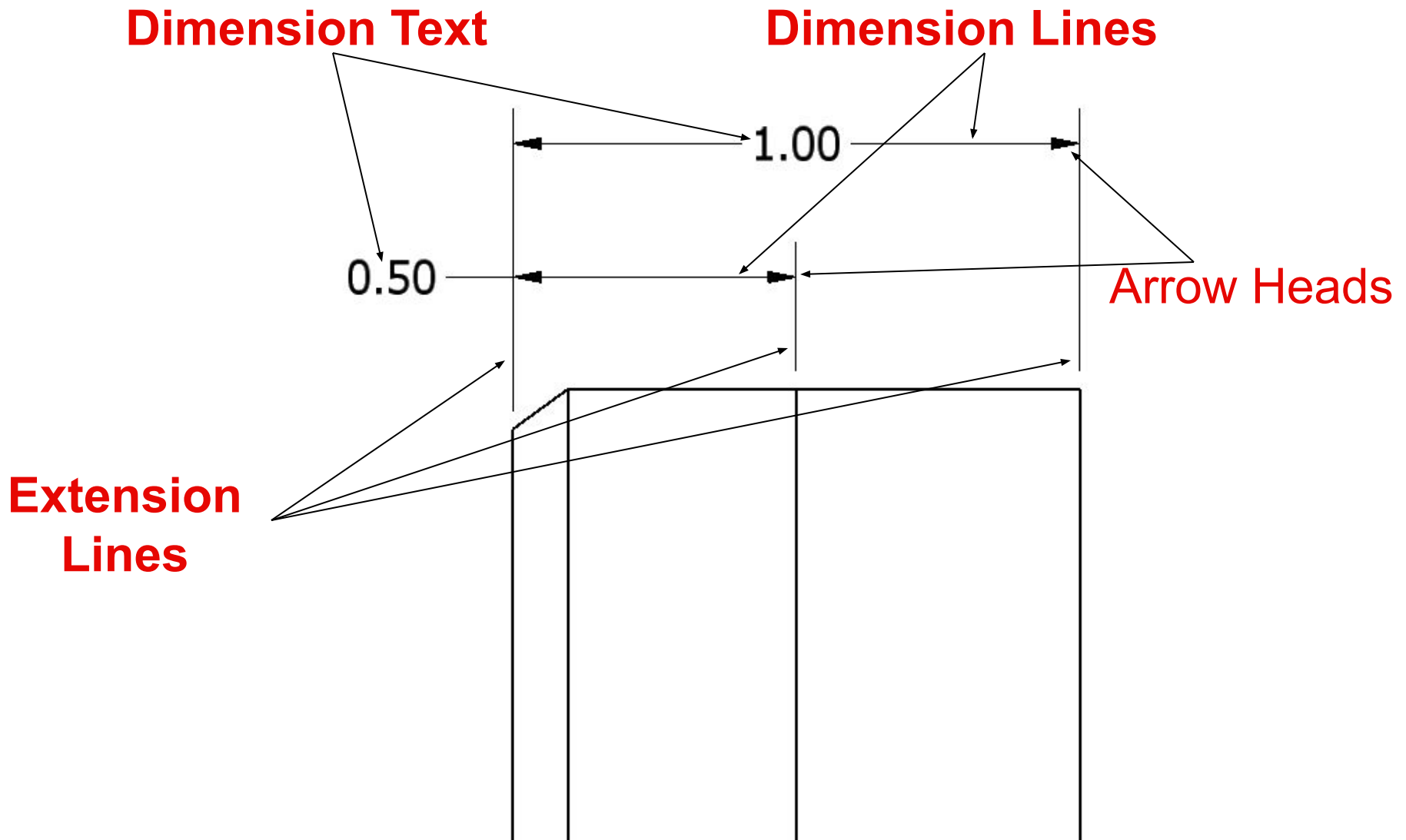
Standards Institutions

- DIN - Deutsches Institut für Normung - The German Standards Institute created many standards used world wide such as the standards for camera film.
- JIS - Japanese Industrial Standard - Created after WWII for Japanese standards.
- CEN - European Standards Organization

Standards Institutions

- The United States military has two organizations that develop standards.
 - DOD - Department Of Defense
 - MIL - Military Standard
- ANSI/ASME Y14.1 and Y14.5 – international drafting standards for engineering drawings

Linear dimensions are comprised of four components:

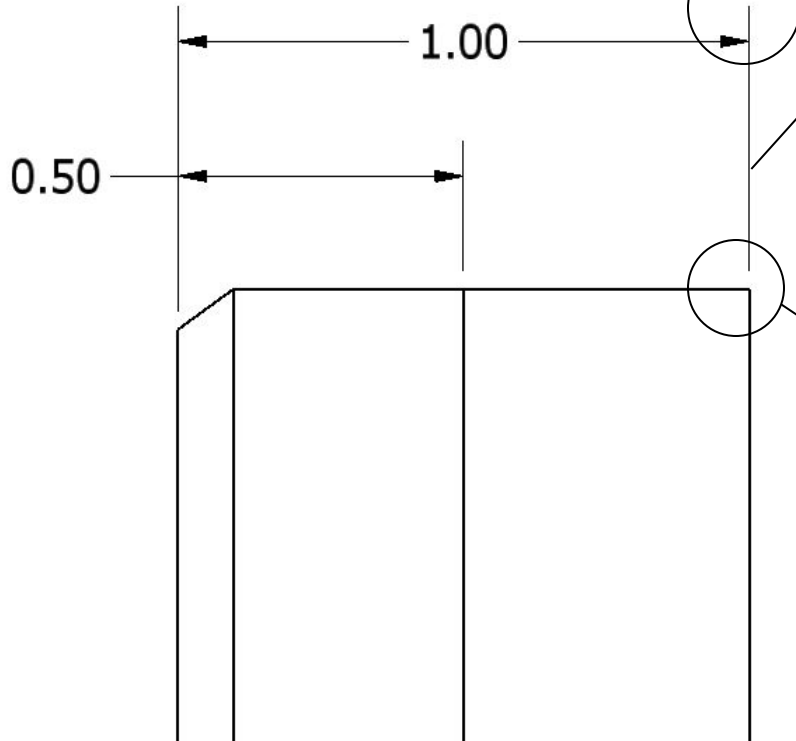


Extension Lines

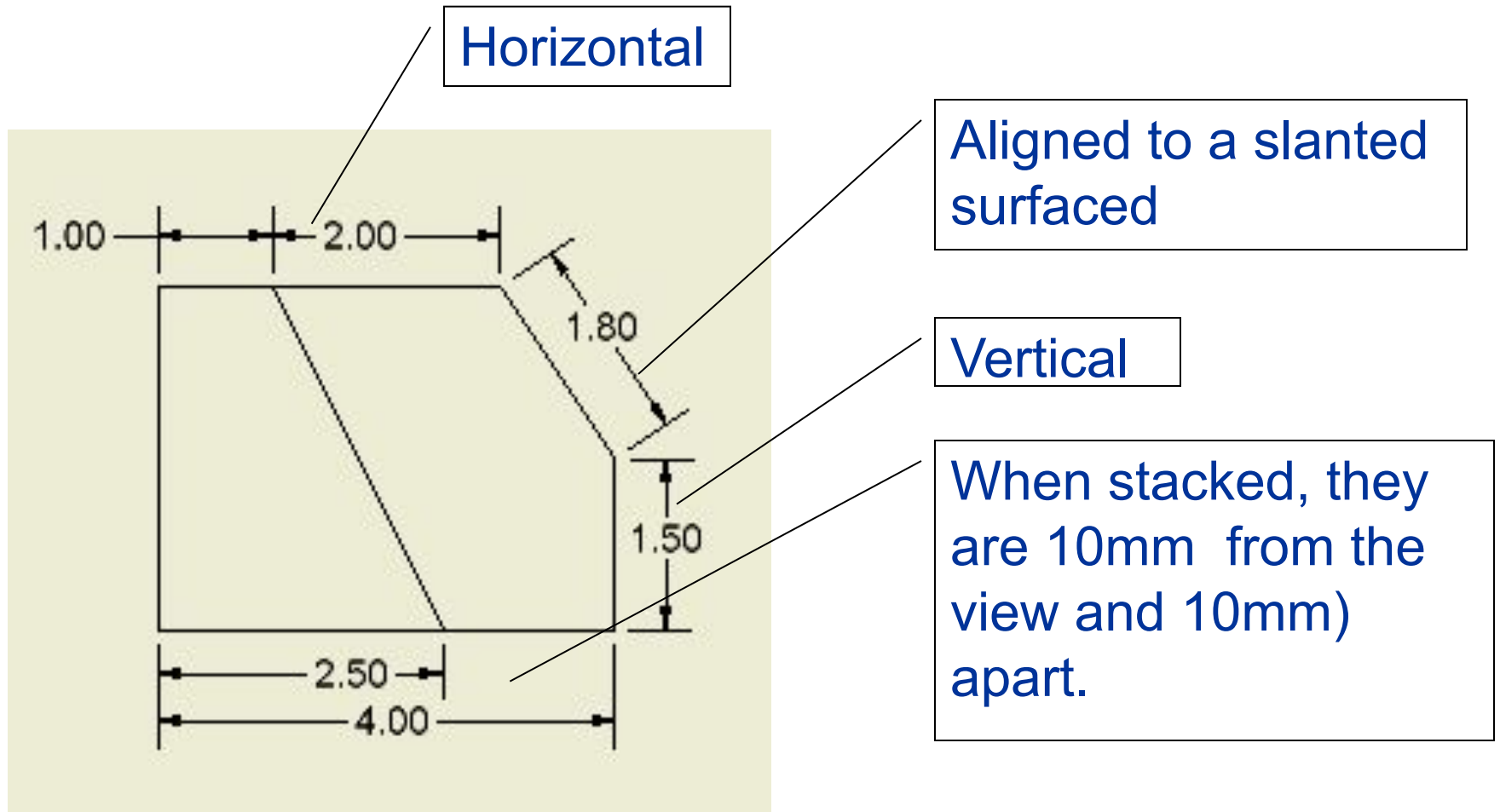
Continue 3mm past the dimension line

Extended from the view to indicate the edges referenced and hold the dimension line

2mm gap from the view so they are not confused with the visible lines

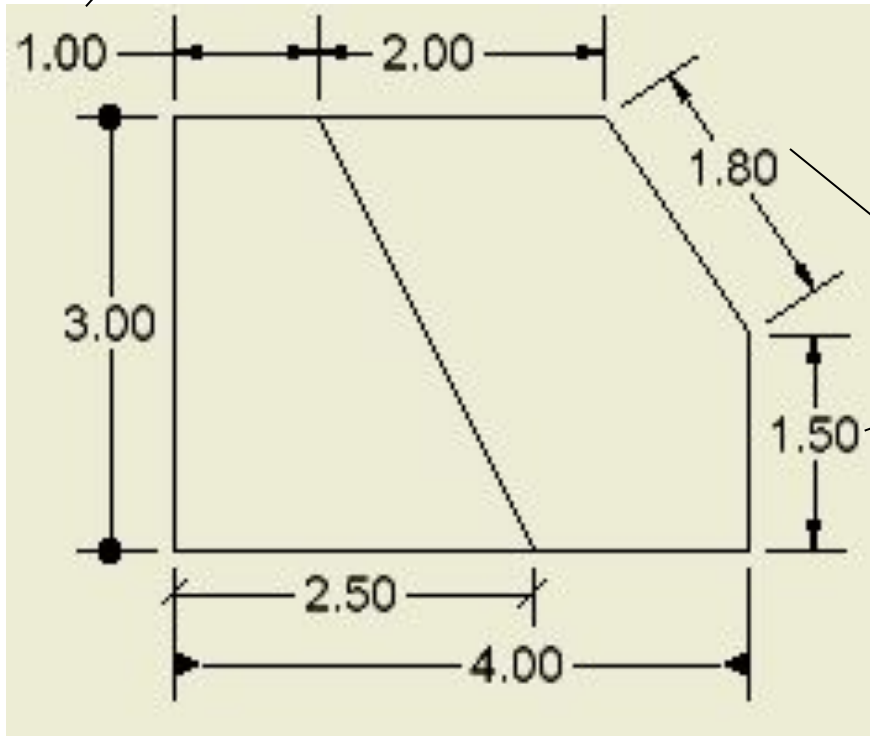


Dimension Lines



Dimension Text

If the dimension text will not fit between the extension lines, it may be placed outside them.



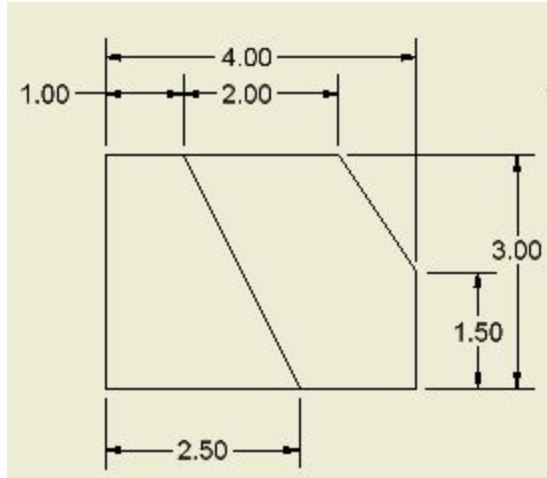
Dimension text is placed in the middle of the line both horizontally and vertically.

Dimensioning Methods

- Dimensions are represented on a drawing using one of two systems, unidirectional or aligned.
- The *unidirectional* method means all dimensions are read in the same direction.
- The *aligned* method means the dimensions are read in alignment with the dimension lines or side of the part, some read horizontally and others read vertically.

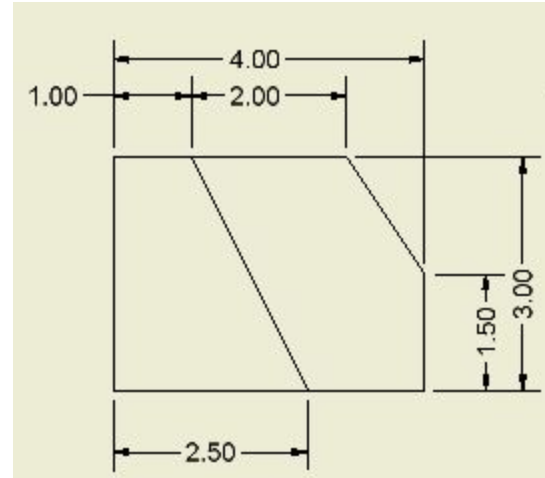
Dimension Text

Unidirectional vs. Aligned



Unidirectional

dimensions are placed so they can be read from the bottom of the drawing sheet. This method is commonly used in mechanical drafting.



Aligned dimensions are placed so the horizontal dimensions can be read from the bottom of the drawing sheet and the vertical dimensions can be read from the right side of the drawing sheet. This method is commonly used in architectural and structural drafting.

Types of Dimensions

- There are two classifications of dimensions: size and location.
- *Size* dimensions are placed in direct relationship to a feature to identify the specific size.
- *Location* dimensions are used to identify the relationship of a feature to another feature within an object.

Dimensioning Checklist

- Each dimension should be written clearly with only one way to be interpreted.
- A feature should be dimensioned only once.(not repeated)
- Dimension should be in fraction only
- Dimension and extension lines should not cross.
- Each feature should be dimensioned.
- Dimension features or surfaces should be done to a logical reference point.

Dimension Checklist

- Dimension circles should have diameters and arcs with a radius.
- A center line should be extended and used as an extension line.
- Dimension features on a view should clearly show its true shape.
- Enough space should be provided to avoid crowding and misinterpretation.

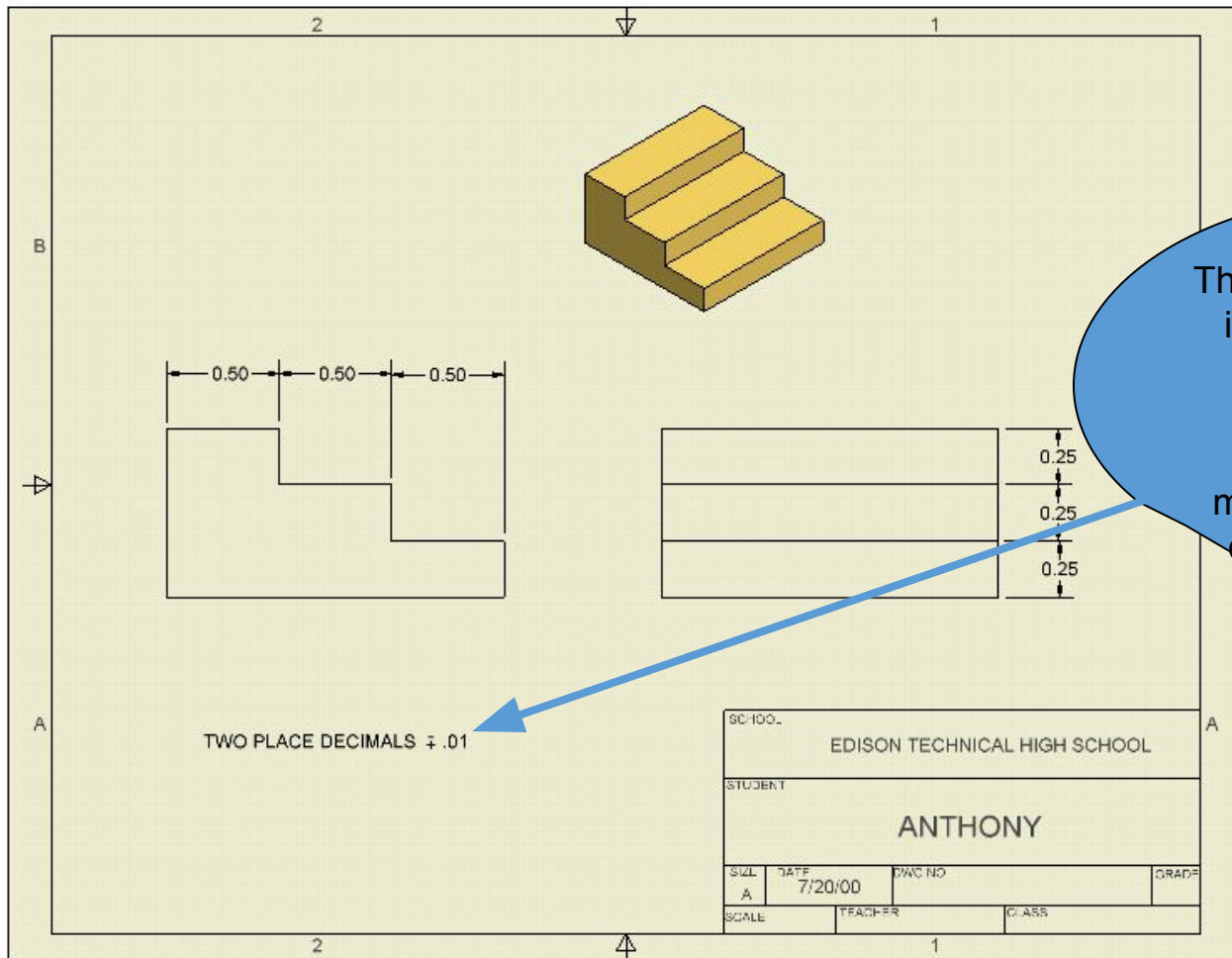
Dimension Checklist

- Extension lines and object lines should not overlap.
- Dimensions should be placed outside the part.
- Center lines or marks should be used on all circles and holes.

Linear Dimensioning

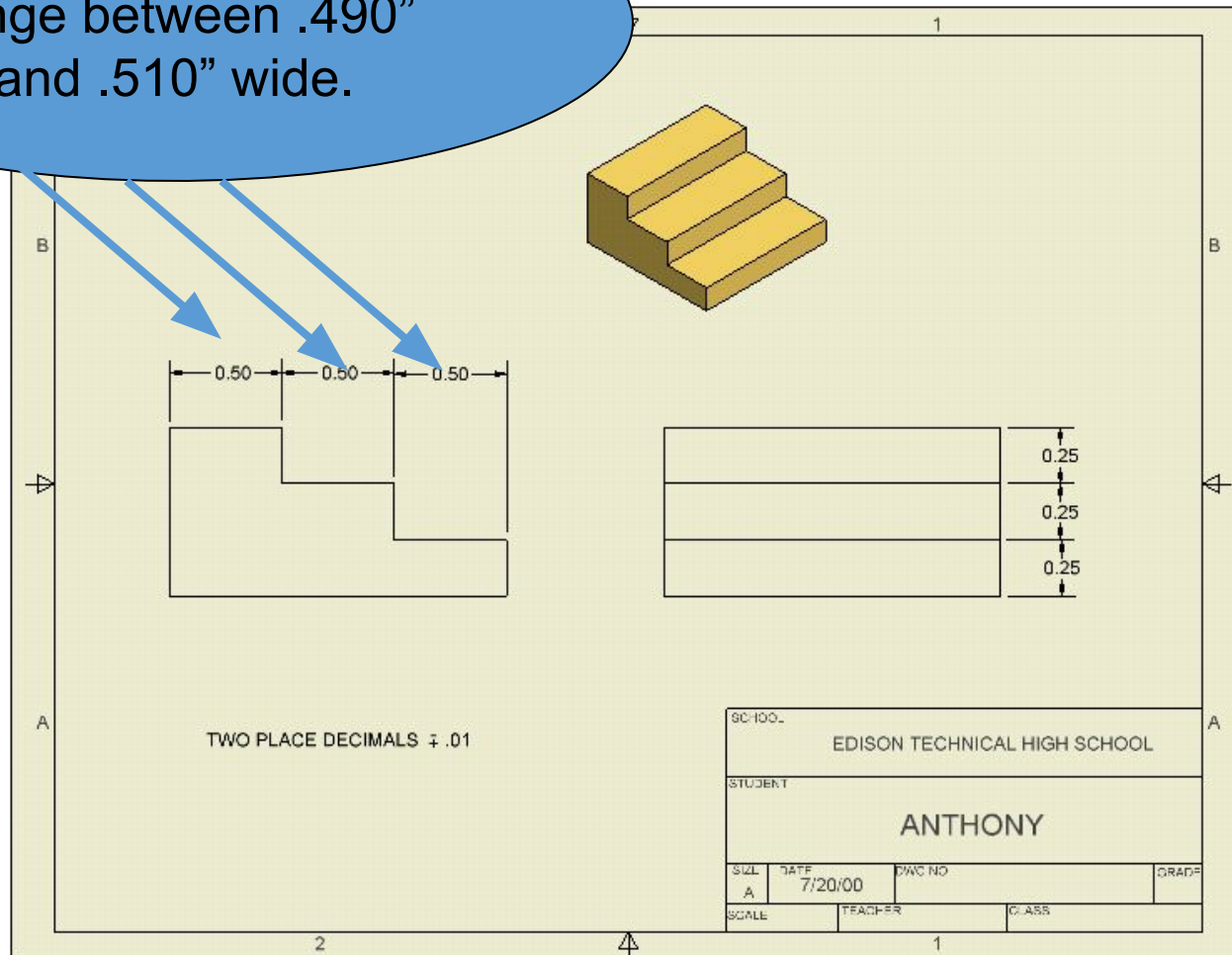
- Dimensioning from feature to feature is known as *Chain Dimensioning*. It is commonly used and easy to lay out. It does have possible consequences in the manufacturing of a part. Tolerances can accumulate, making the end product larger or smaller than expected.

Chain Dimensioning



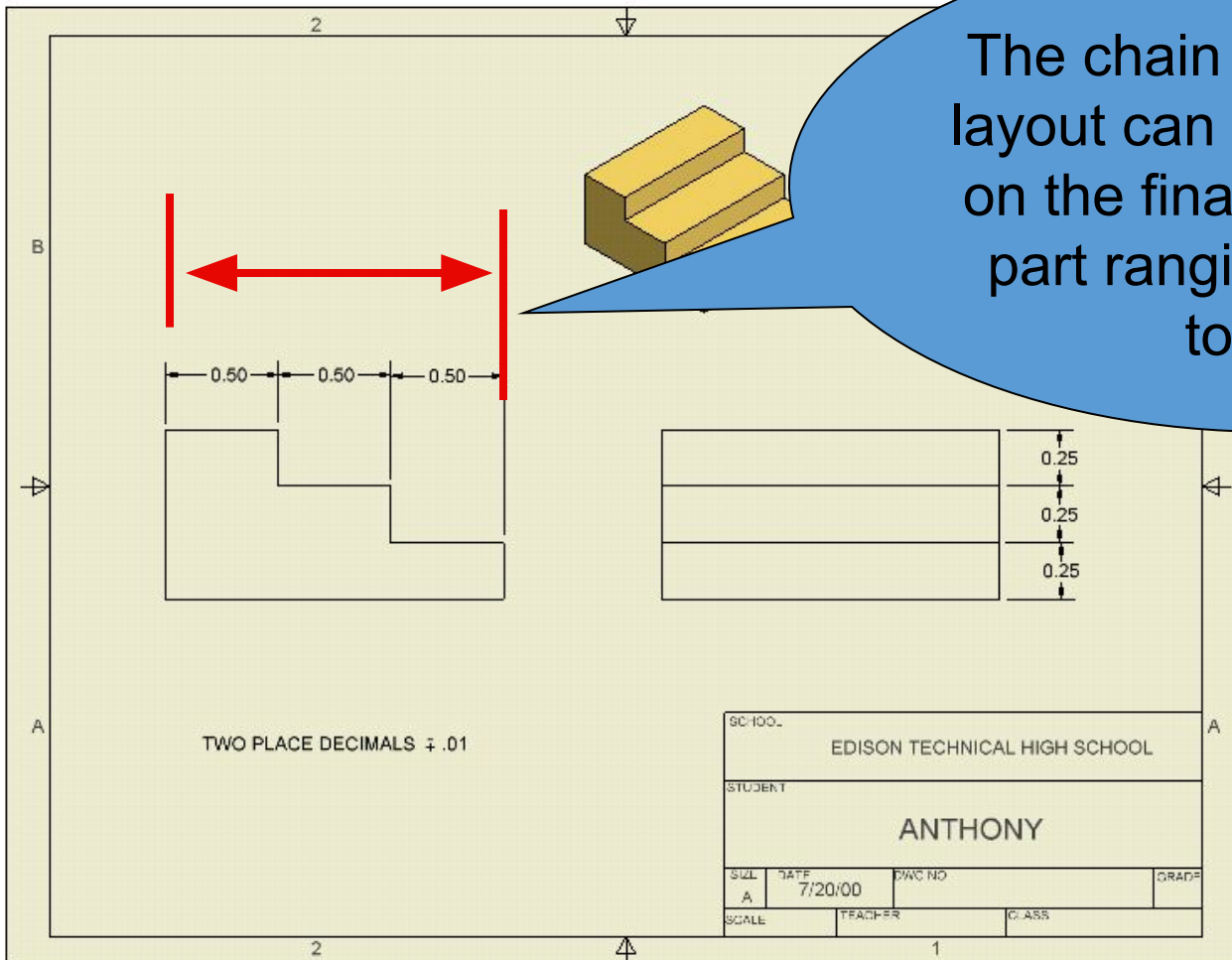
Chain Dimensioning

Each of these steps can range between .490" and .510" wide.



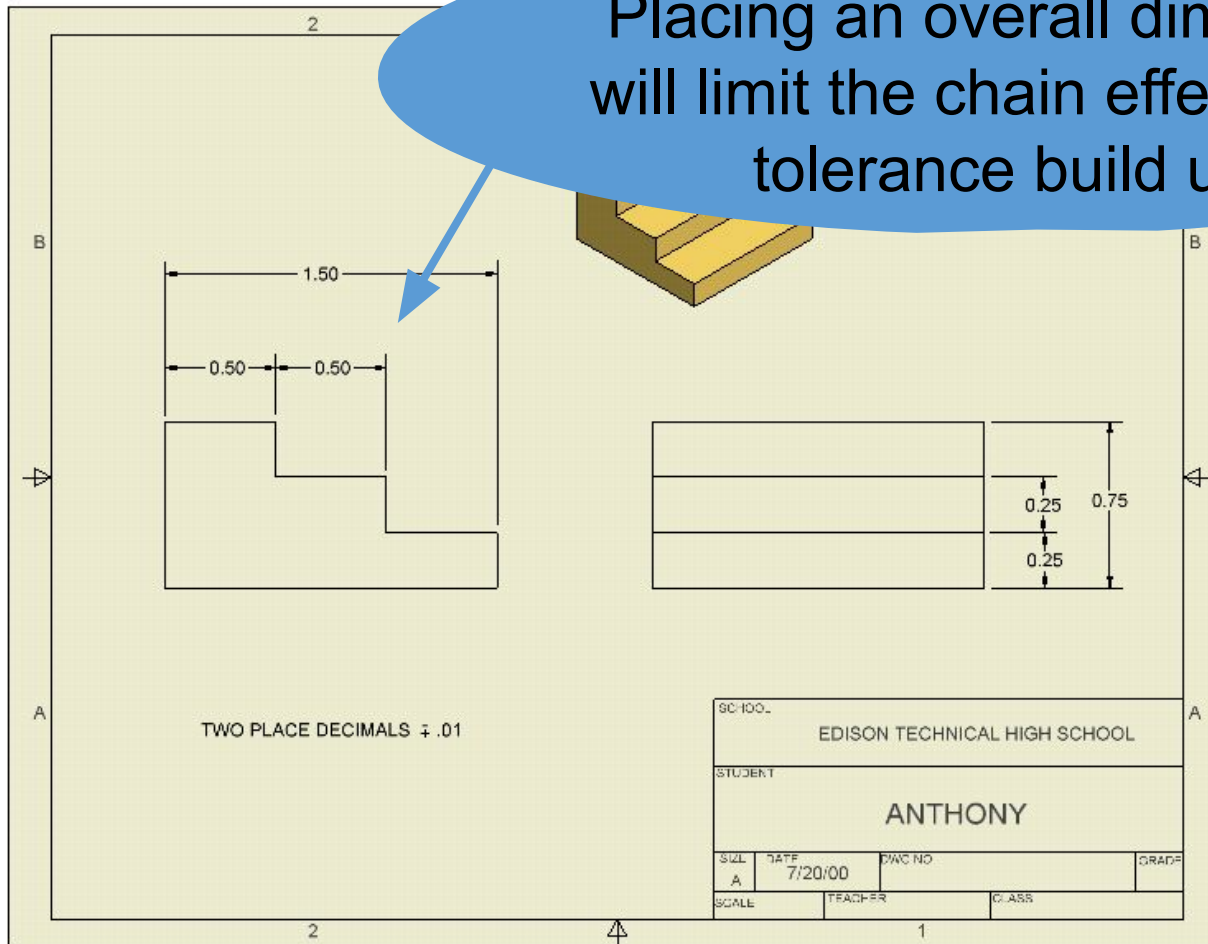
Chain Dimensioning

The chain dimensioning layout can have an effect on the final length of the part ranging from 1.47 to 1.53.



Chain Dimensioning

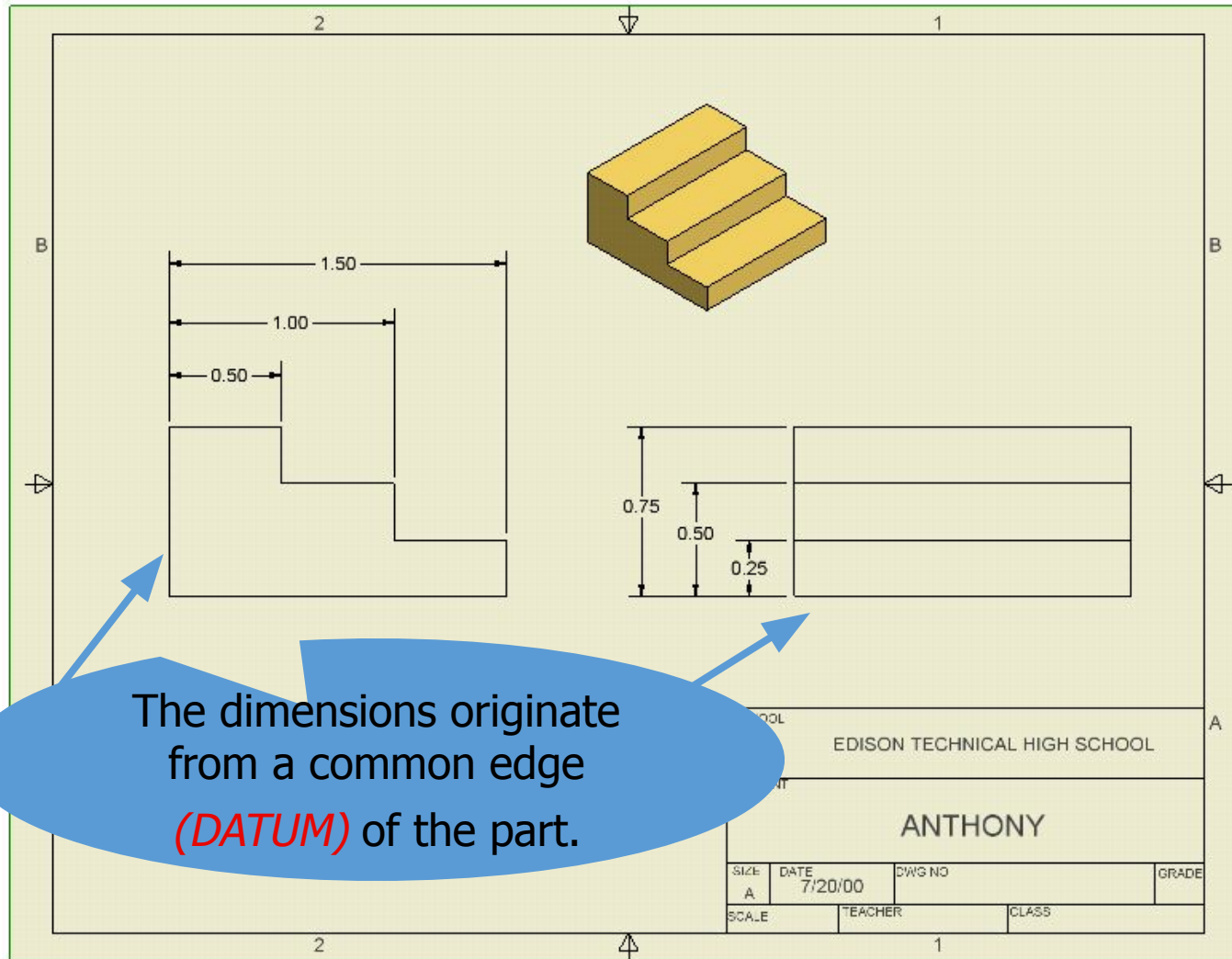
Placing an overall dimension will limit the chain effect of the tolerance build up.



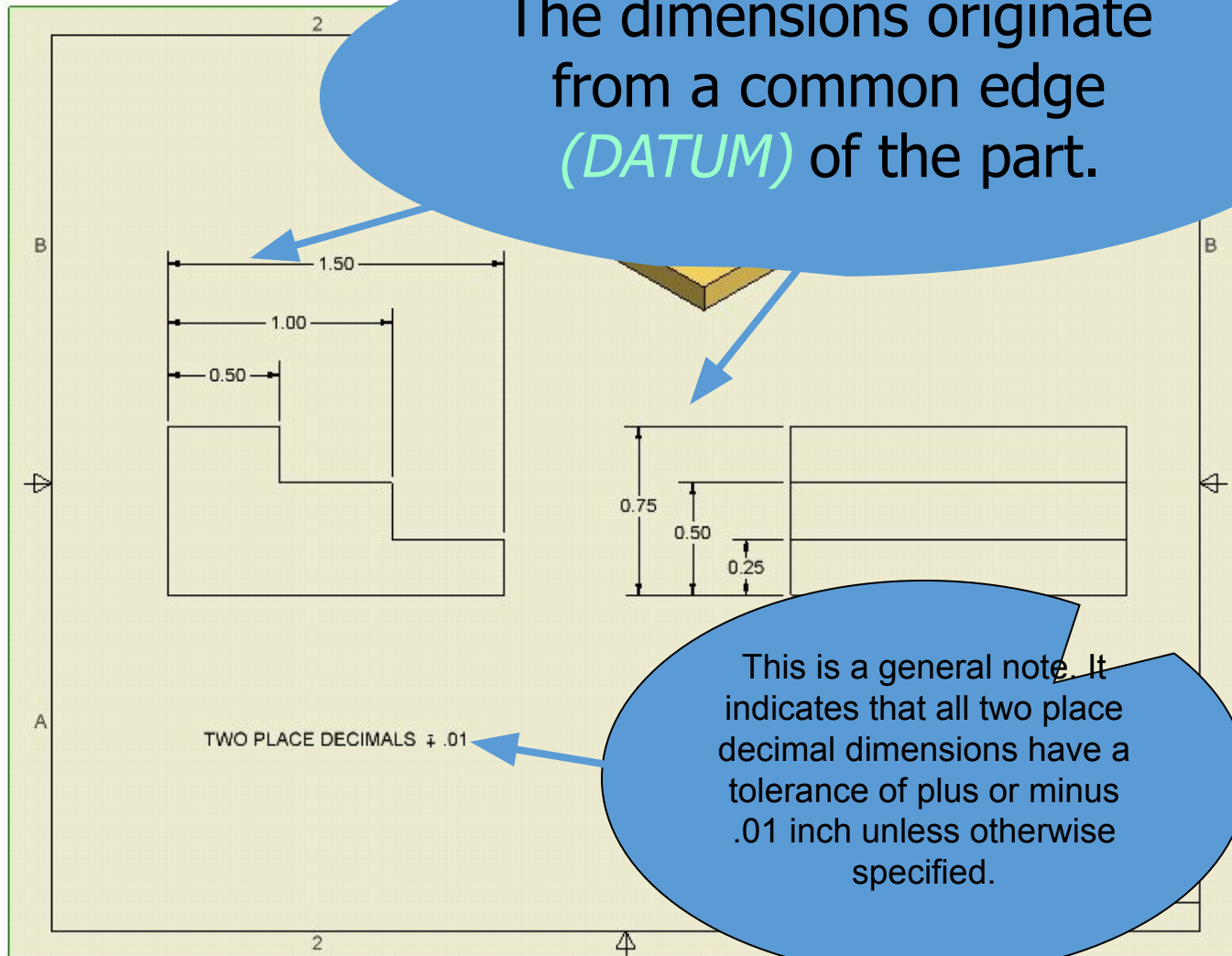
Linear Dimensioning

- The accuracy of the final product is determined by the dimensions on the drawing. If all the dimensions originate from a common corner of the part, the object will be more accurate. This is referred to as *Datum Dimensioning*. Datums insure the tolerance or errors in manufacturing do not accumulate.

Datum Dimensioning



Datum Dimensioning



The dimensions originate from a common edge (*DATUM*) of the part.

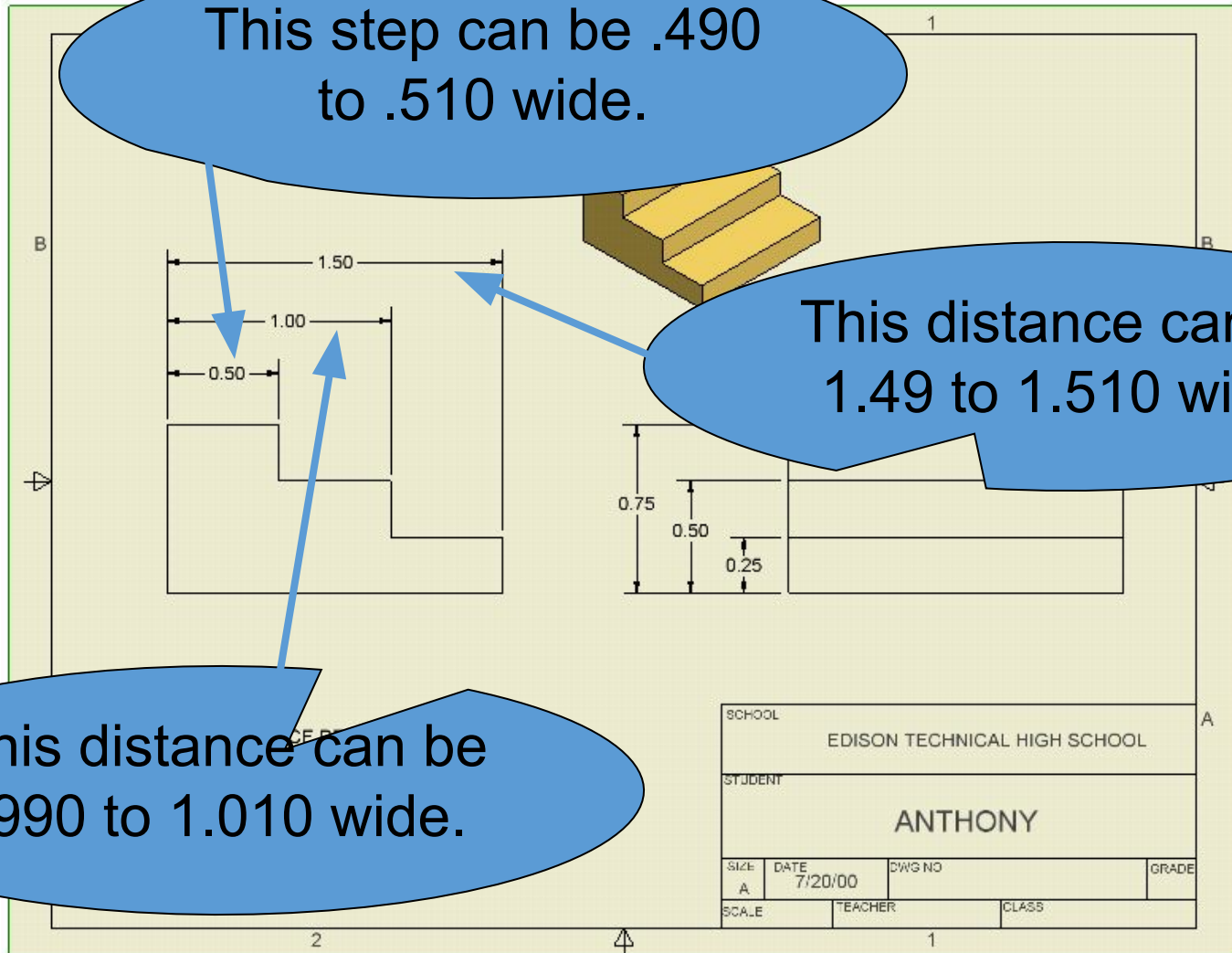
This is a general note. It indicates that all two place decimal dimensions have a tolerance of plus or minus .01 inch unless otherwise specified.

Datum Dimensioning

This step can be .490 to .510 wide.

This distance can be 1.49 to 1.510 wide.

This distance can be .990 to 1.010 wide.



Dimensioning Symbols

• Degree Symbol

() Reference Symbol

∅ Diameter Symbol

R Radius Symbol

└─ Counter Bore or Spot Face Symbol

∨ Counter Sink Symbol

⌞ Depth or Deep Symbol

X Places or By Symbol

± Plus/Minus Symbol

⌀ Center Line Symbol

□ Square(shape) Symbol

⌒ Arc Symbol

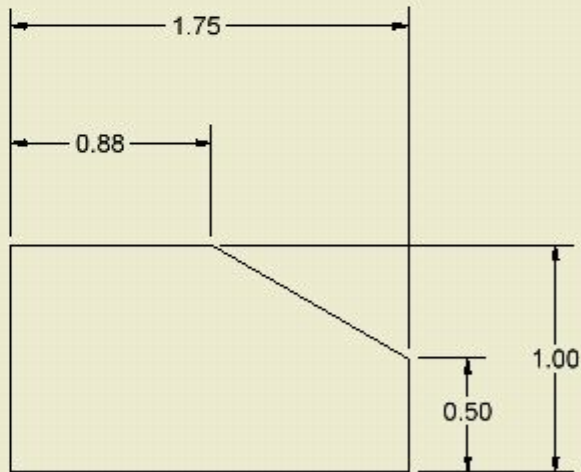
▴ Slope Symbol

Dimensioning Angles

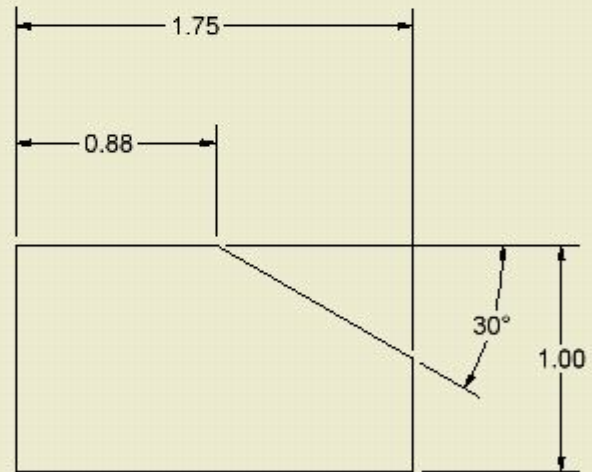
- Angled surface may be dimensioned using *coordinate method* to specify the two location distances of the angle.
- Angled surfaces may also be dimensioned using the *angular method* by specifying one location distance and the angle.

Dimensioning Angles

Coordinate Method



Angular Method



Dimensioning Arcs and Circles

- Arcs and circles are dimensioned in views that show the arc or circle.
- Arcs are dimensioned with a leader to identify the radius; in some cases, a center mark is included.
- Circles should have a center mark and are dimensioned with a leader to identify the diameter.

Dimensioning Curved Features and Arcs

The arrow can be inside for small arcs.

Small arcs do not need center marks. Arrow can be outside.

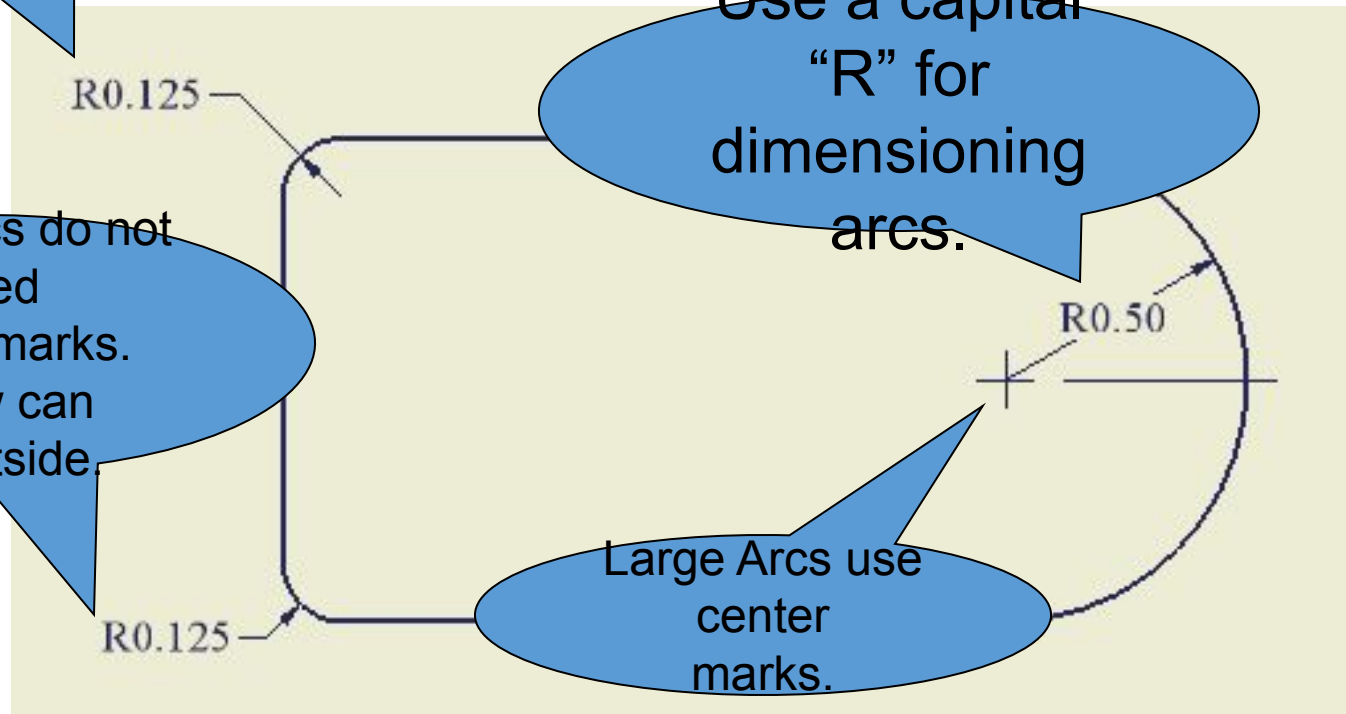
R0.125

R0.125

Use a capital "R" for dimensioning arcs.

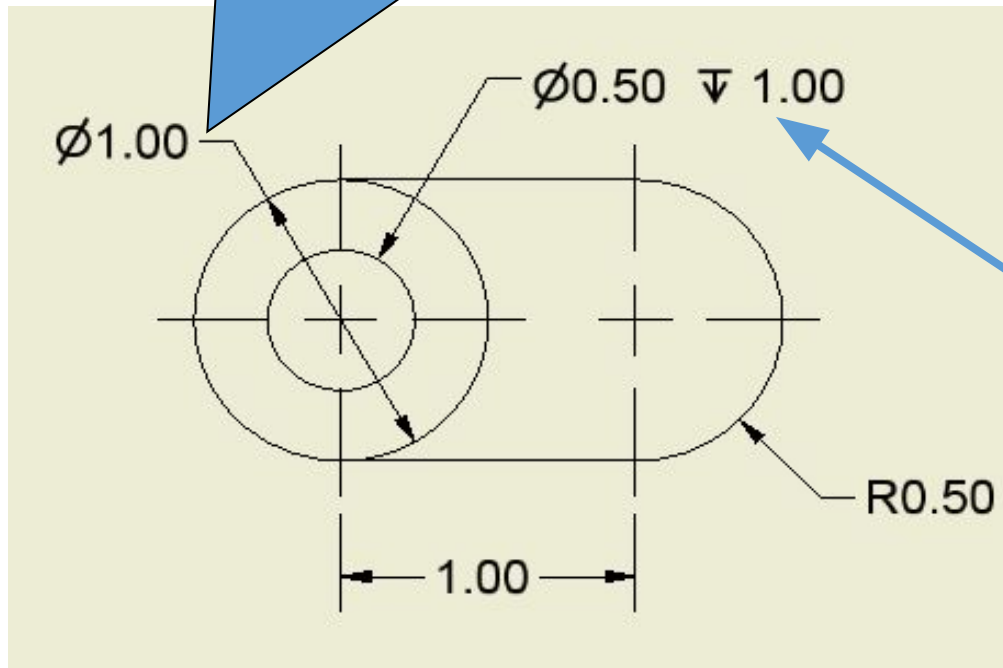
R0.50

Large Arcs use center marks.



Diameters

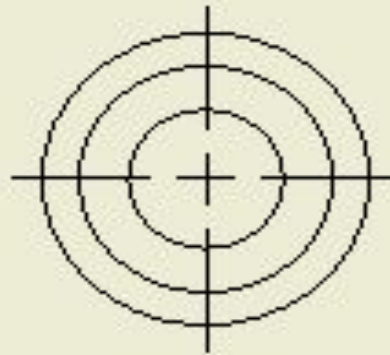
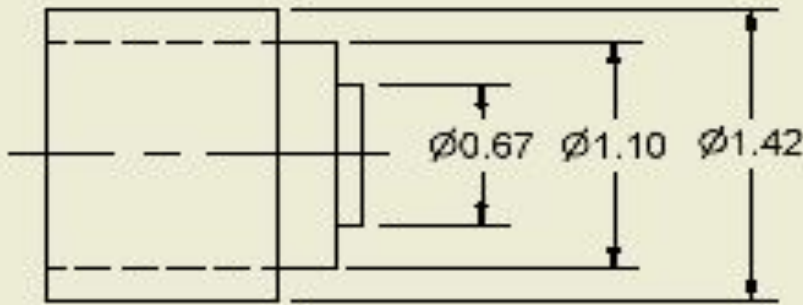
A full circular object should be dimensioned using its diameter. Holes should use hole notes.



This specification calls for a hole with a .5 diameter and 1.00 deep.

Diameters

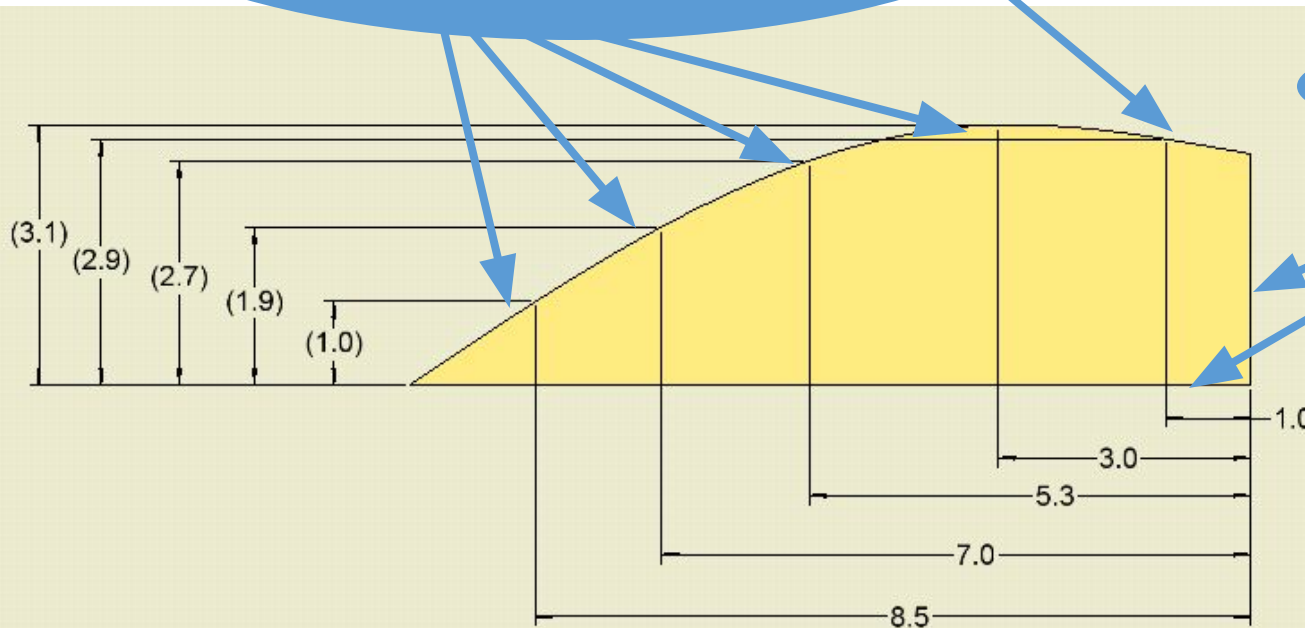
Cylindrical parts may show their diameters in this manner. Dimensioning on the right side view could be too crowded.



Note that the diameter symbol is used so it is not confused with a linear dimension.

Dimensioning Curved Features

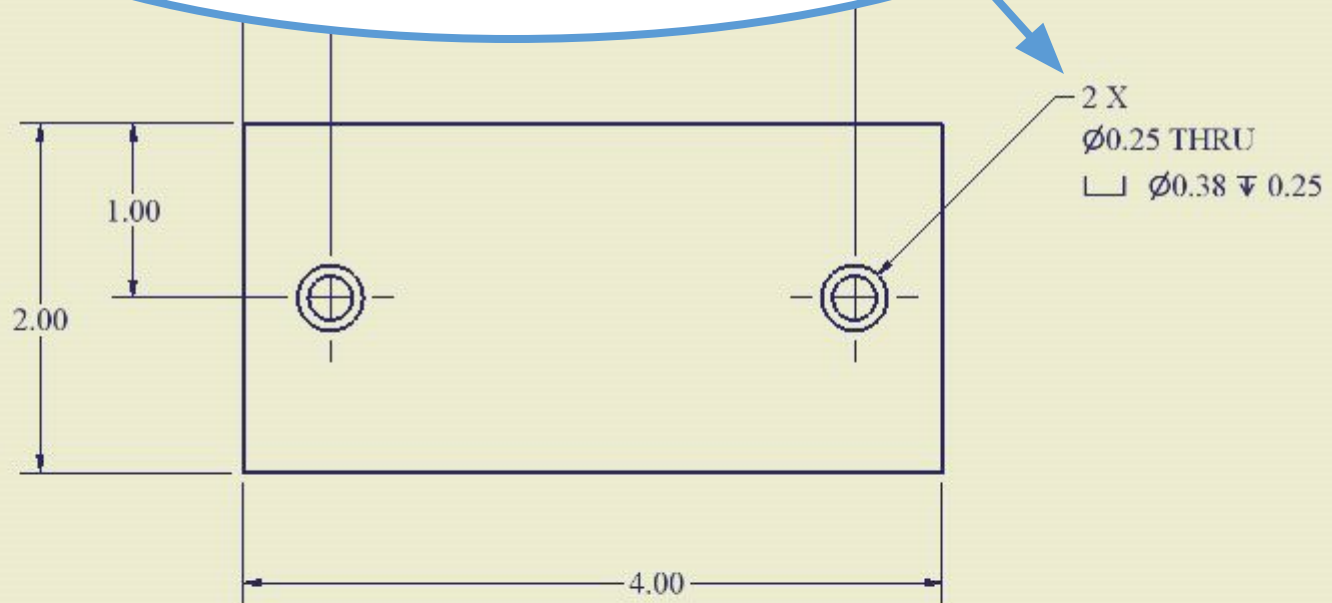
Points are placed along the contour and are dimensioned from the datum.



Reference Dimensions

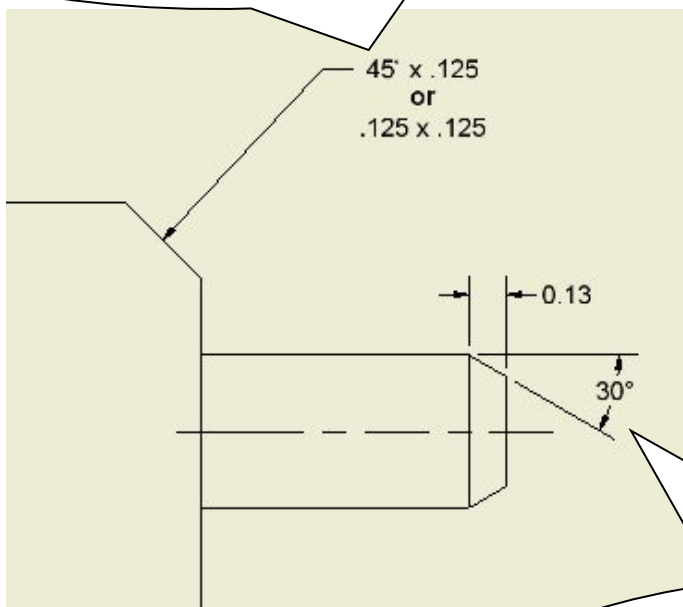
Designates more than one of the
same feature.

In this case, it is identifying there
are
two identical holes.

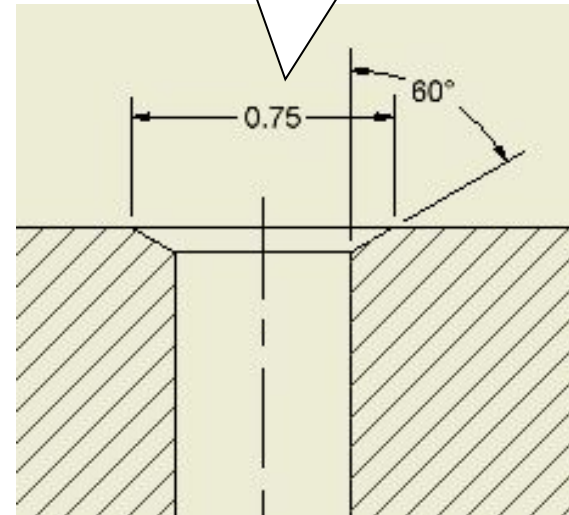


Chamfers

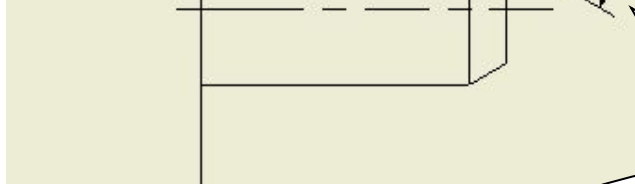
External chamfer for 45 degree chamfers only. There are two options.



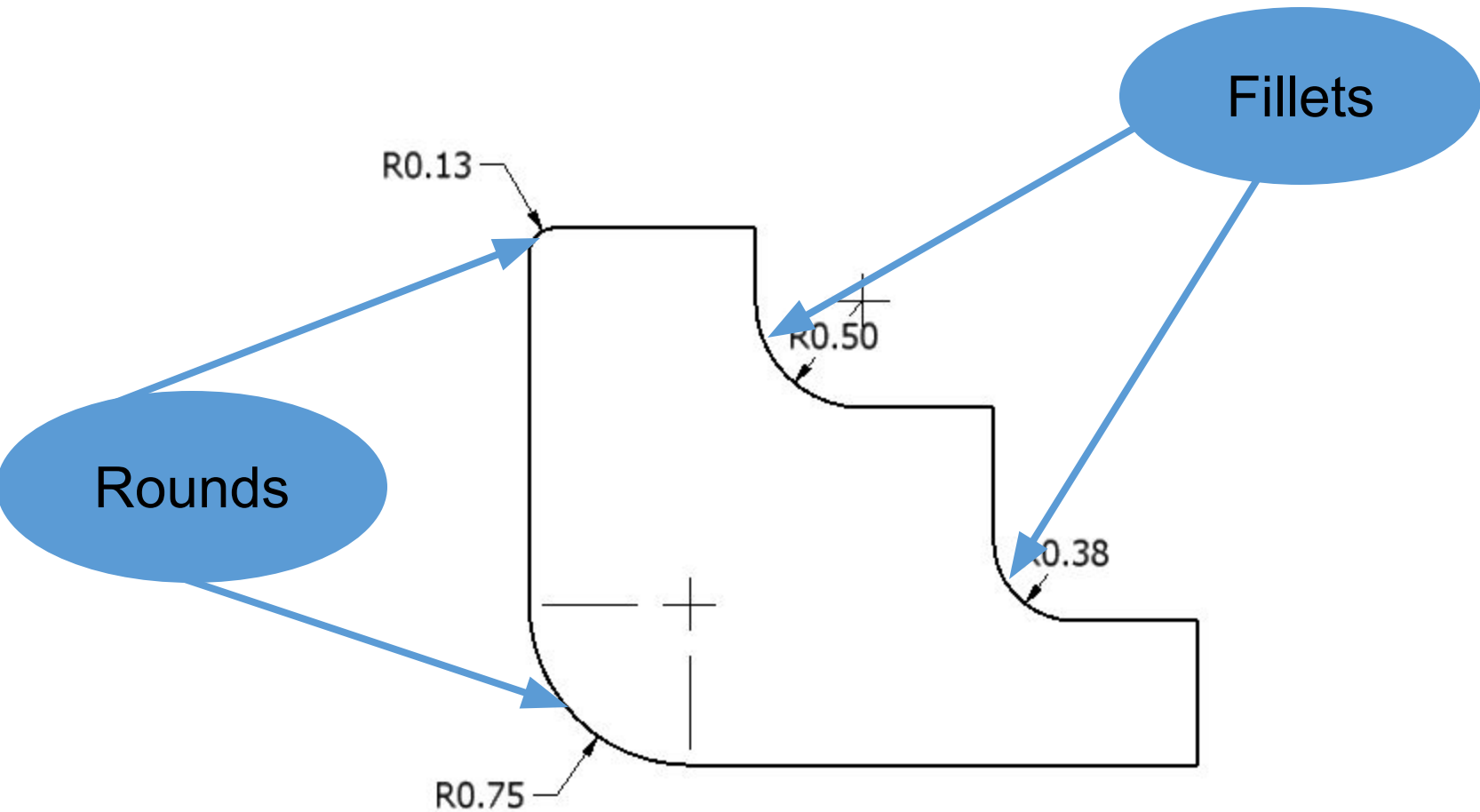
Internal chamfers.



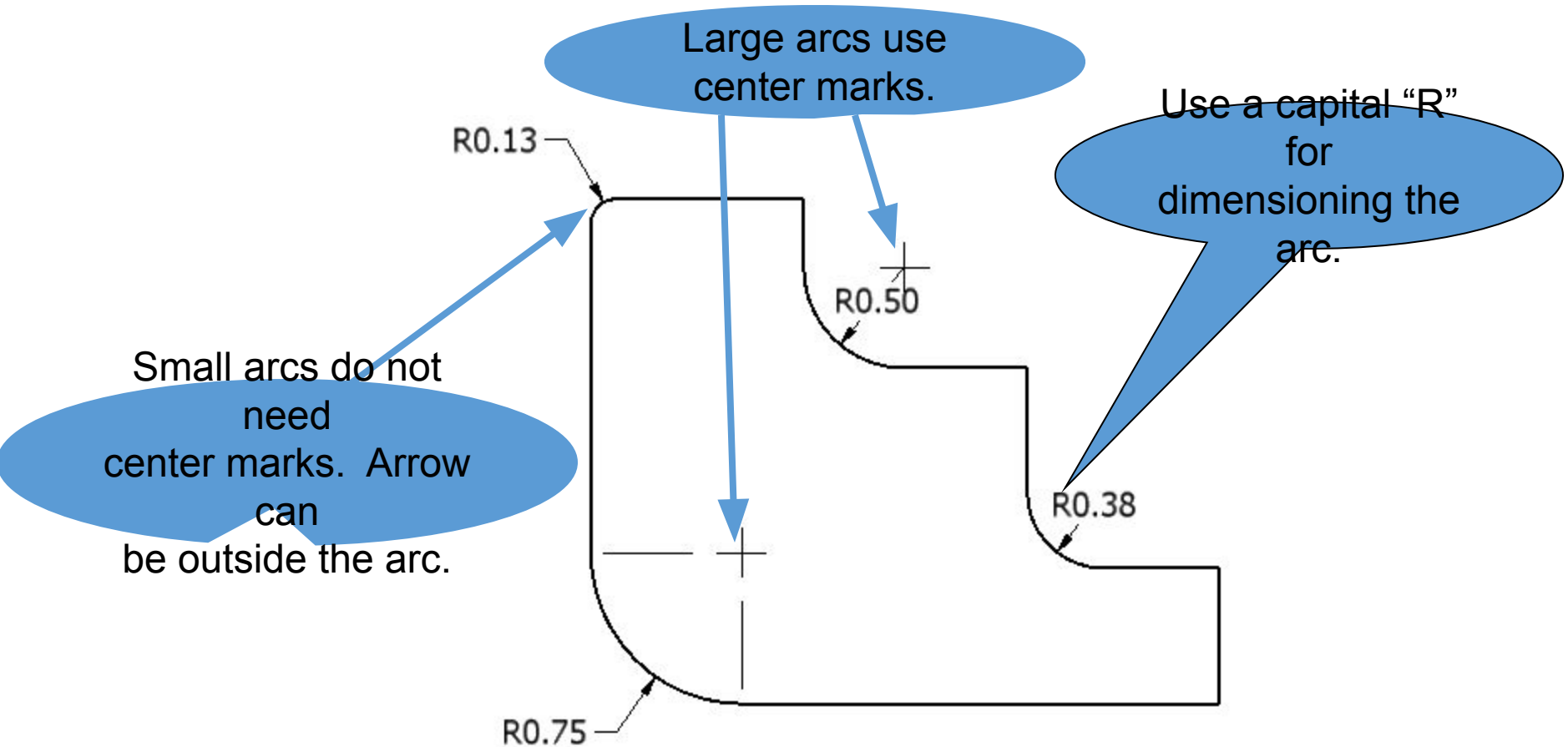
External chamfer for angles other than 45 degrees.



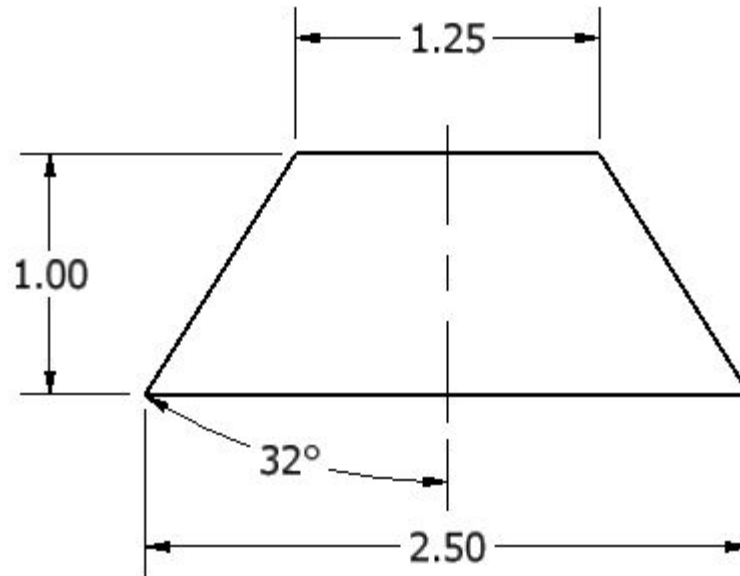
Fillets and Rounds



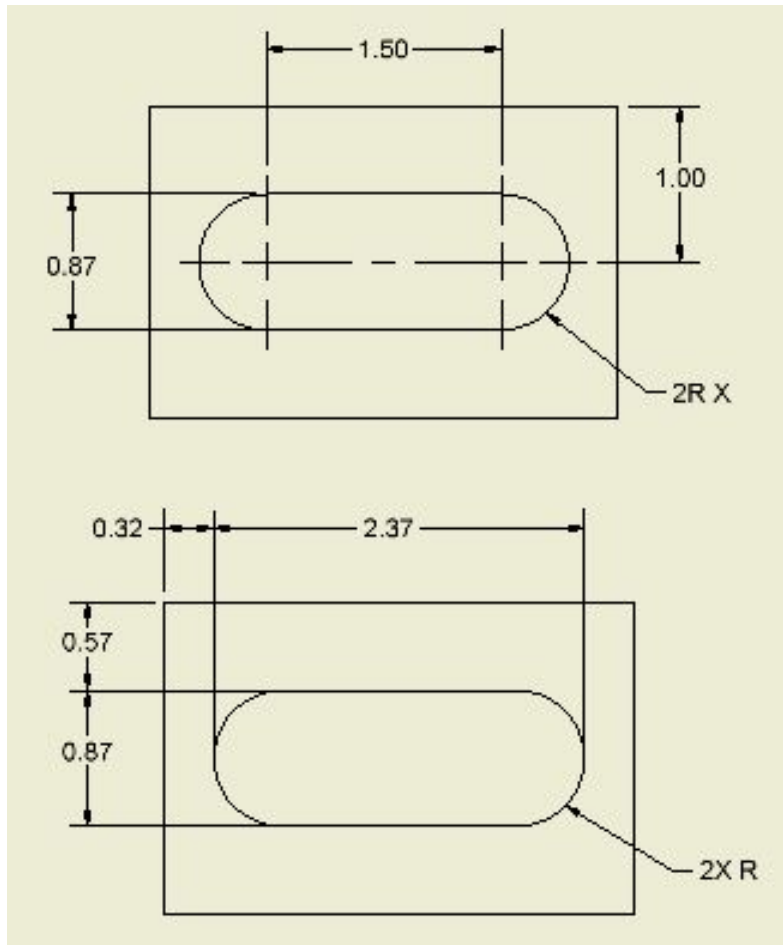
Fillets and Rounds



Conical Tapers



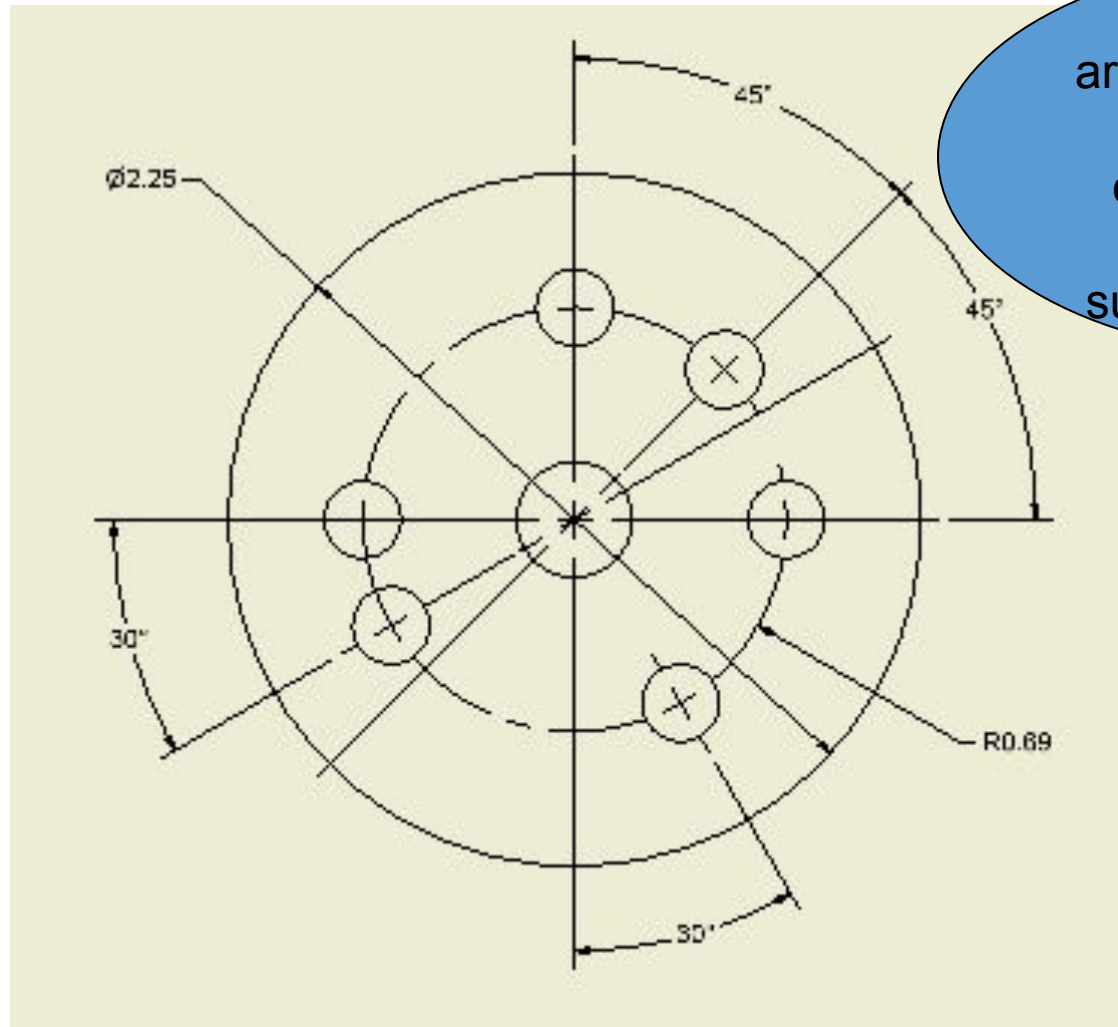
Slot Dimensioning



The two methods shown on the left are the acceptable methods for dimensioning slotted holes.

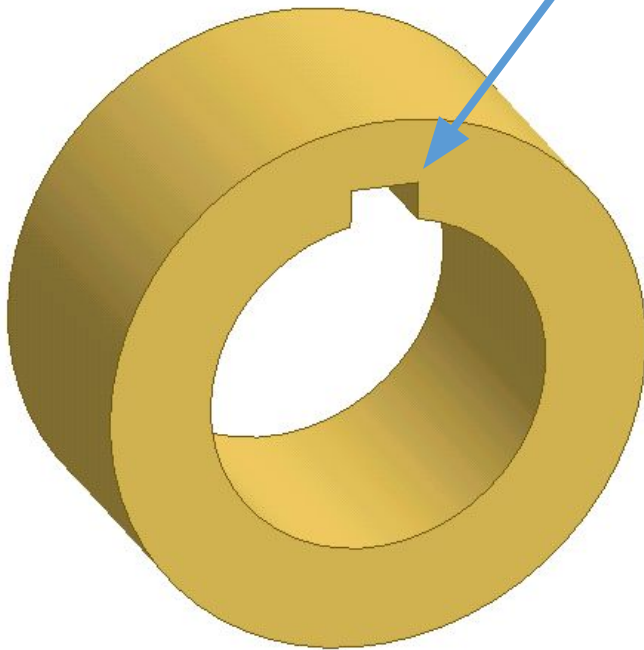
Dimensioning Radial Patterns

Angles and radius values are used to locate the centers of radial patterned features, such as the holes on this plate.

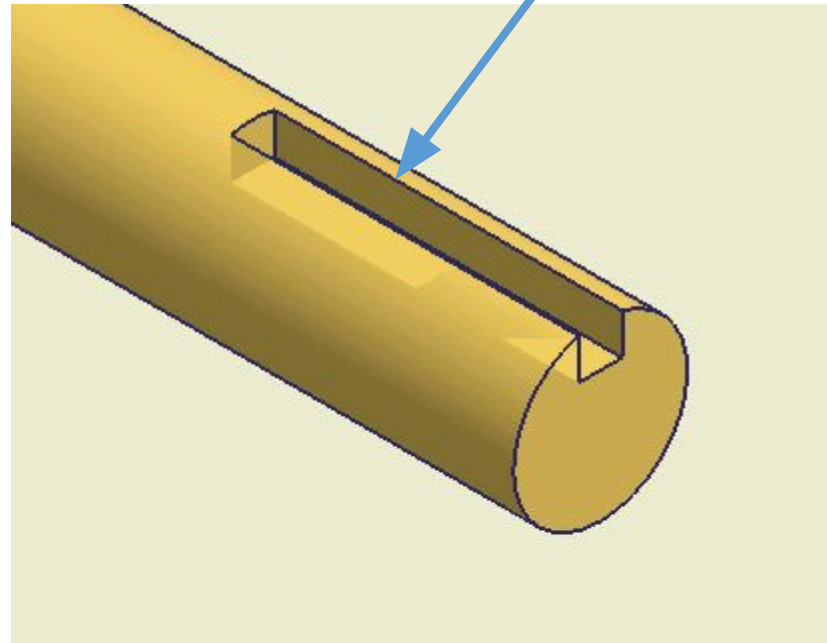


Keyway and Keyseat

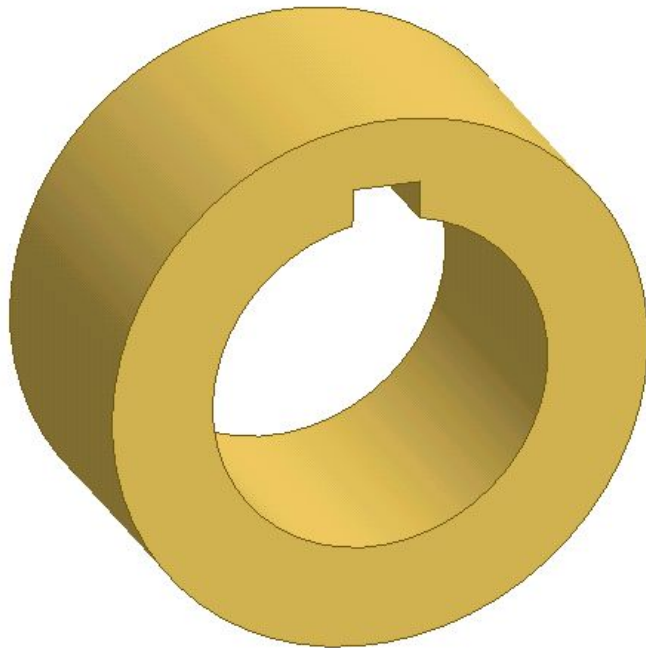
Keyway



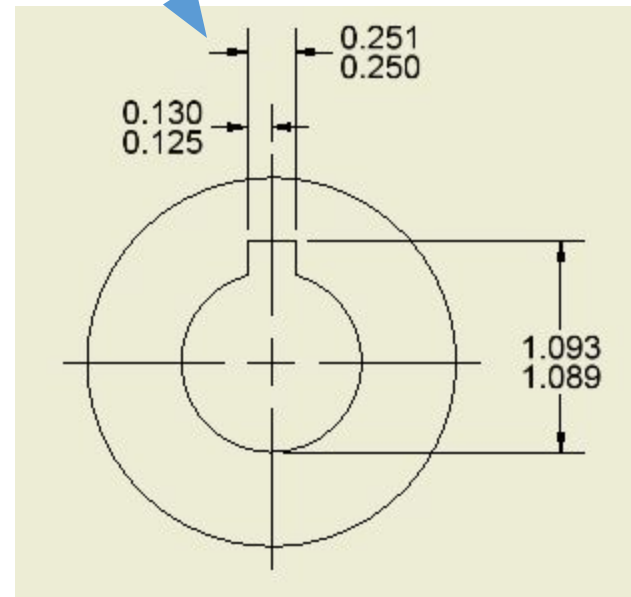
Keyseat



Keyways

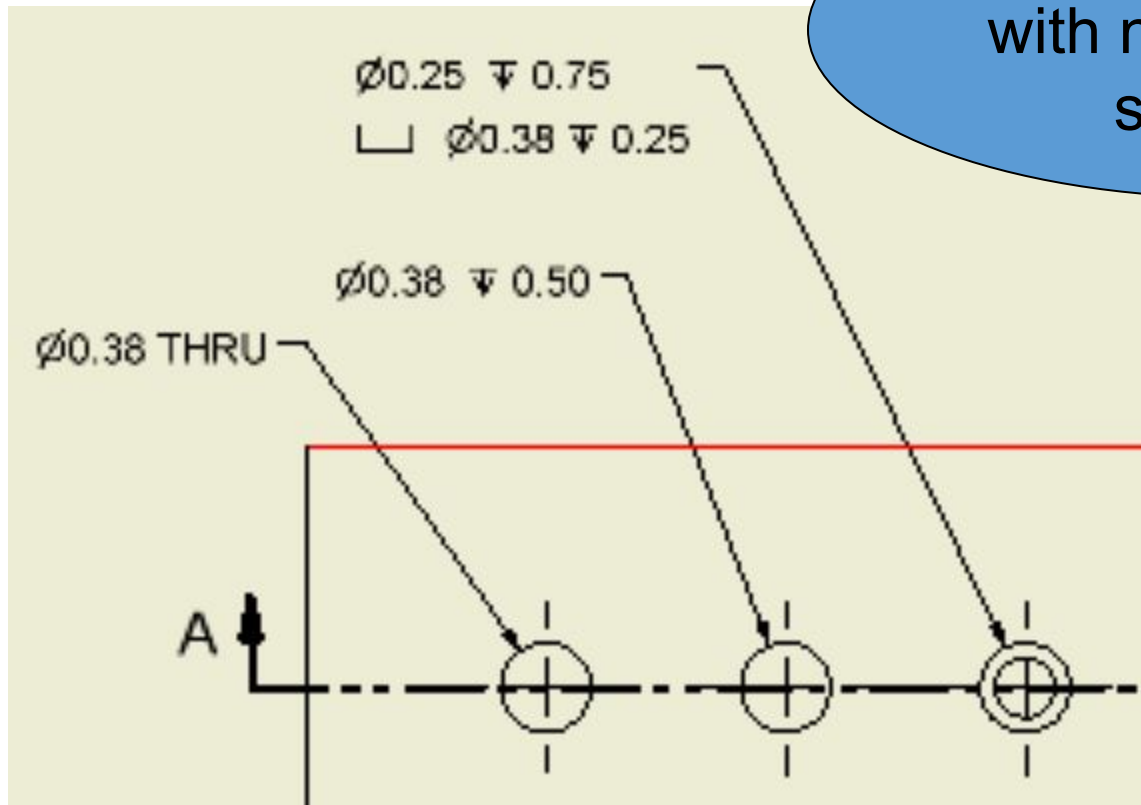


Keyway Dimensions



Hole Dimensioning

Holes are specified with numbers and symbols.



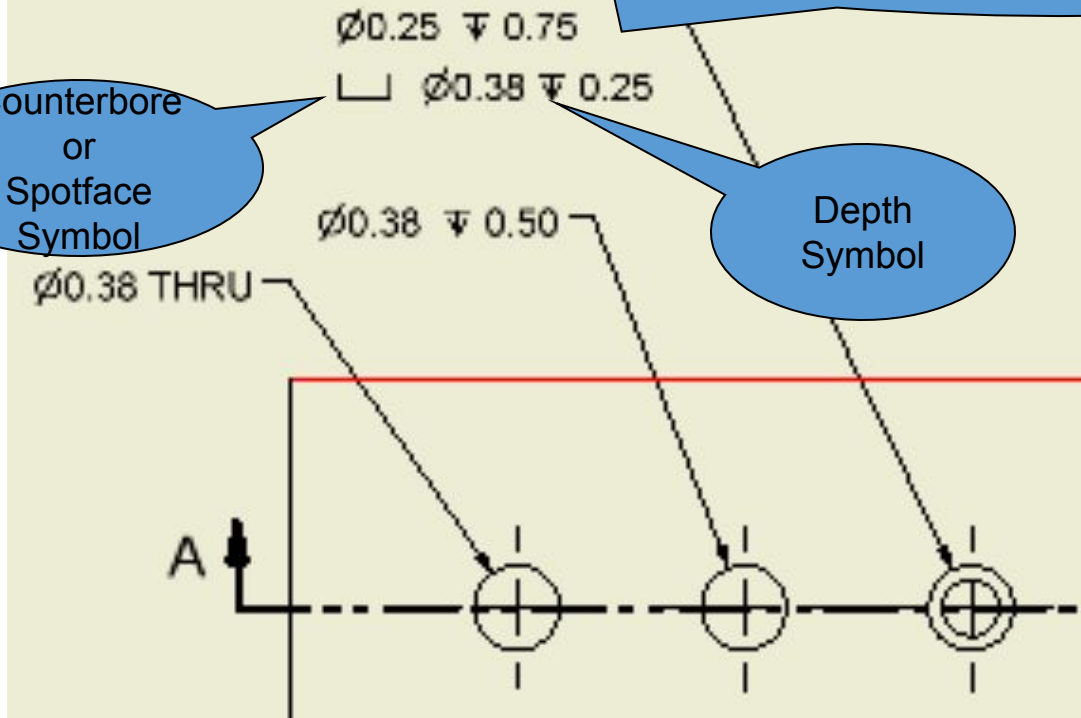
Reading a Hole Note

The Hole Diameter is .25" and will be drilled
.75" deep.

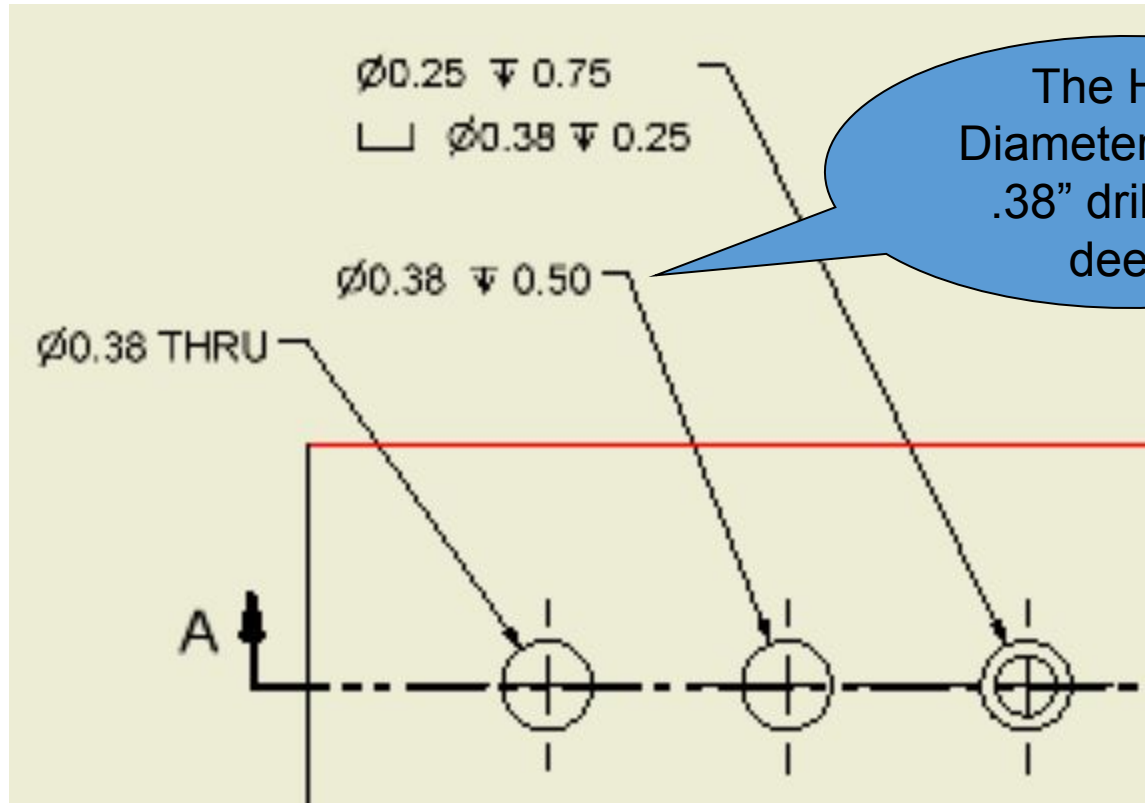
The Hole will be Counterbored to
a .38" diameter and to a depth of .25"

Counterbore
or
Spotface
Symbol

Depth
Symbol

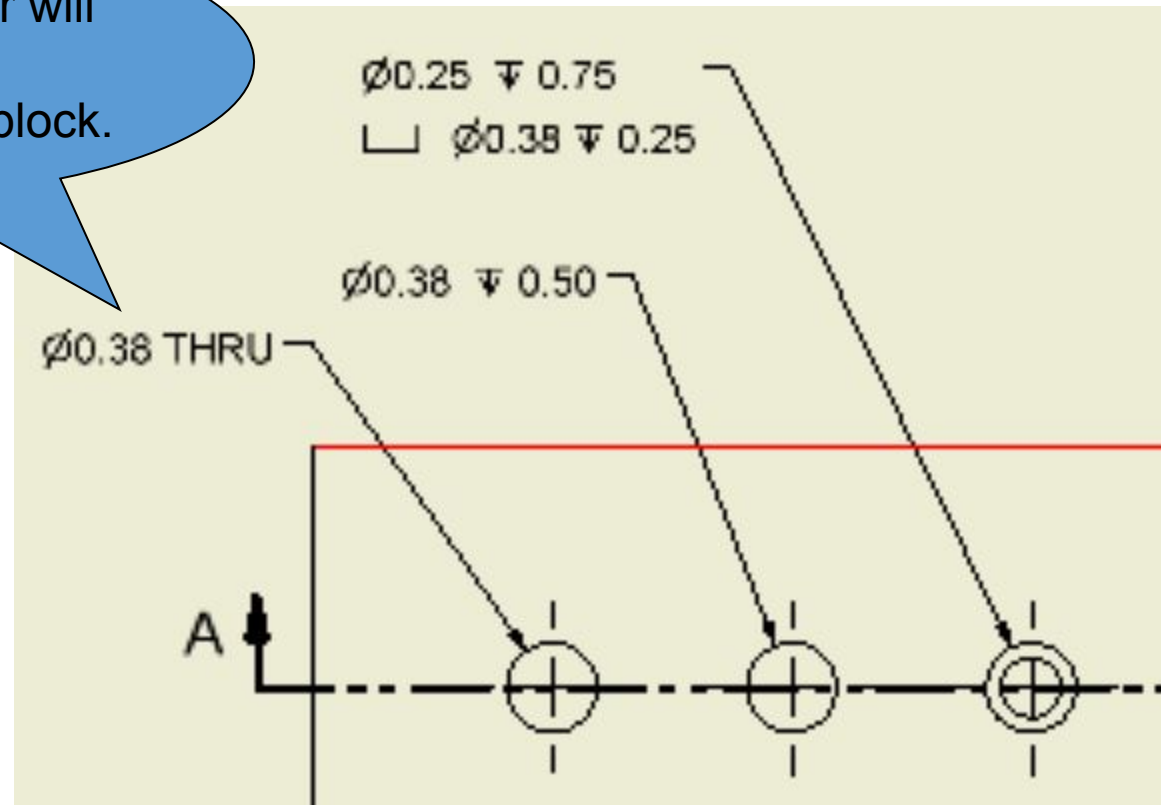


Reading a Hole Note

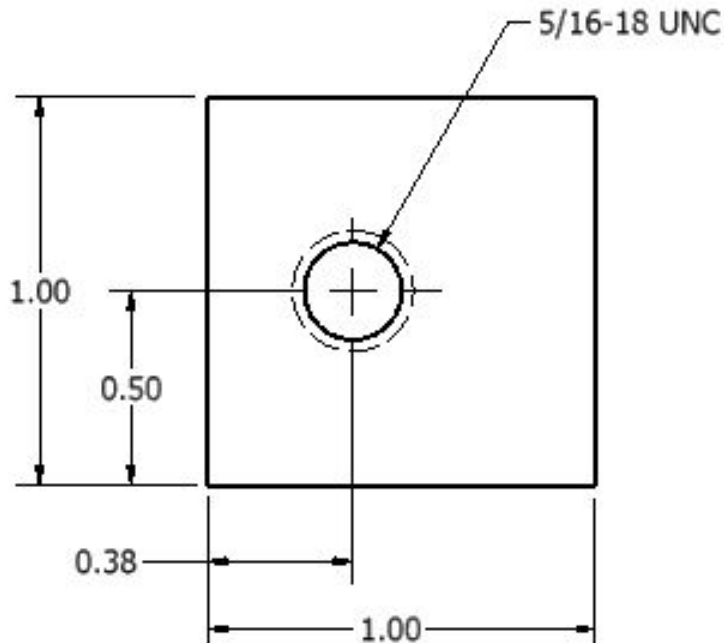


Reading a Hole Note

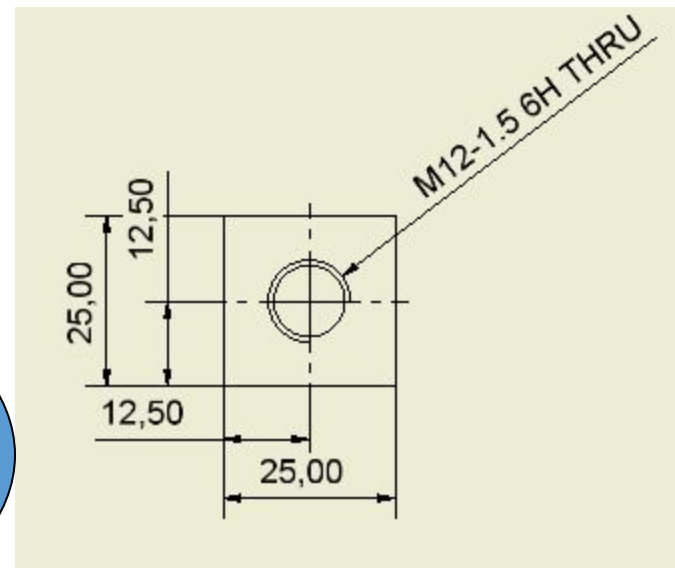
The Hole Diameter will be .38" through the whole block.



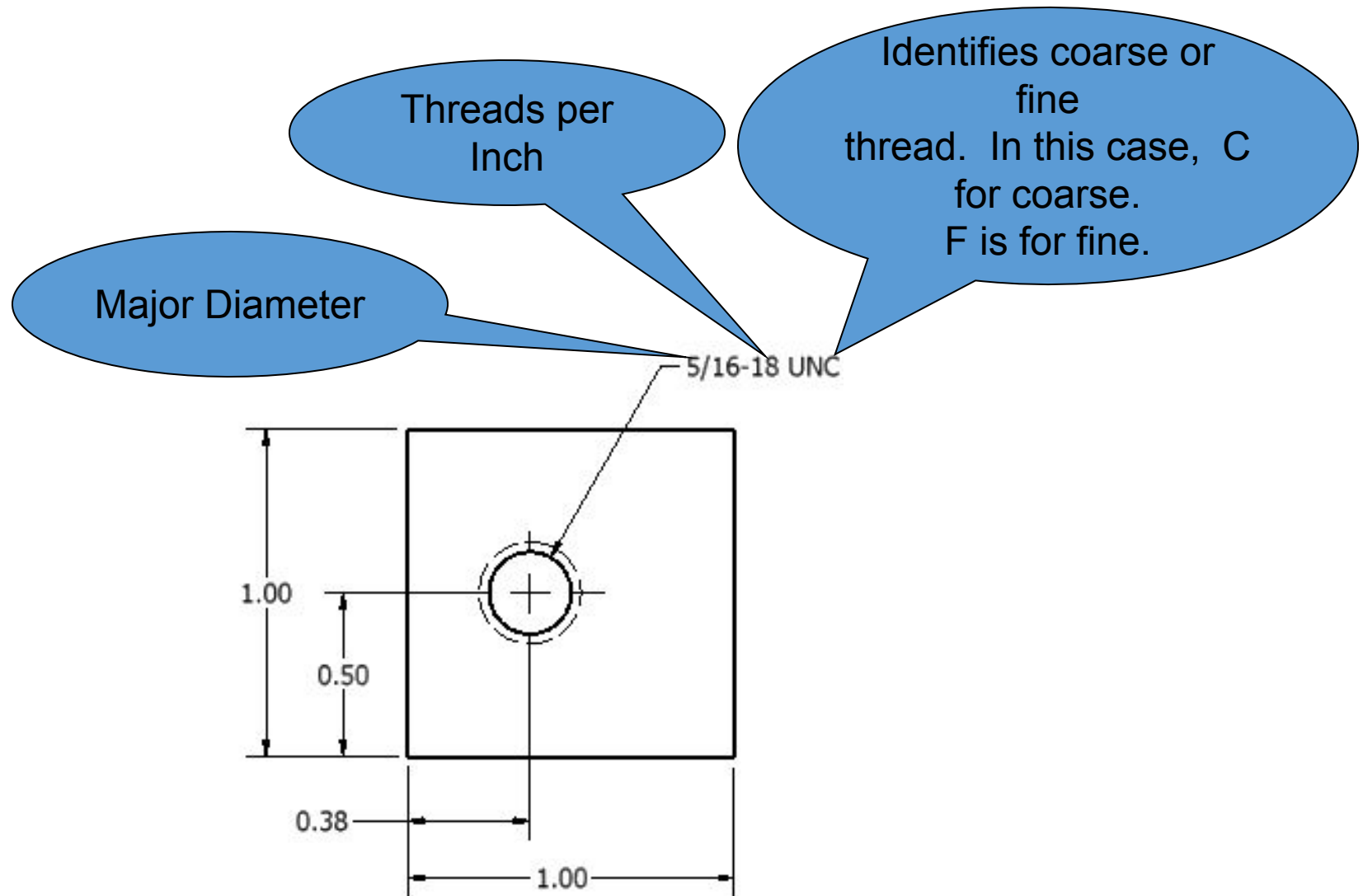
Reading Thread Notes



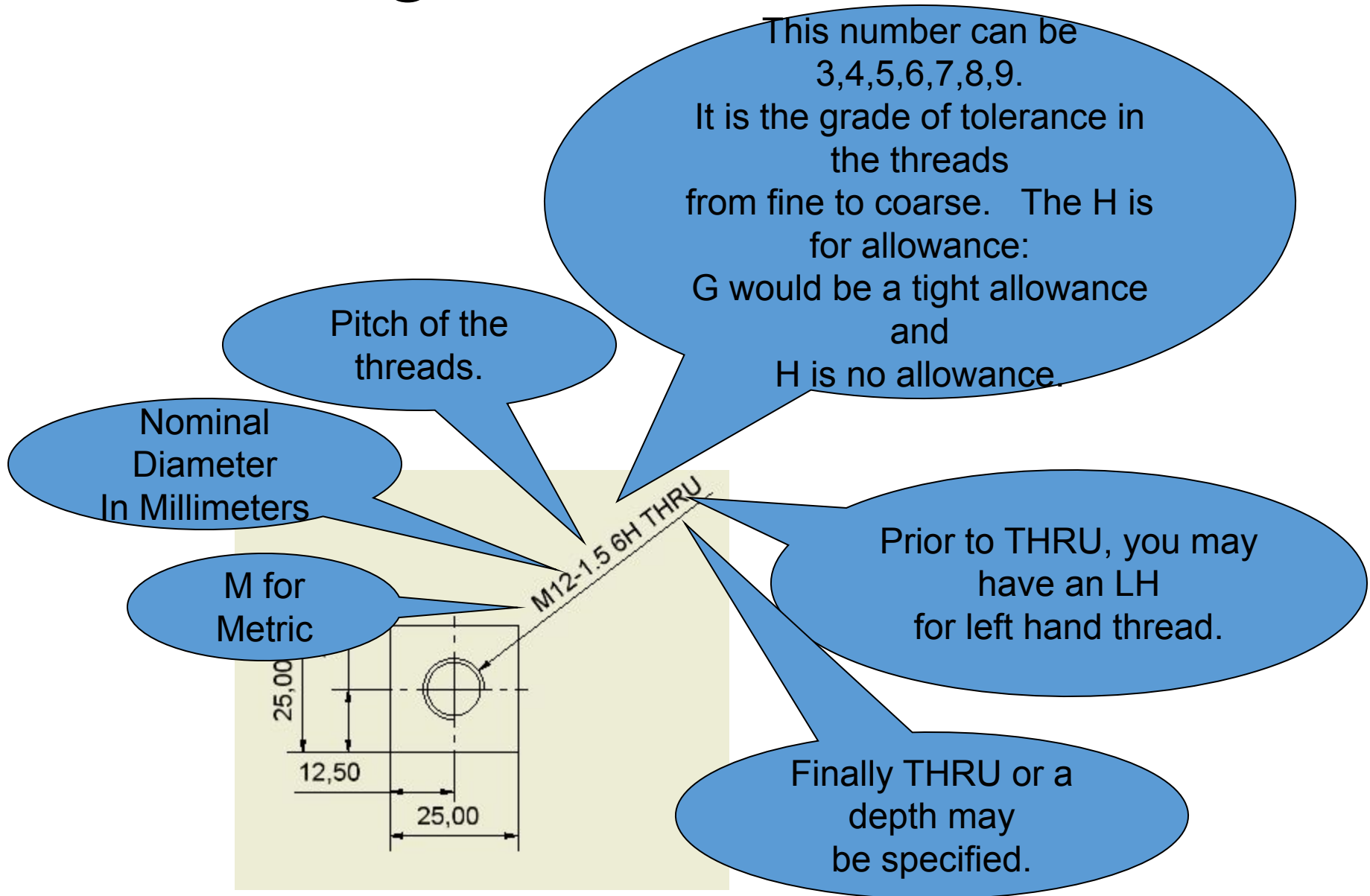
Threads are dimensioned with the use of local notes. We will discuss two methods: the ISO and the Unified National Thread method.



Reading a Unified National Thread Note



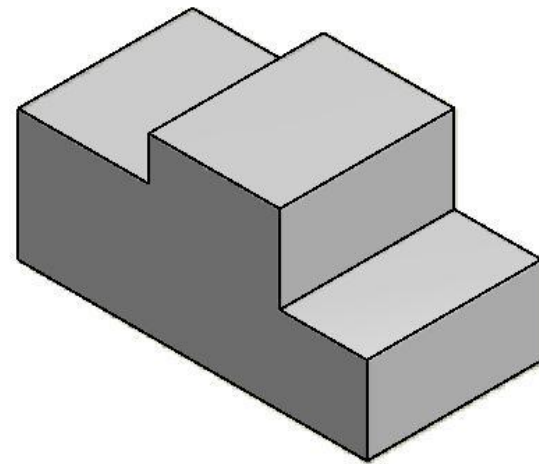
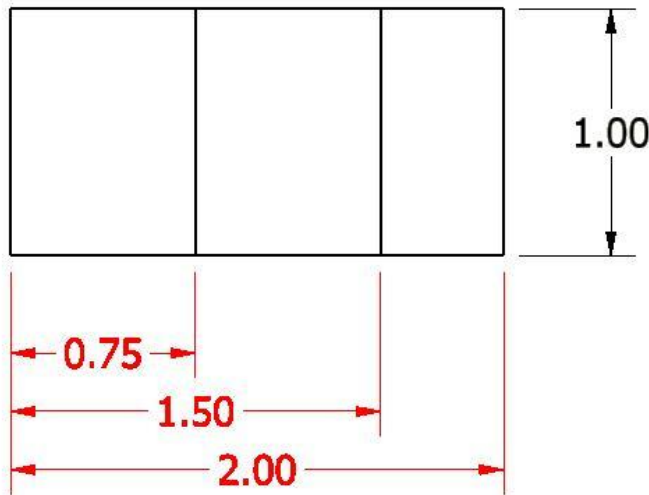
Reading a ISO Thread Notes



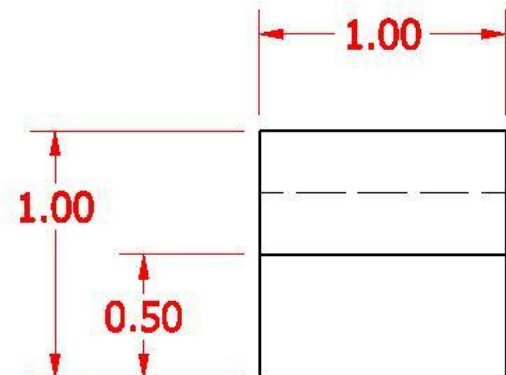
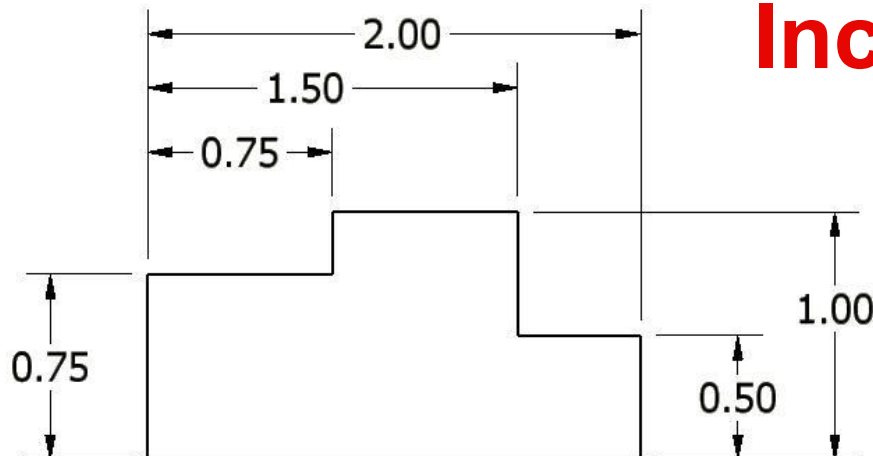
Dimension Guidelines



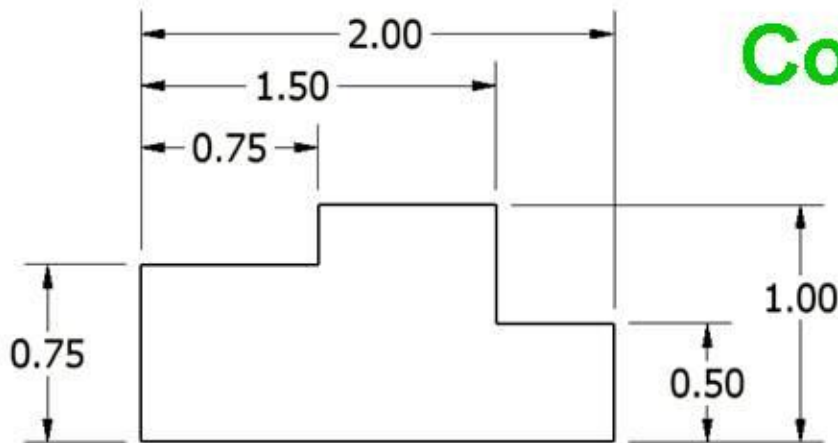
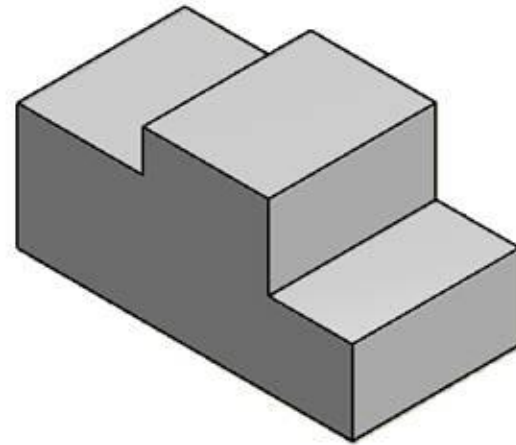
1. Dimensions should NOT be duplicated, or the same information given in two different ways.



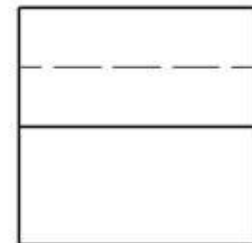
Incorrect



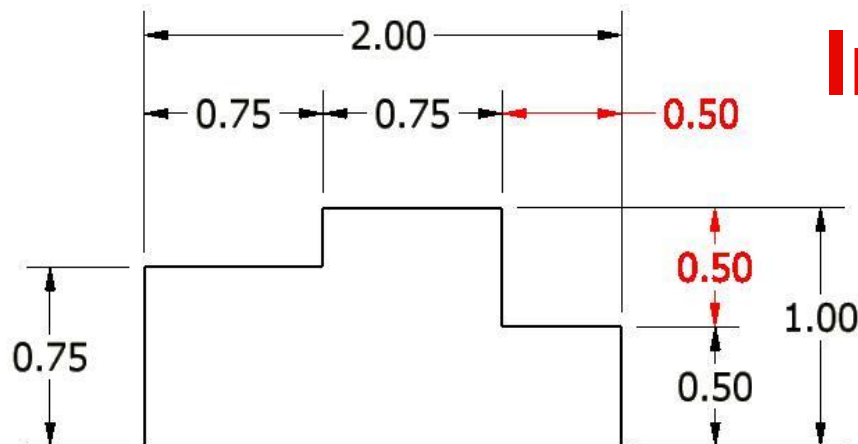
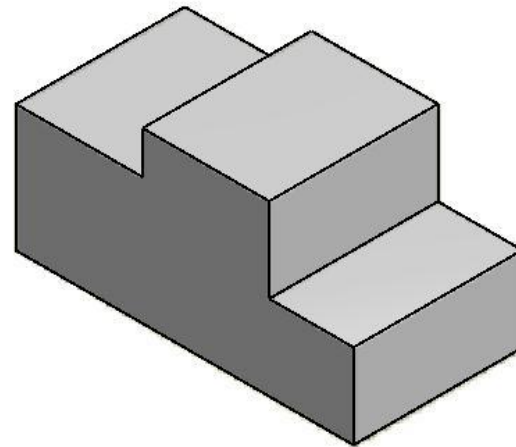
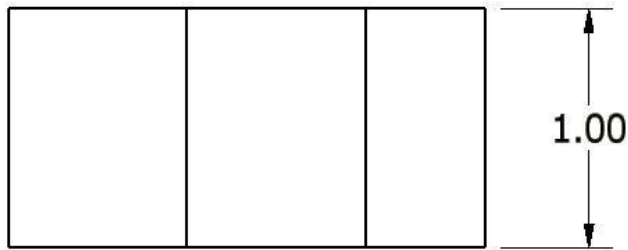
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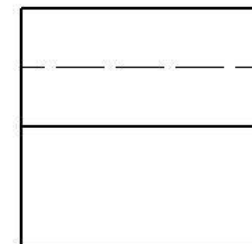
Correct



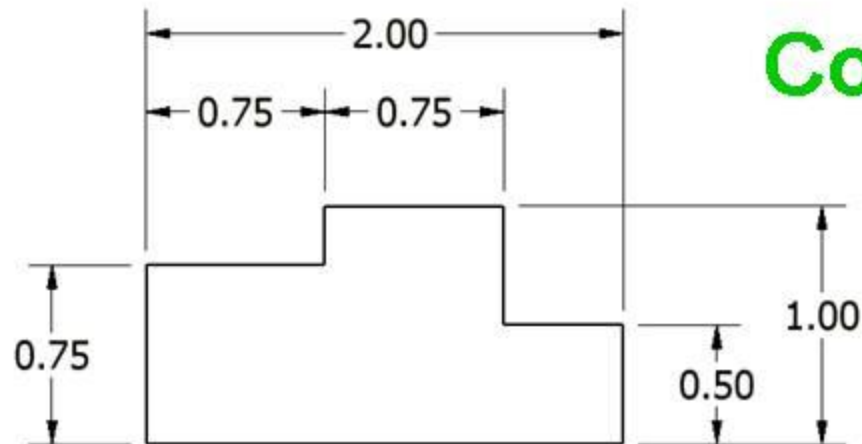
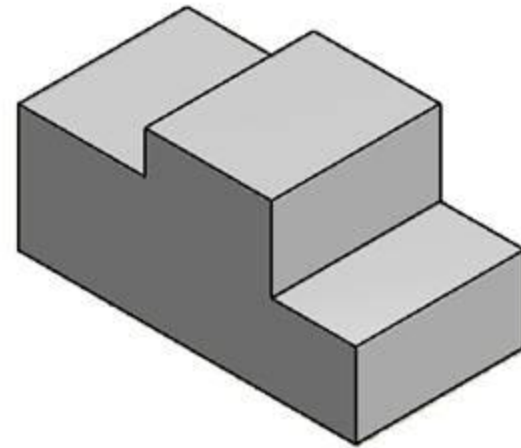
2. No unnecessary dimensions should be used – only those needed to produce or inspect the part.



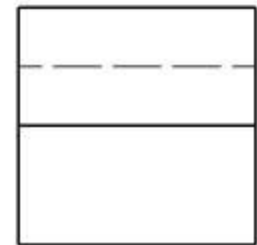
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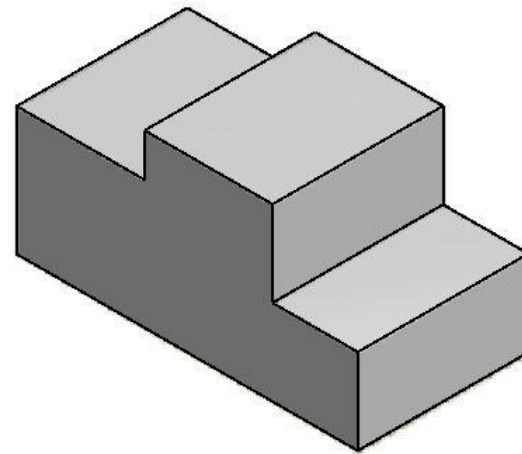
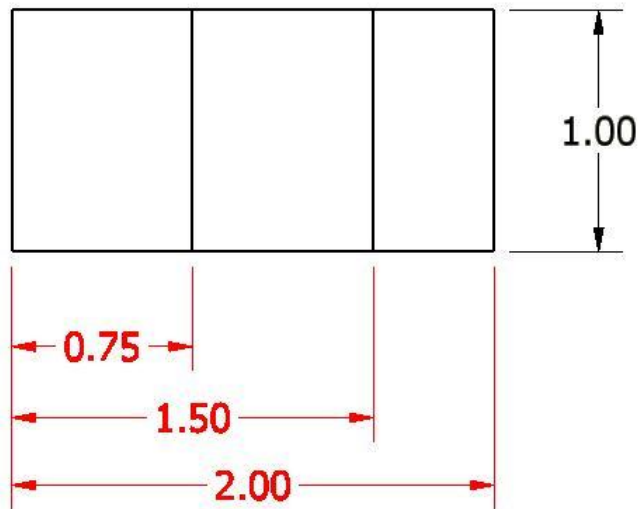
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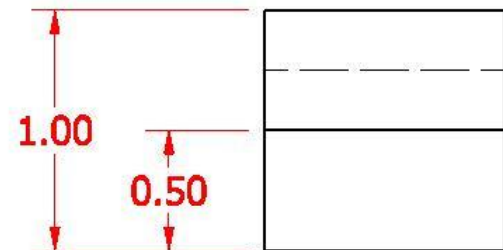
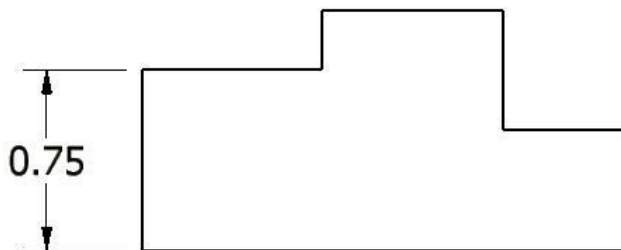
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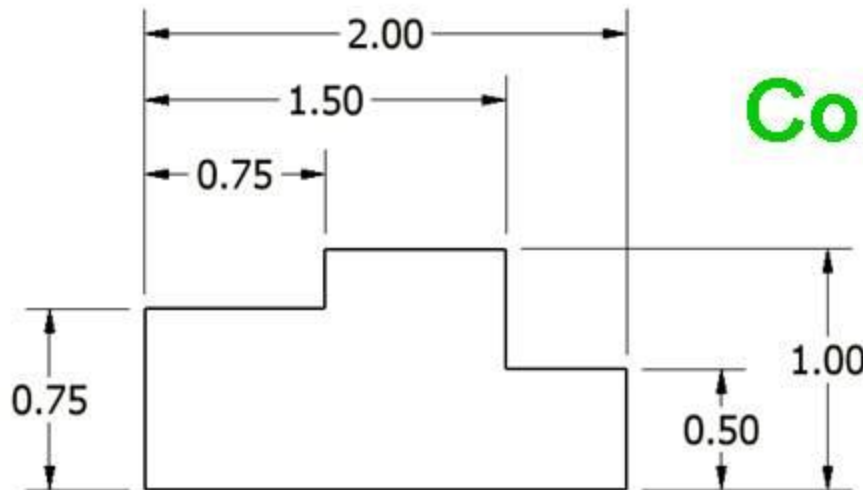
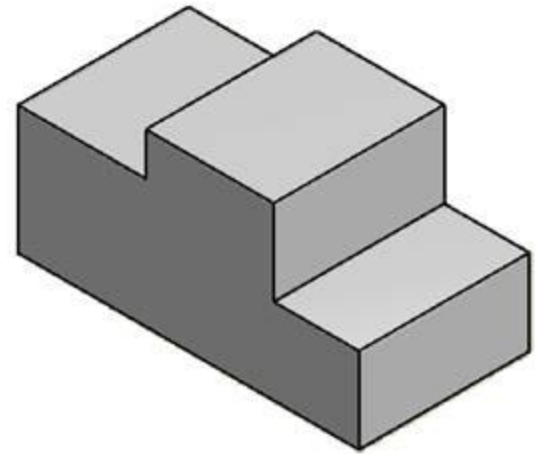
3. Dimensions should be attached to the view that best shows the ***contour*** of the feature to be dimensioned.



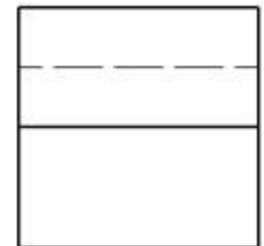
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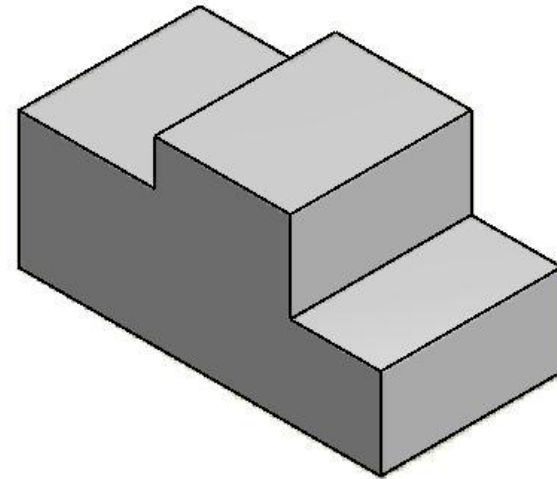
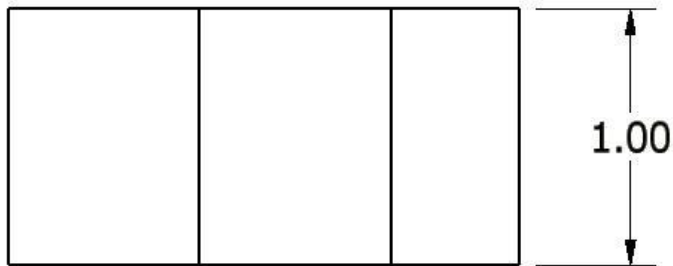
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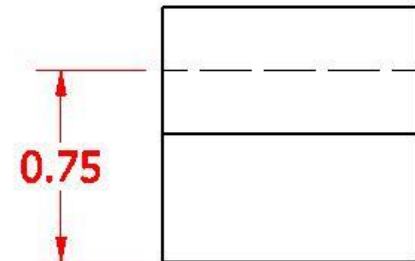
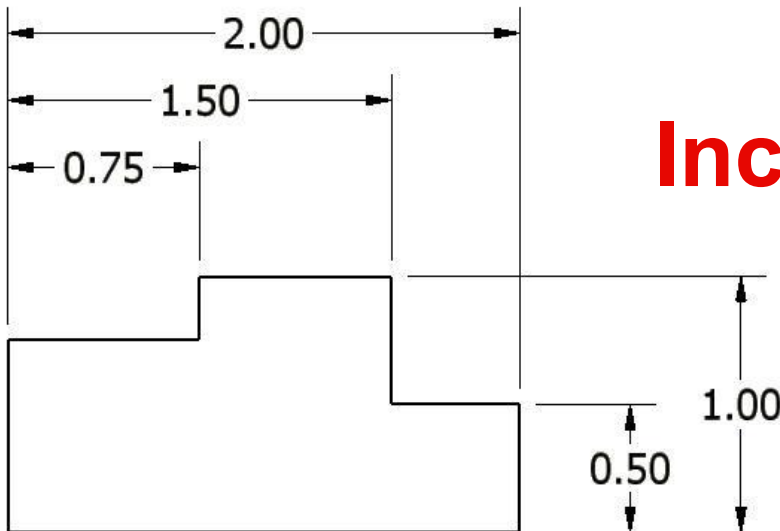
Correct



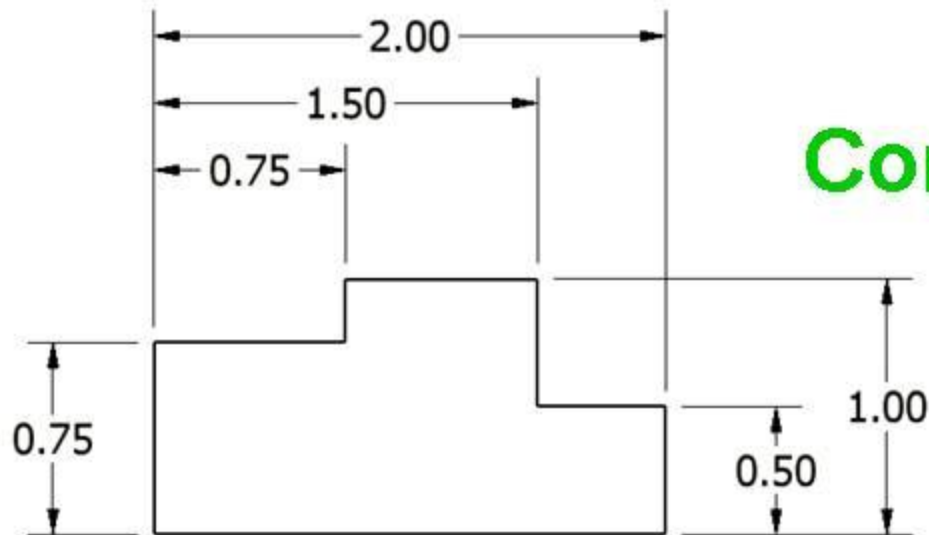
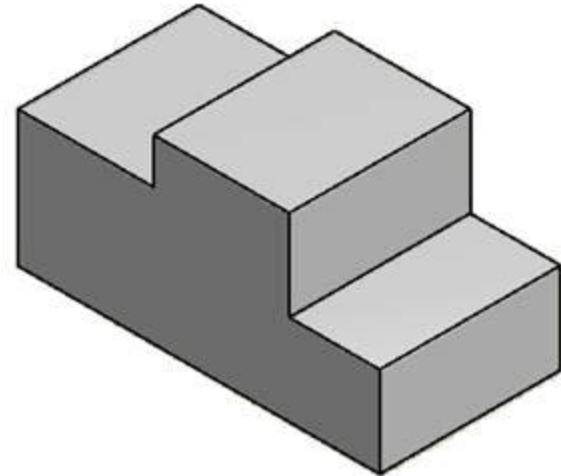
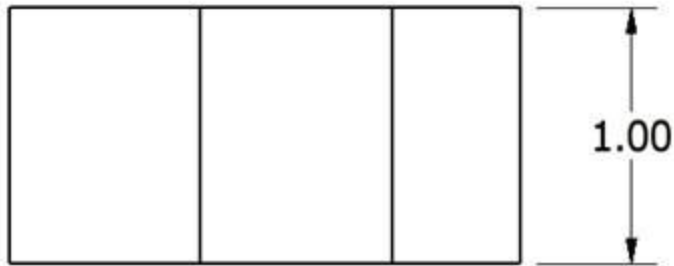
4. Whenever possible, avoid dimensioning to hidden lines and features.



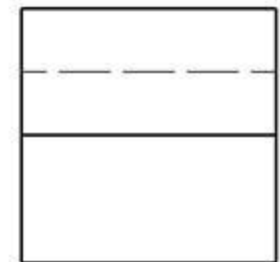
Incorrect



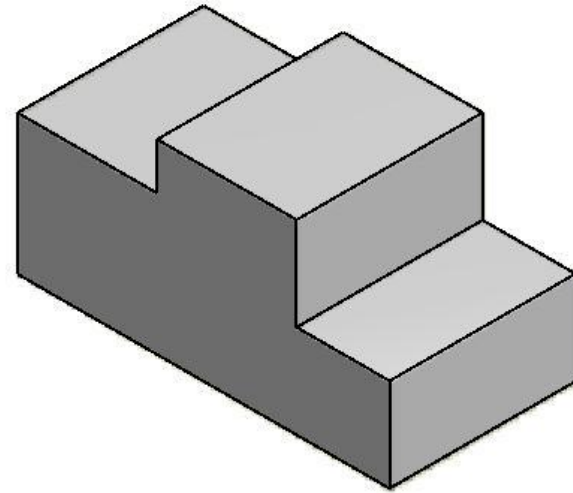
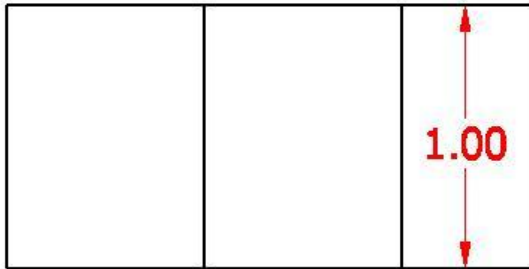
4. Whenever possible, avoid dimensioning to hidden lines and features.



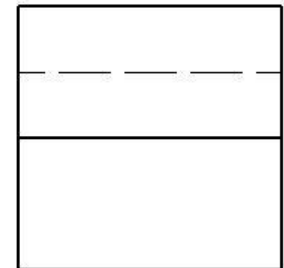
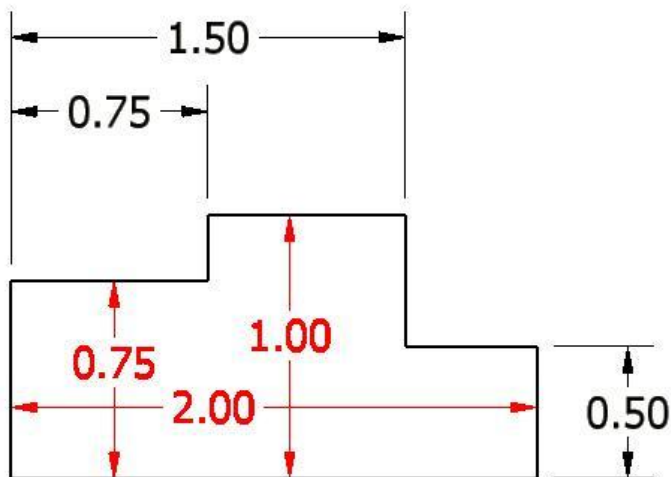
Correct



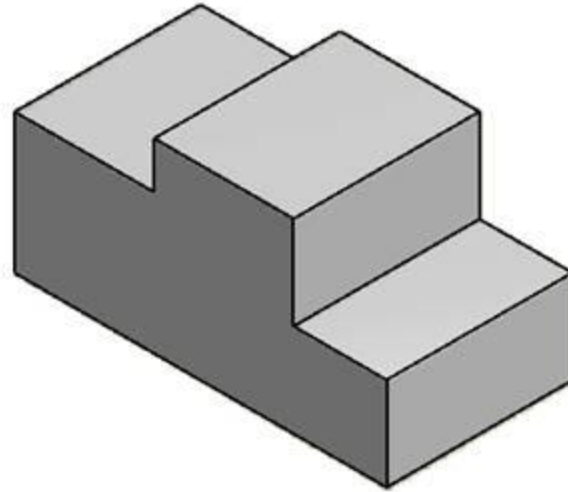
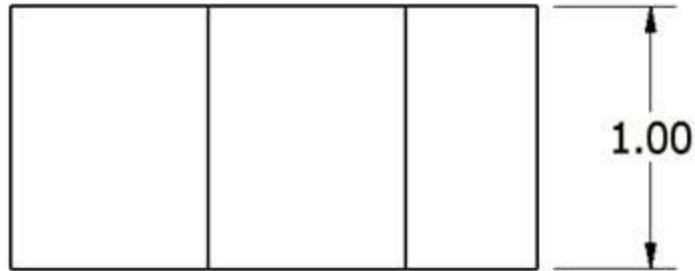
5. Avoid dimensioning over or through the object.



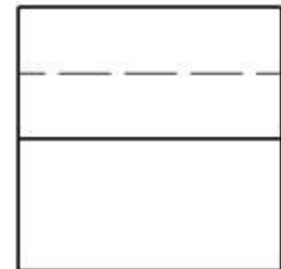
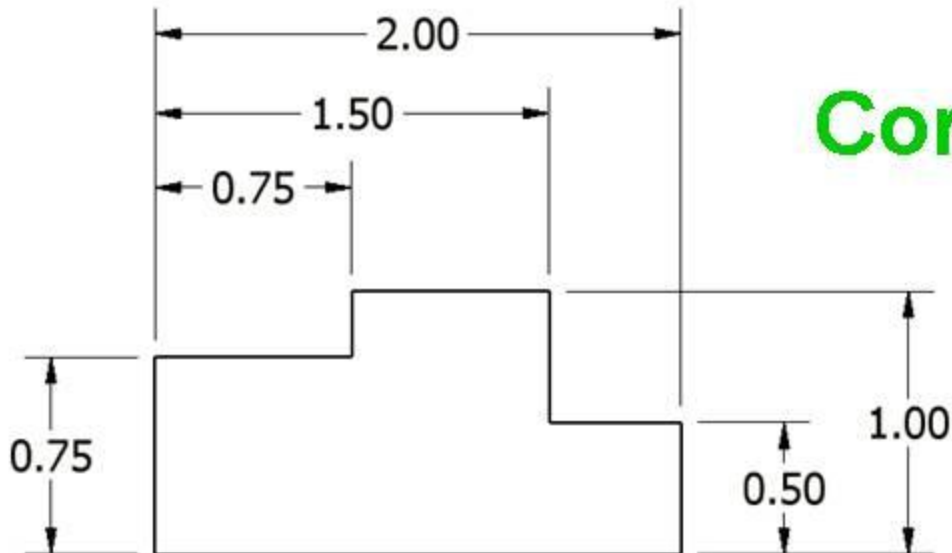
Incorrect



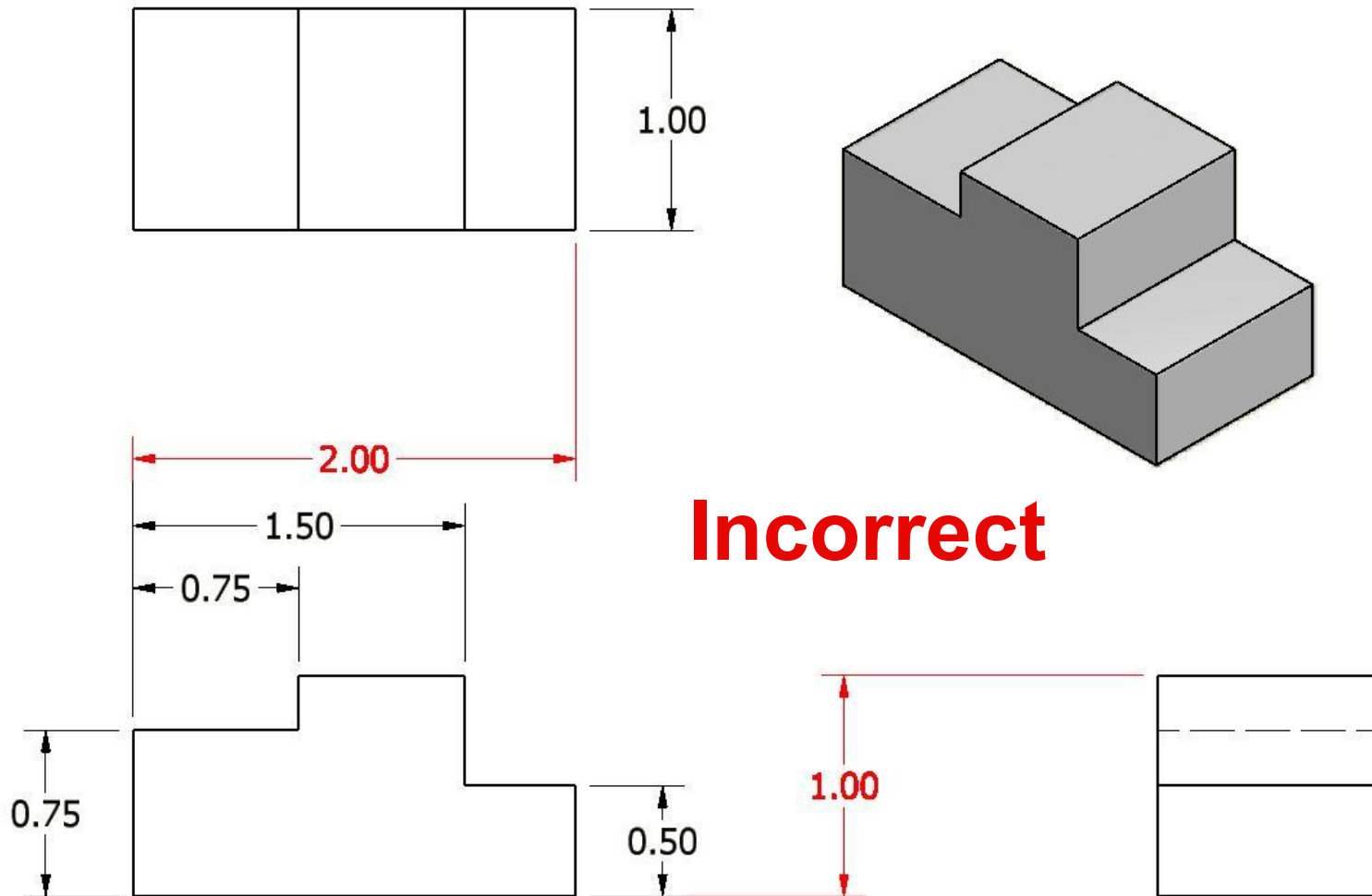
5. Avoid dimensioning over or through the object.



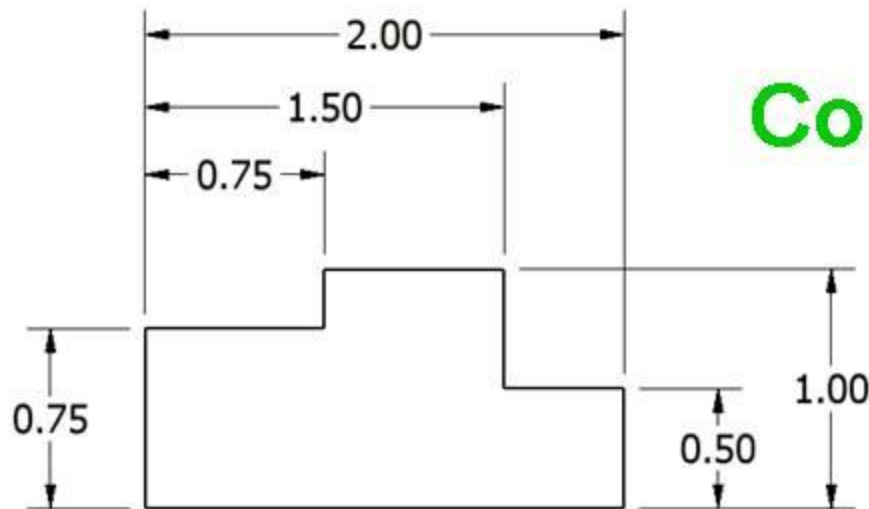
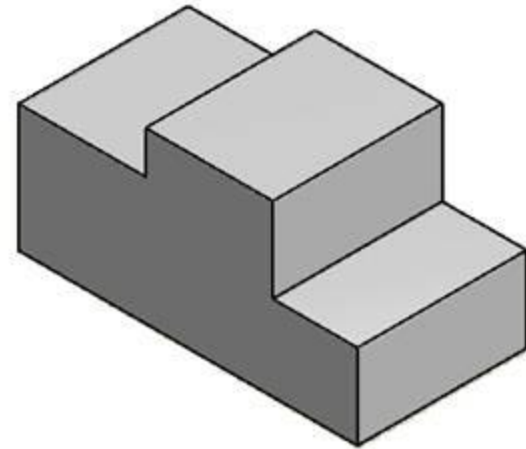
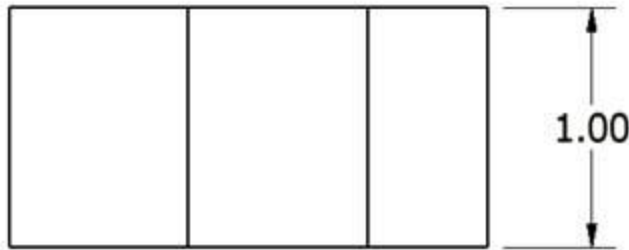
Correct



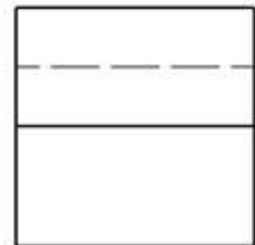
6. A dimension should be attached to only one view; for example, extension lines should not connect two views.



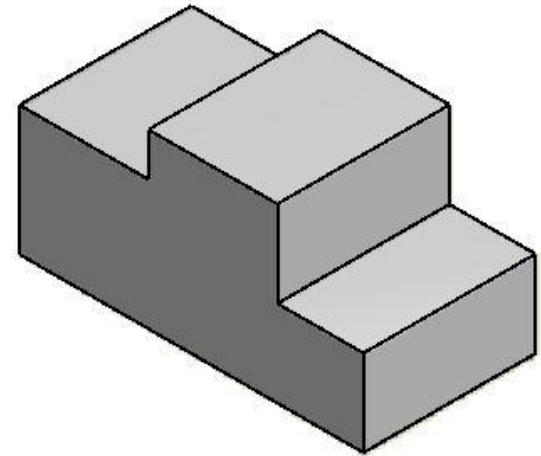
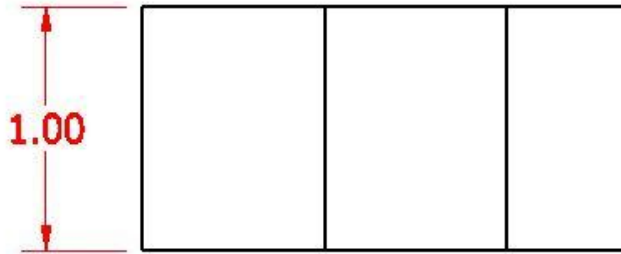
6. A dimension should be attached to only one view; for example, extension lines should not connect two views.



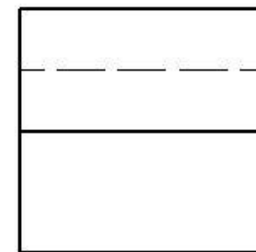
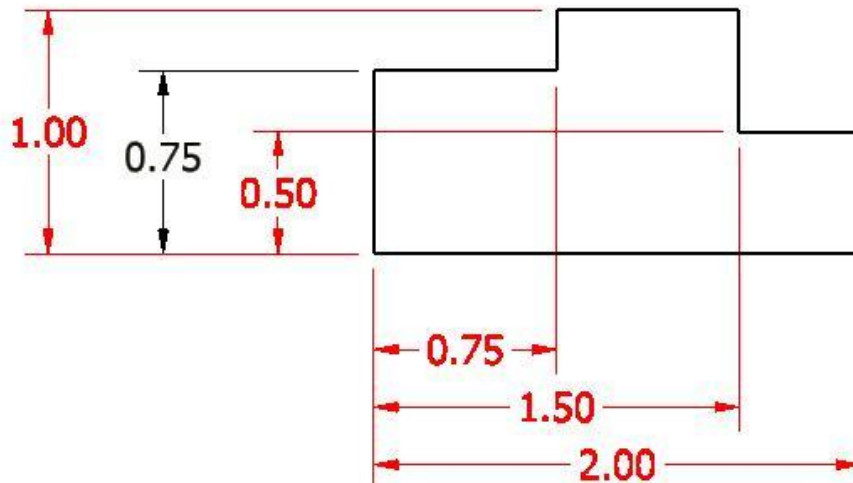
Correct



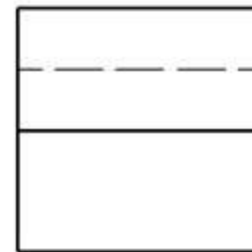
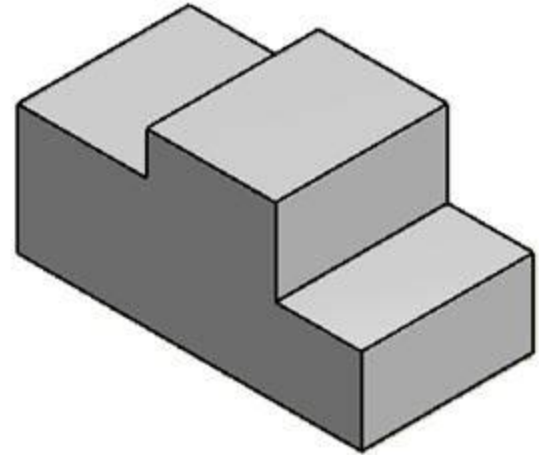
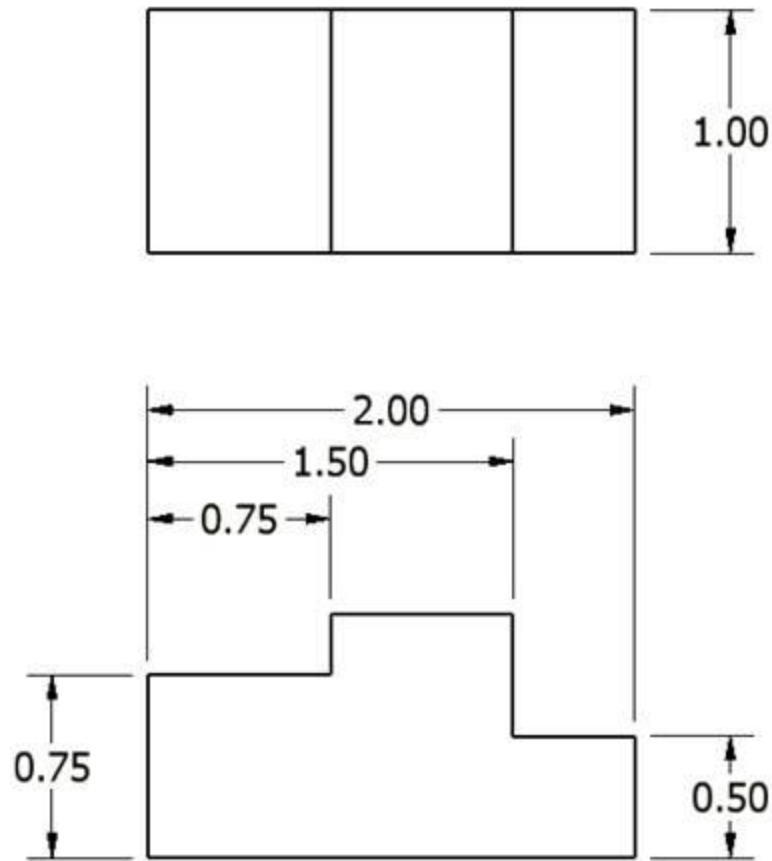
7. Whenever possible, locate dimensions between adjacent views.



Incorrect

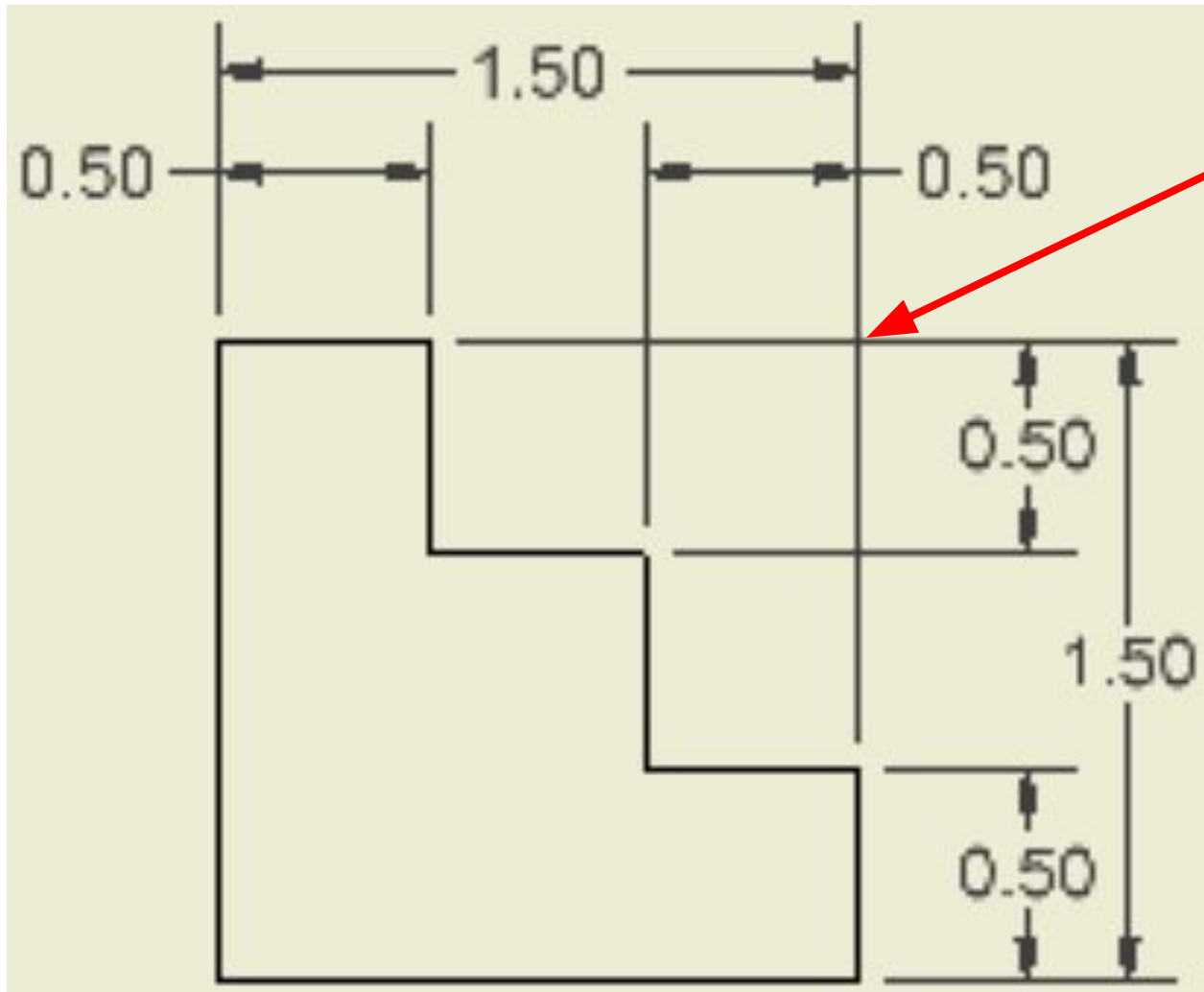


7. Whenever possible, locate dimensions between adjacent views.



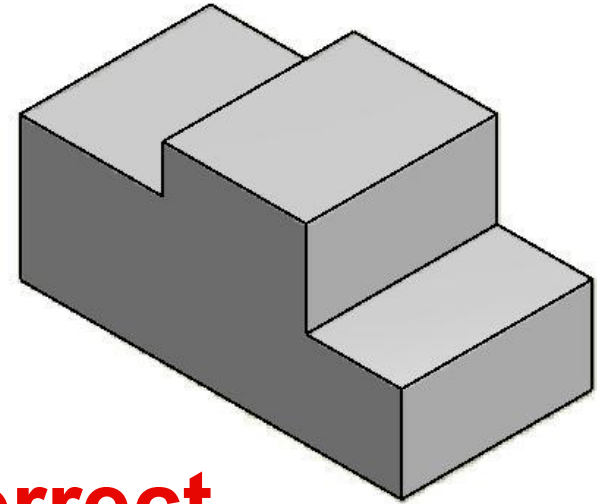
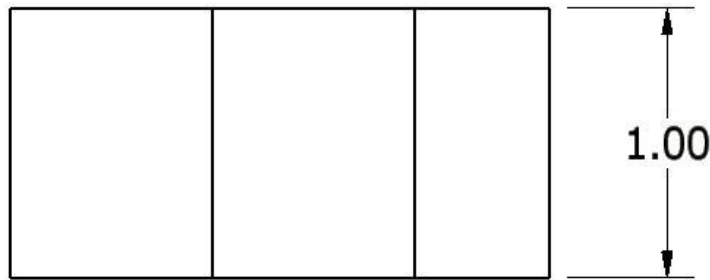
Correct

8. Avoid crossing extension lines, but do not break them when they do cross.

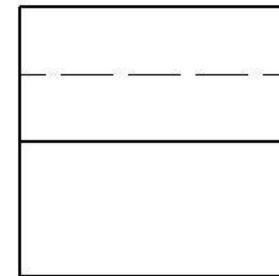
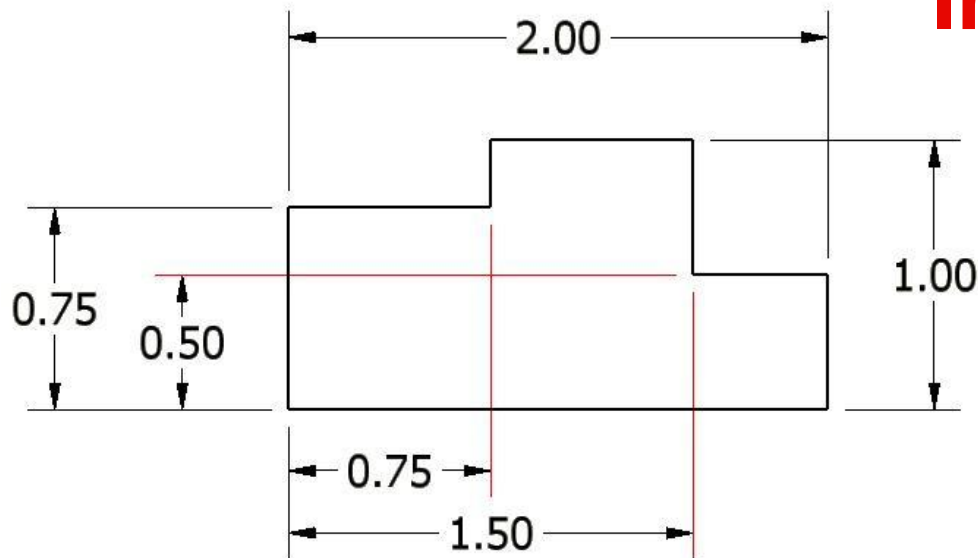


Multiple extension line crossings may be confused for the outside corner of the part.

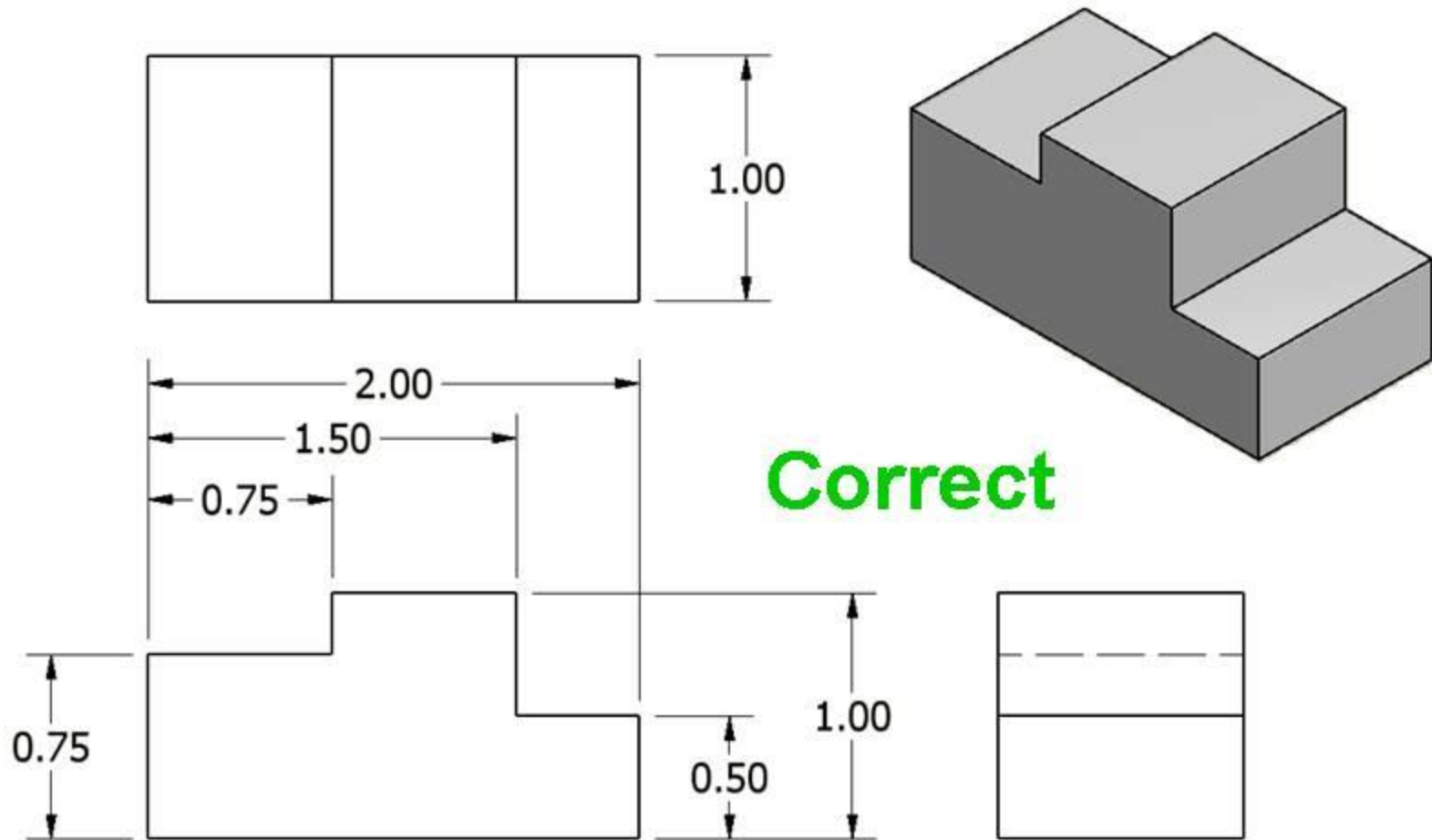
9. Whenever possible, avoid sending extension lines through object views.



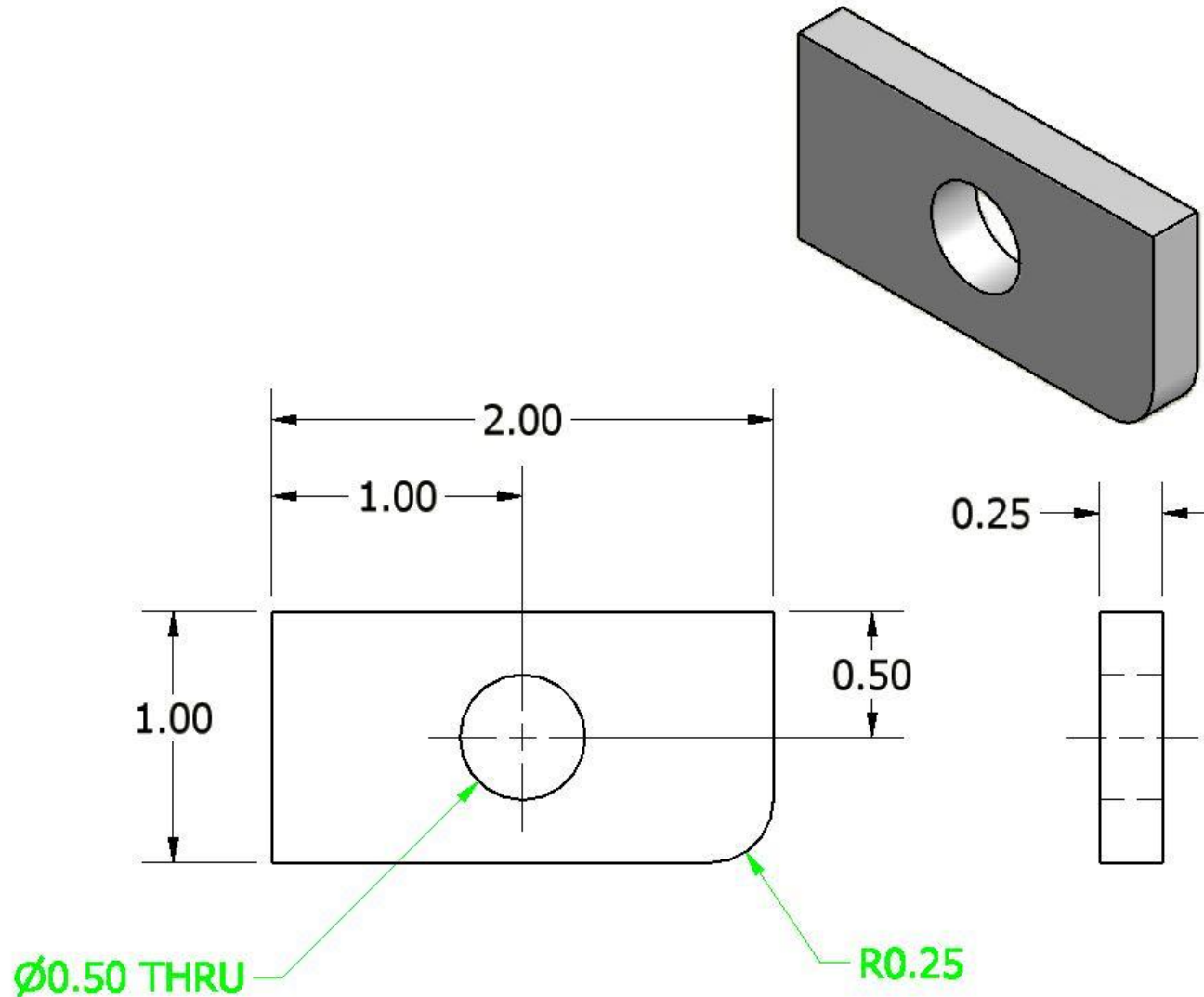
Incorrect



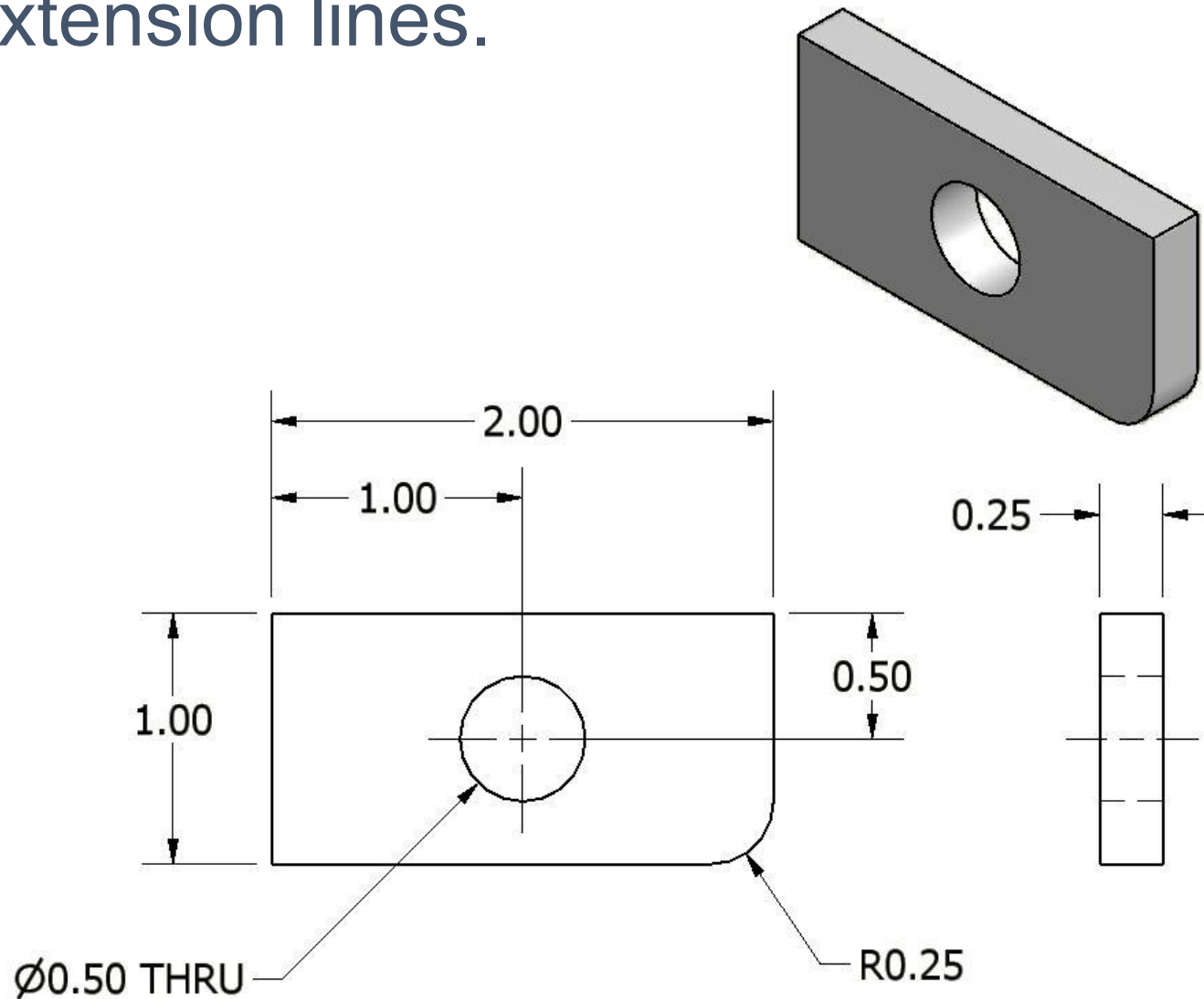
9. Whenever possible, avoid sending extension lines through object views.



10. In general, a circle is dimensioned by its diameter and an arc by its radius.

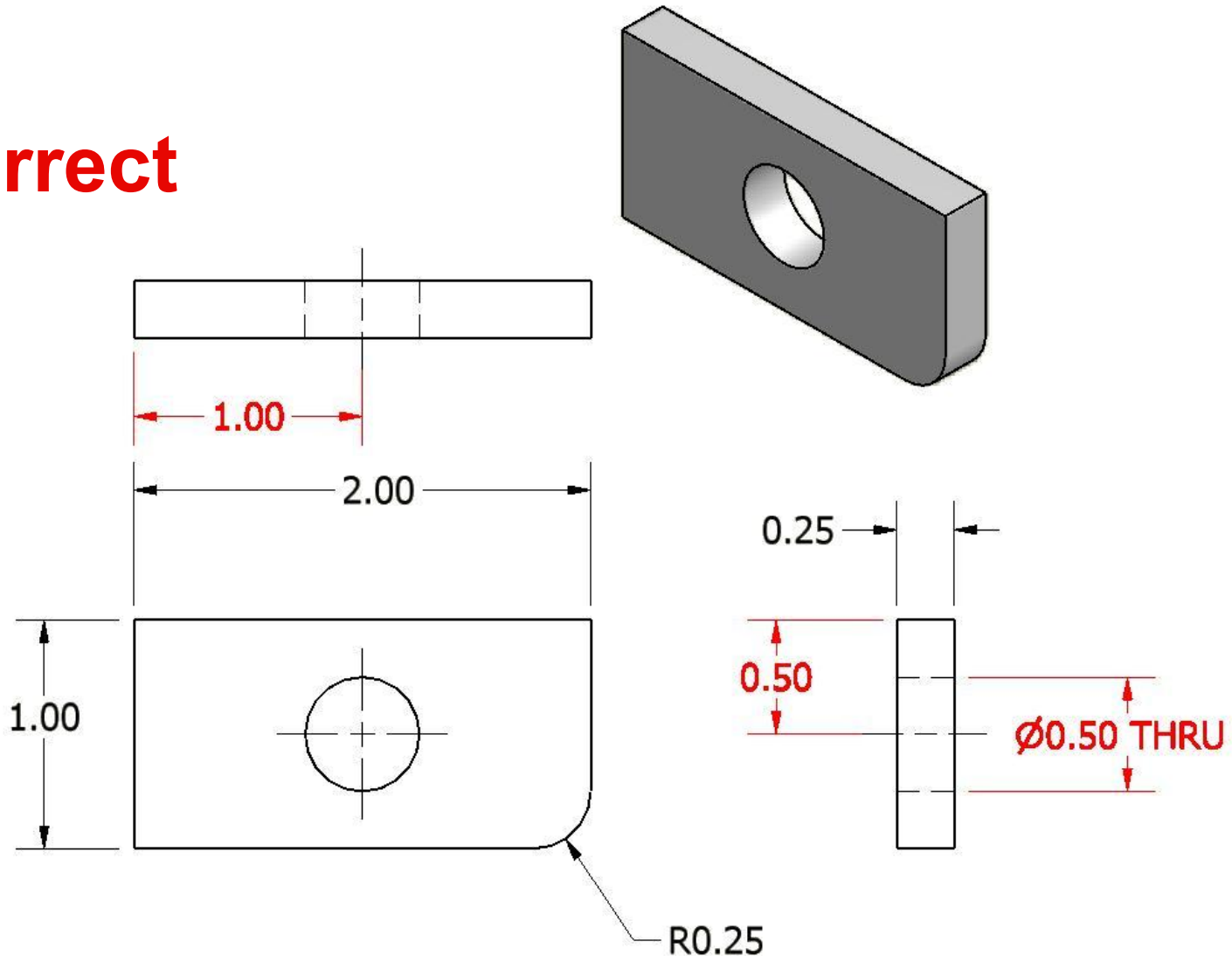


11. Holes are located by their centerlines, which may be extended and used as extension lines.



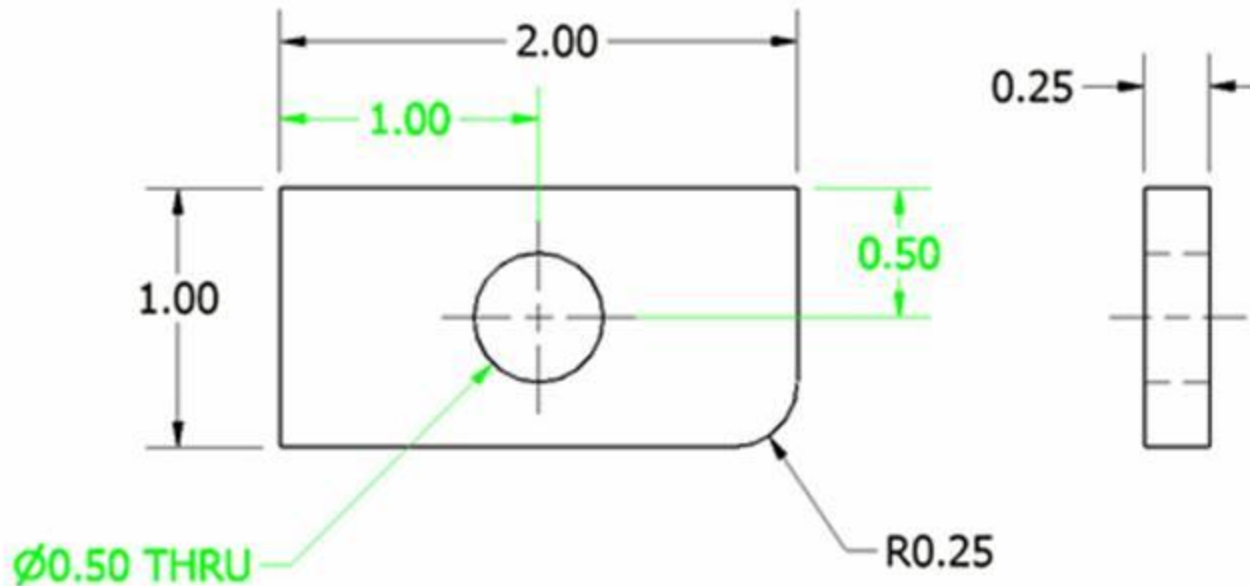
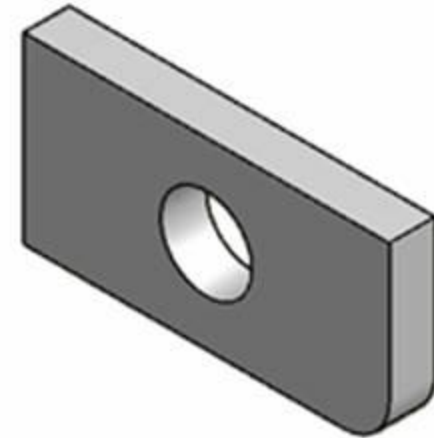
12. Holes should be located and sized in the view that shows the feature as a circle.

Incorrect



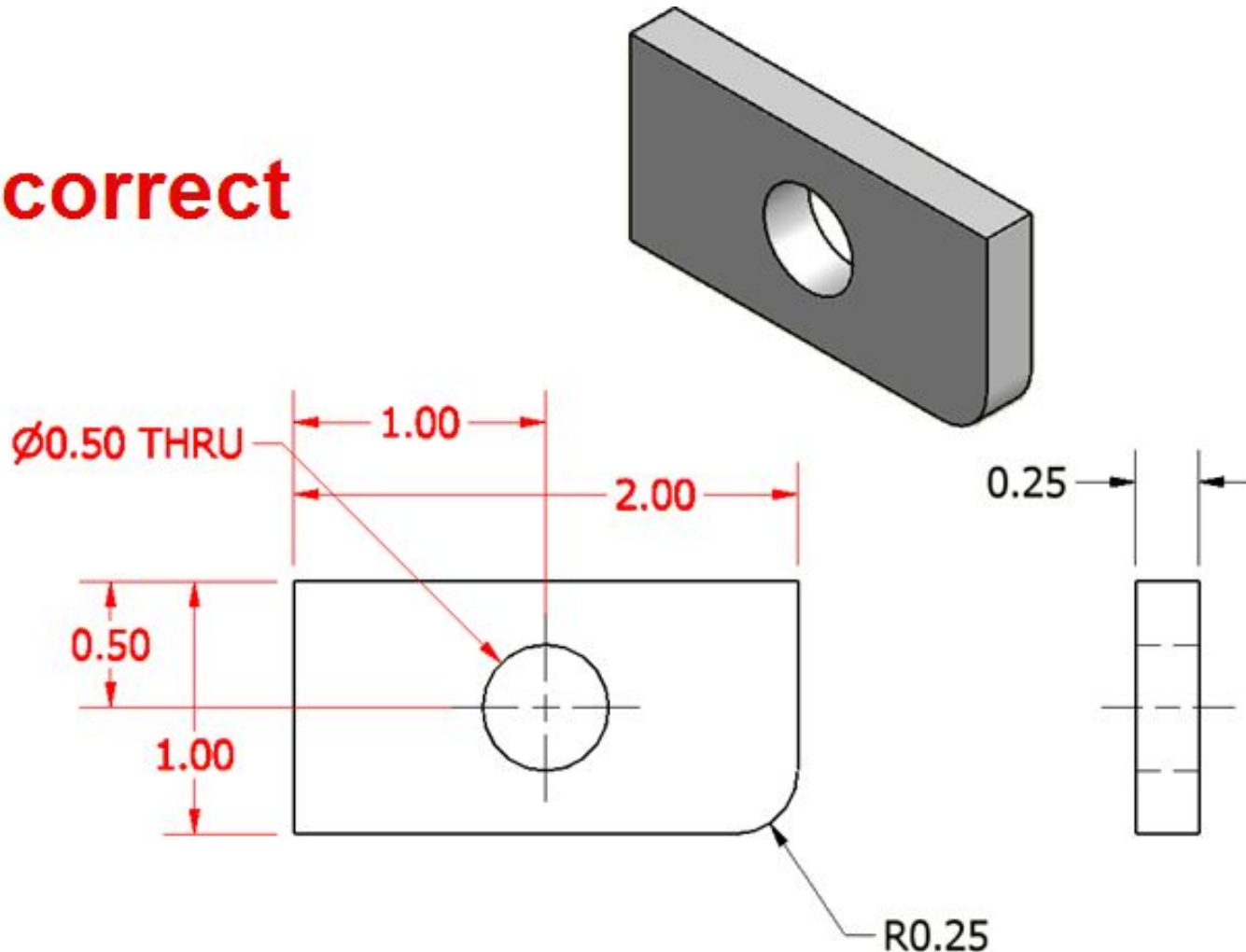
12. Holes should be located and sized in the view that shows the feature as a circle.

Correct



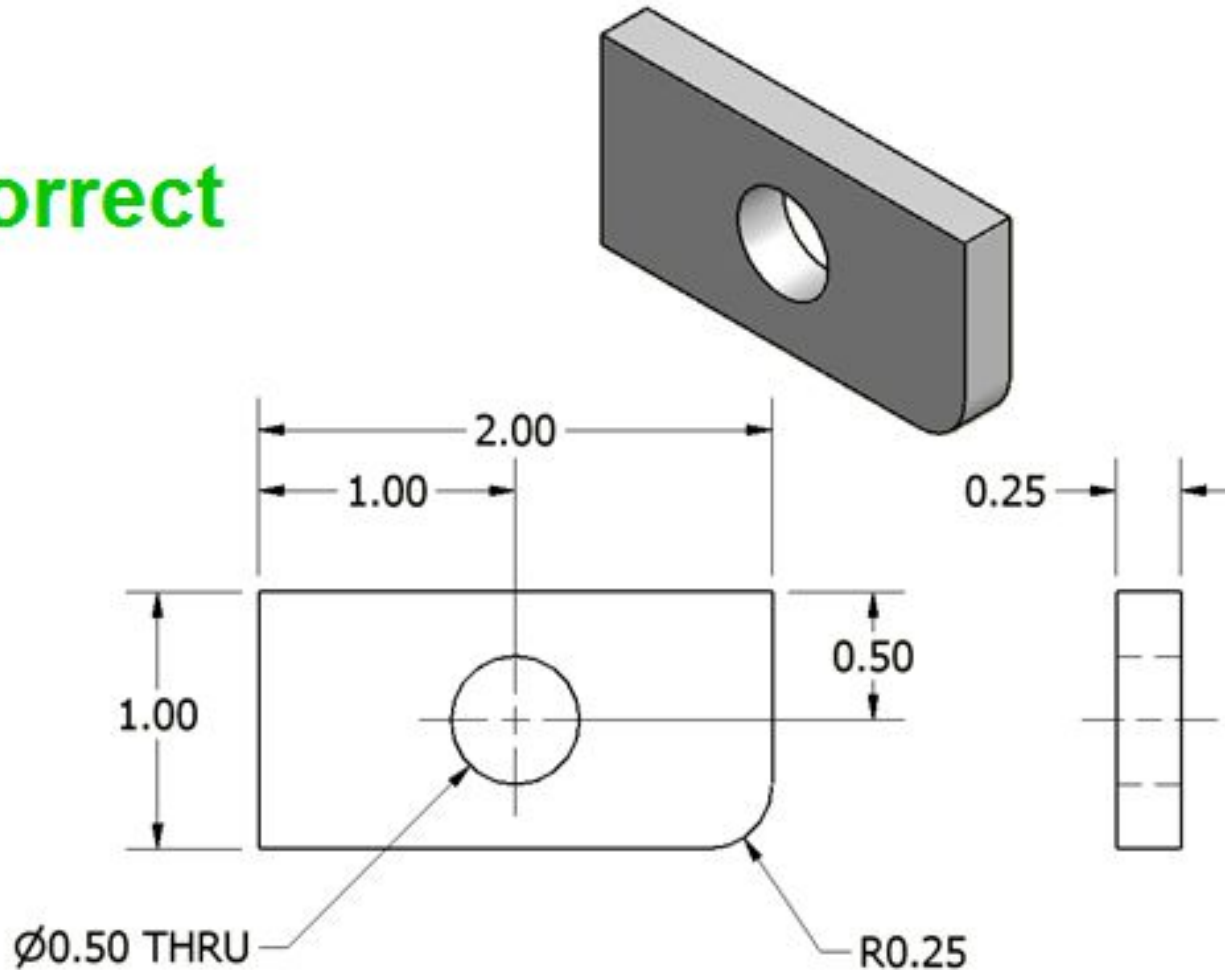
13. Do not cross a dimension line with an extension line, and avoid crossing dimensions with leader lines.

Incorrect



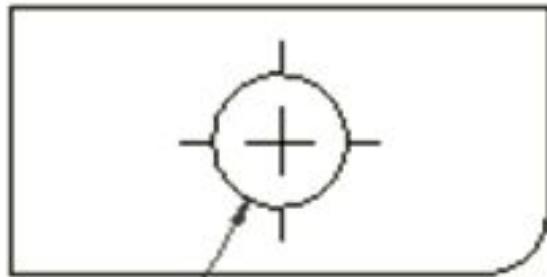
13. Do not cross a dimension line with an extension line, and avoid crossing dimensions with leader lines.

Correct



14. Leader lines point toward the center of the feature, and should not occur horizontally or vertically.

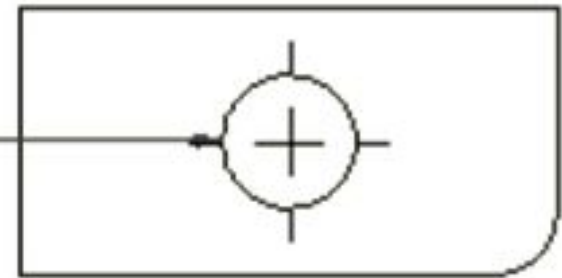
This



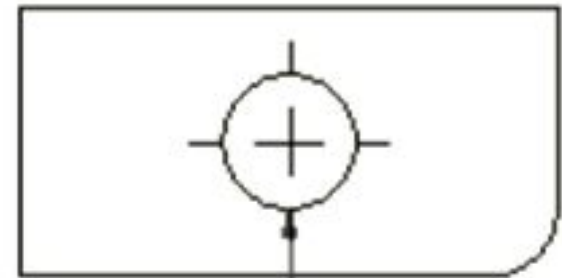
Ø0.50 THRU

Not This

Ø0.50 THRU

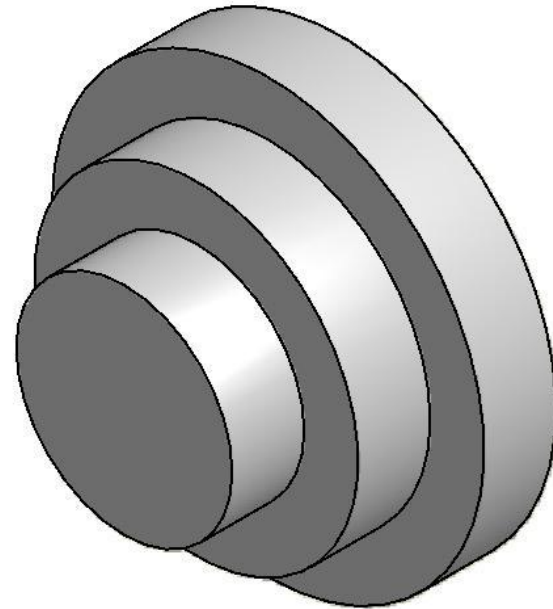
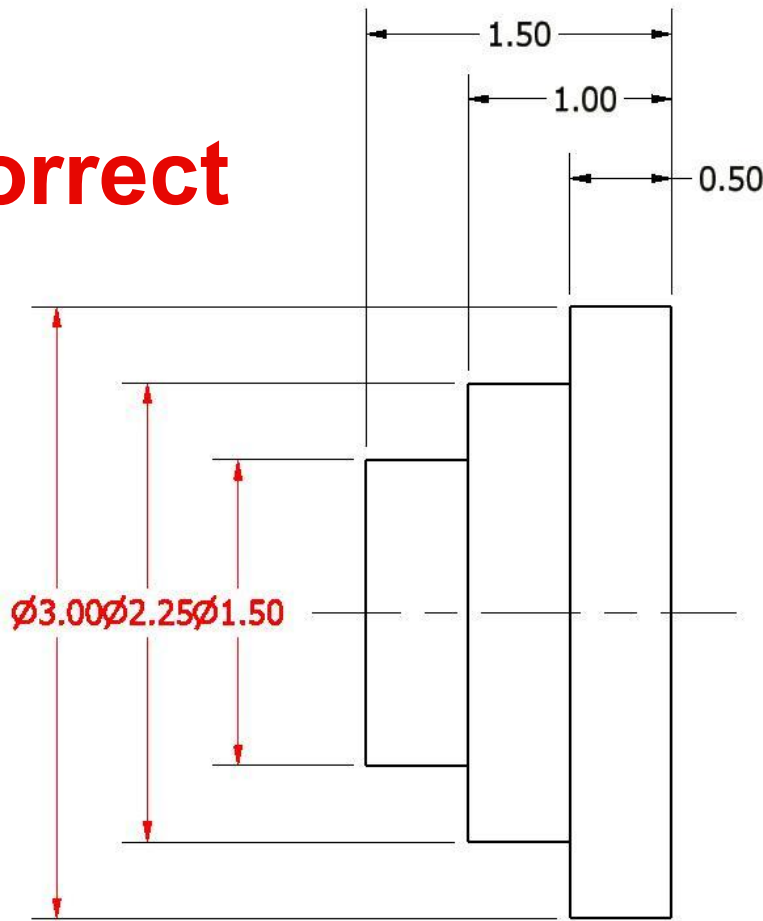


Ø0.50 THRU



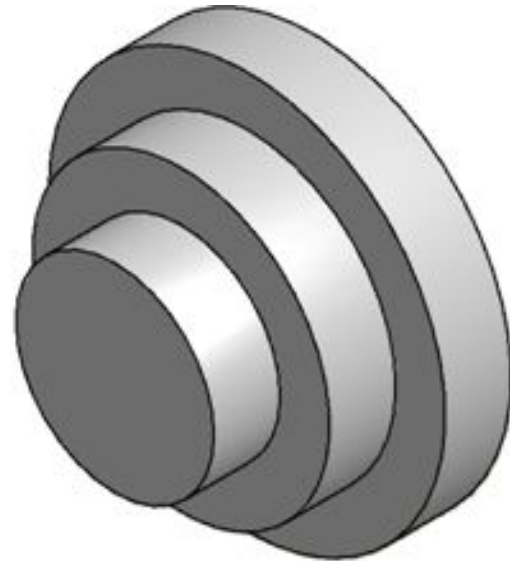
15. Dimension numbers should be centered between arrowheads, except when using stacked dimensions, and then the numbers should be staggered.

Incorrect



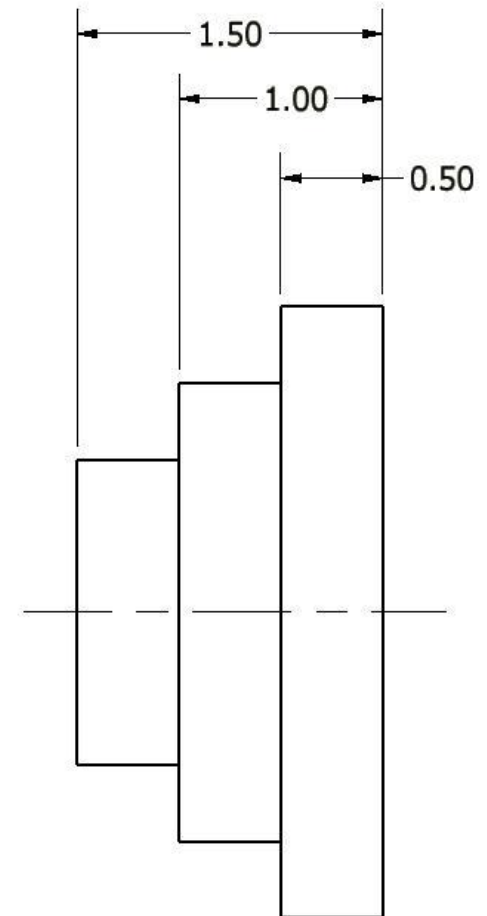
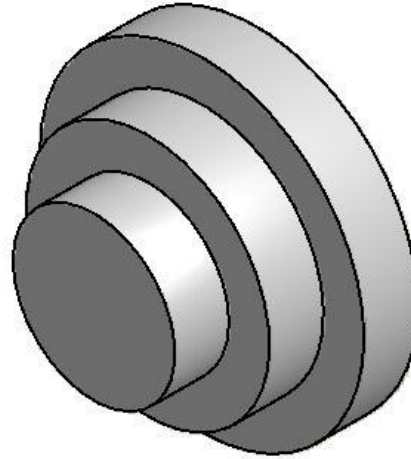
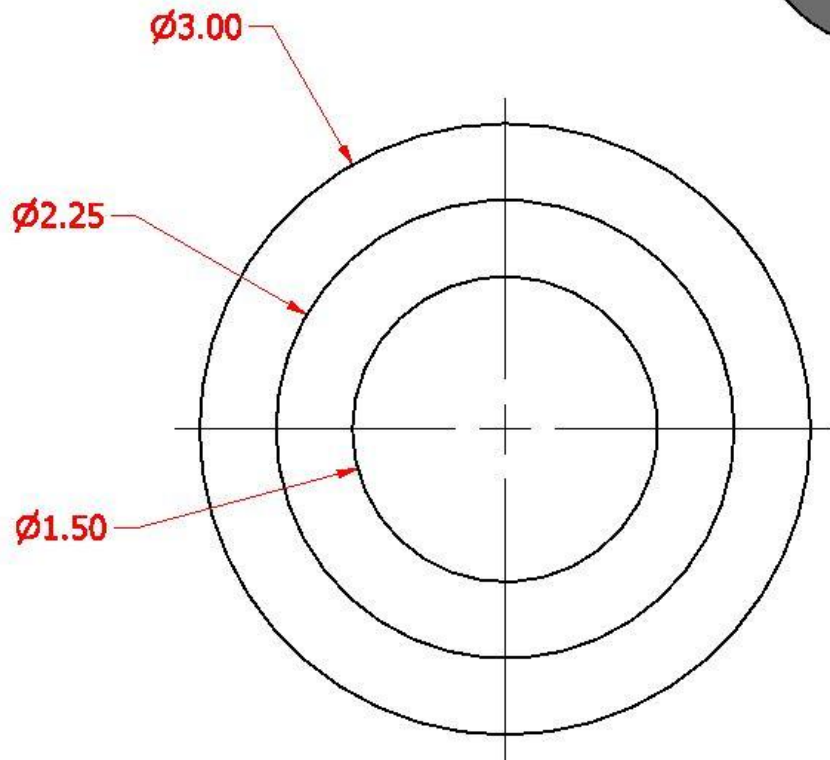
15. Dimension numbers should be centered between arrowheads, except when using stacked dimensions, and then the numbers should be staggered.

Correct



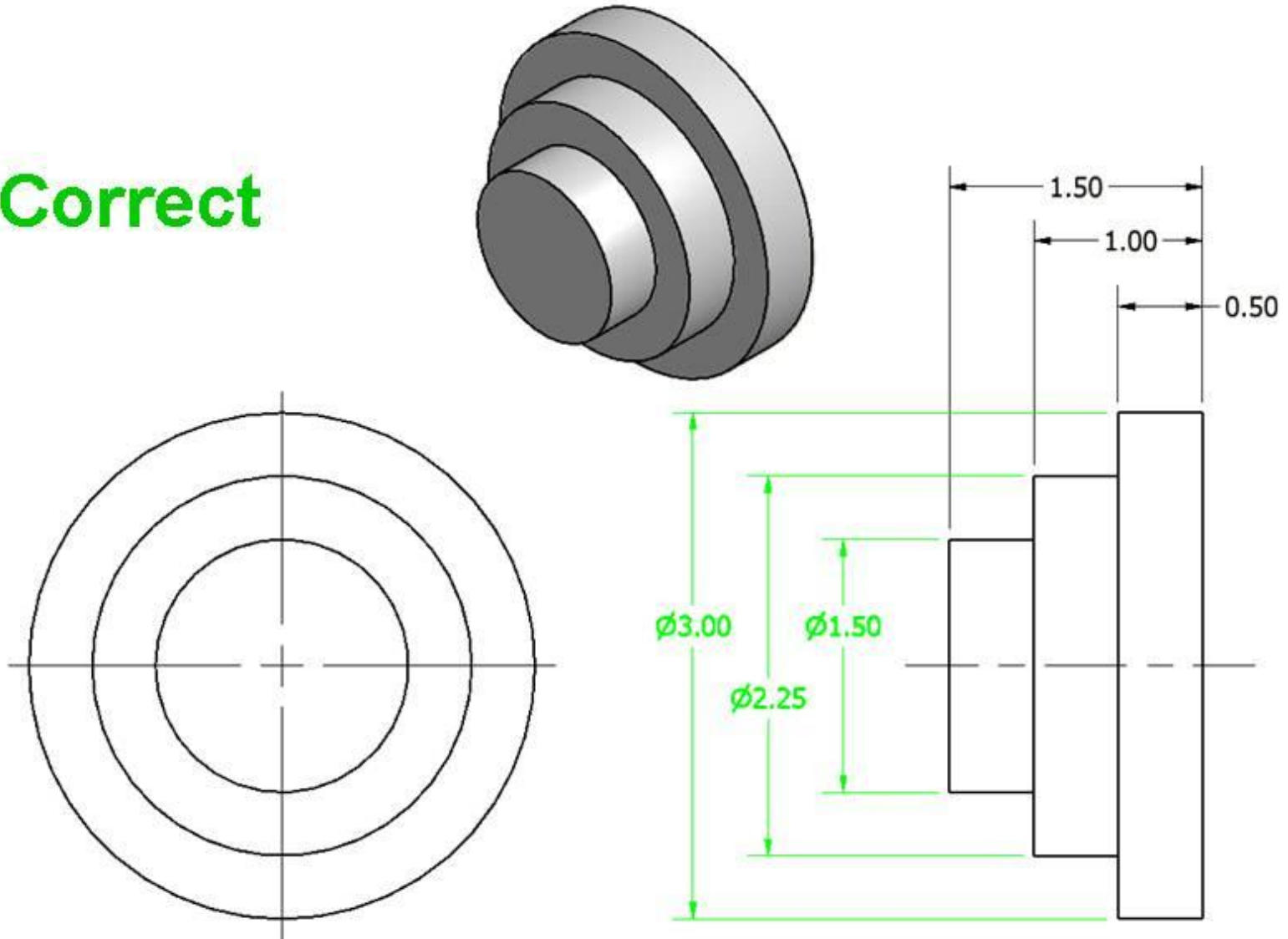
16. Concentric circles are dimensioned in the longitudinal view, whenever practical.

Incorrect



16. Concentric circles are dimensioned in the longitudinal view, whenever practical.

Correct



Tolerances

Variation is Unavoidable

- No two manufactured objects are identical in every way. Some degree of *variation* will exist.
- Engineers apply *tolerances* to part dimensions to reduce the amount of variation that occurs.

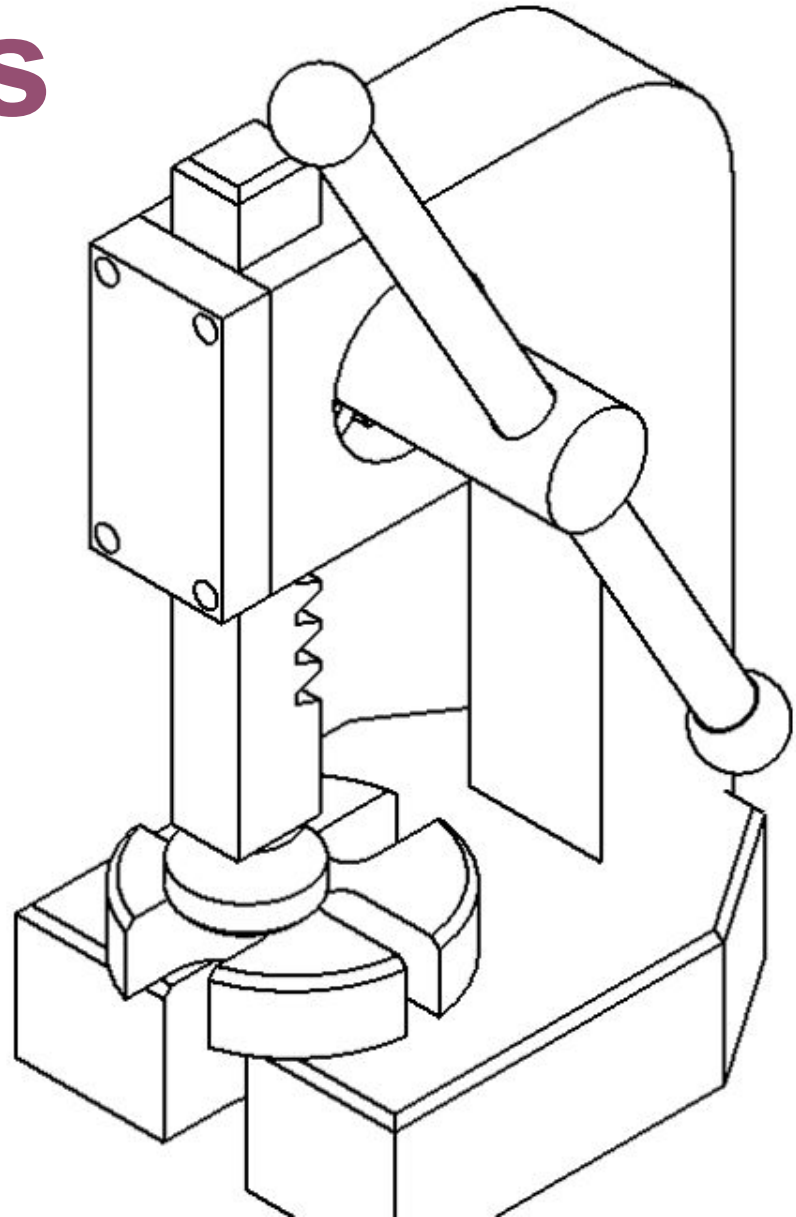
ANSI/ASME Standard

ANSI/ASME Standard Y14.5

Each dimension shall have a tolerance, except those dimensions specifically identified as reference, maximum, minimum, or stock. The tolerance may be applied directly to the dimension or indicated by a general note located in the title block of the drawing.

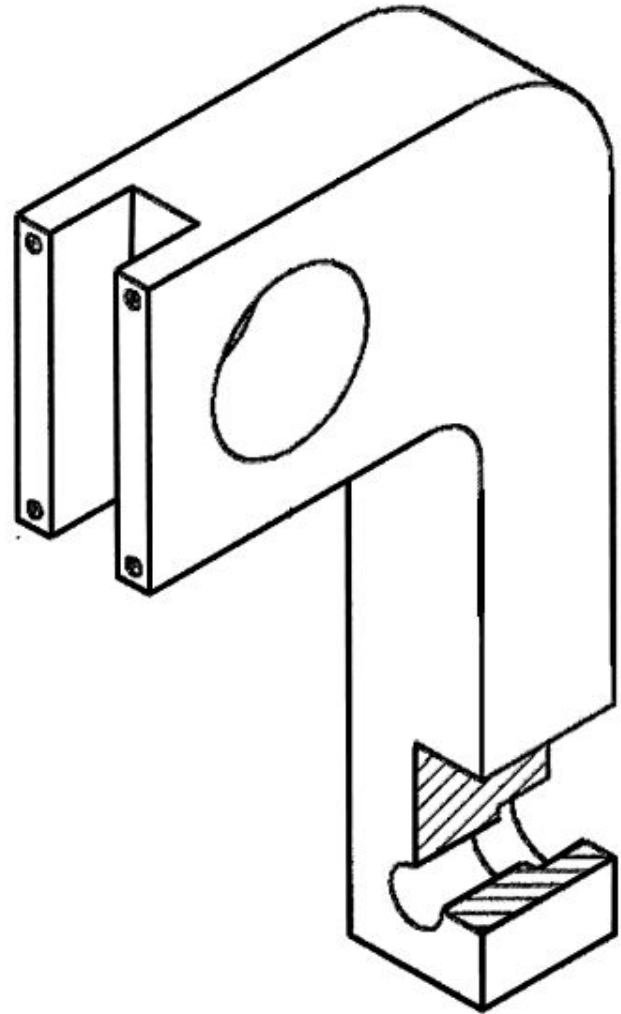
Tolerances

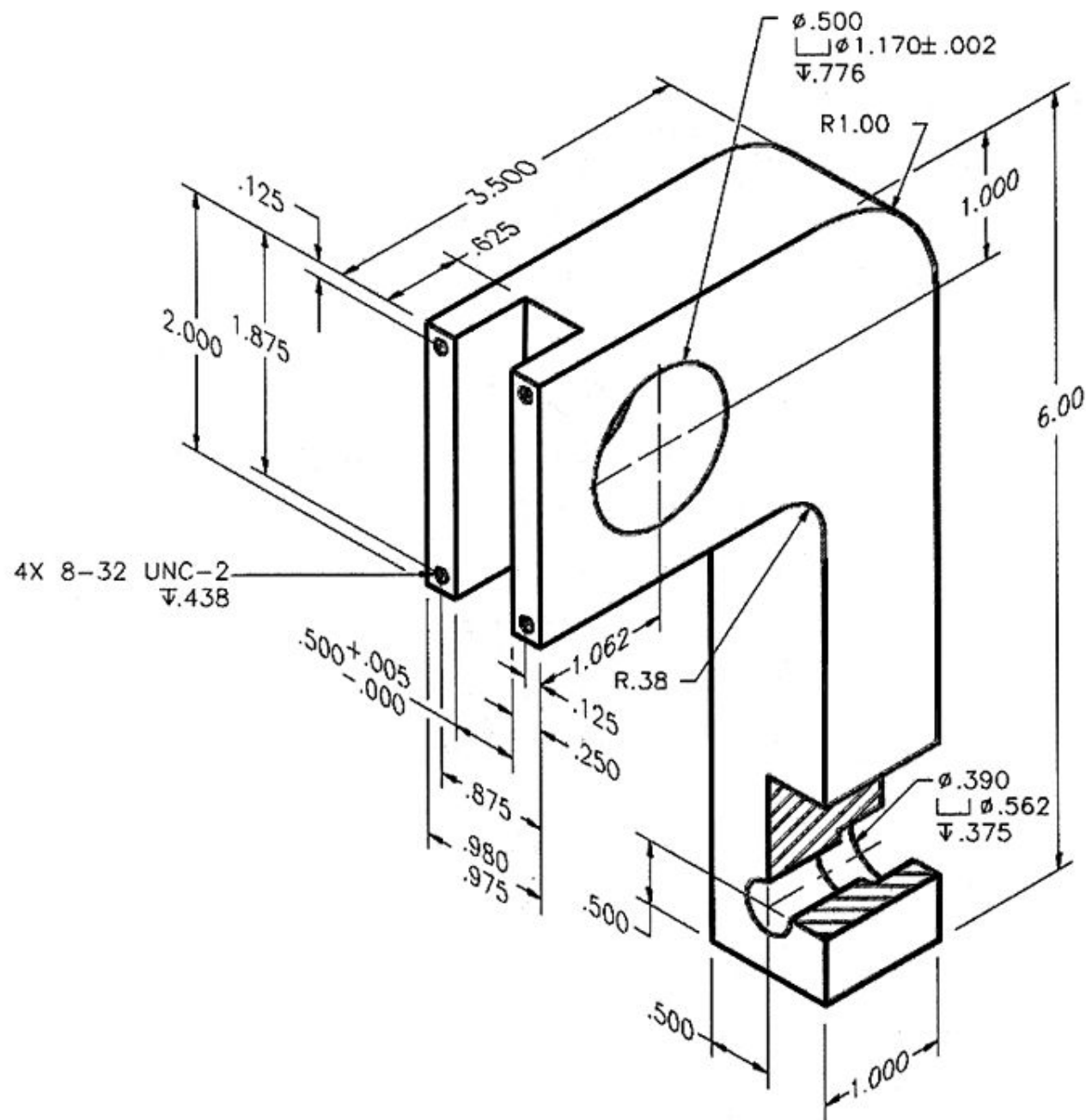
A *tolerance* is an acceptable amount of dimensional variation that will still allow an object to function correctly.



Tolerances

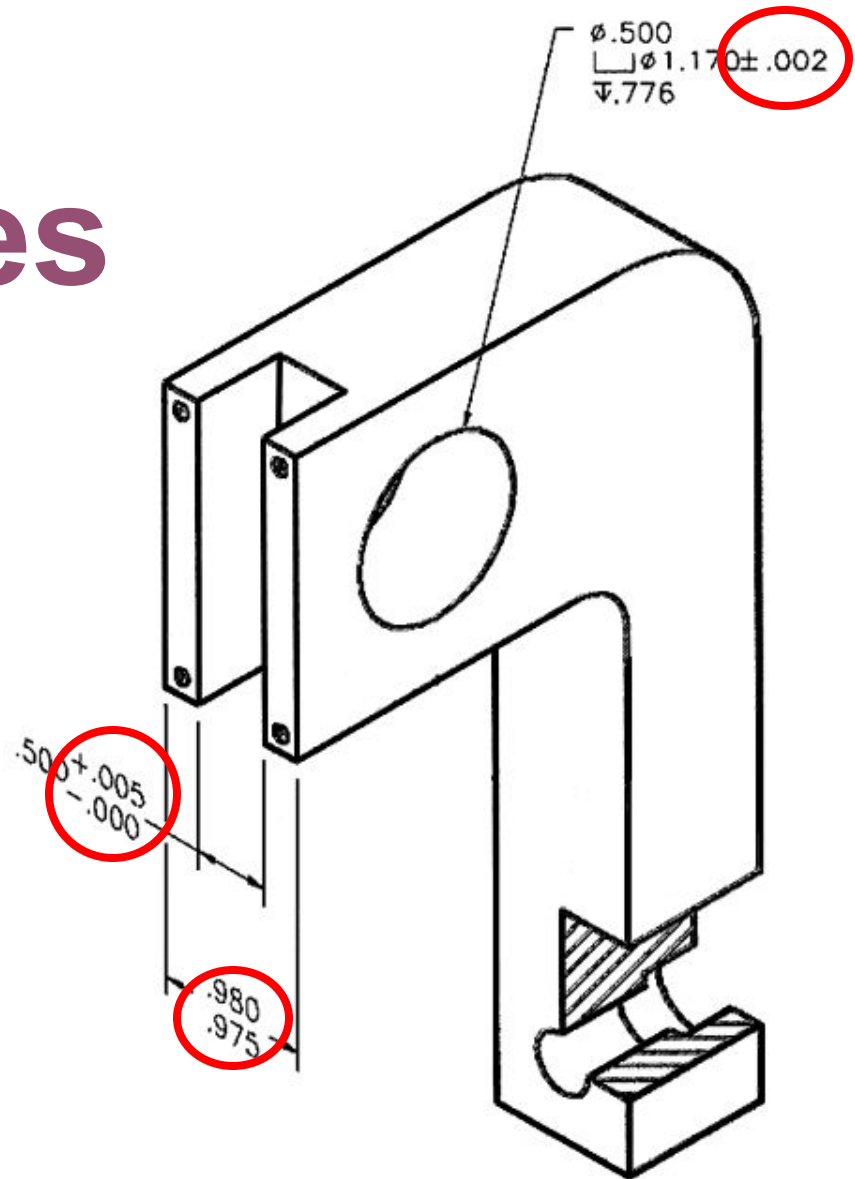
A *tolerance* is an acceptable amount of dimensional variation that will still allow an object to function correctly.





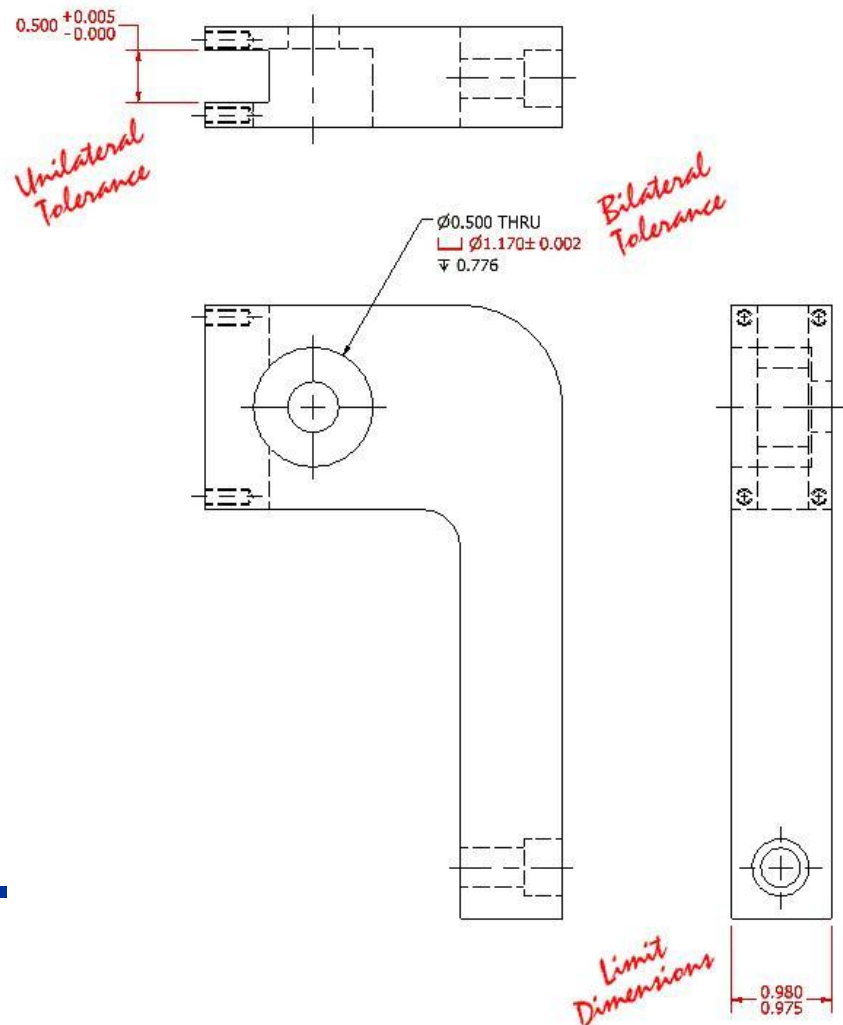
Tolerances

Three basic tolerances that occur most often on working drawings are: *limit dimensions*, *unilateral*, and *bilateral tolerances*.



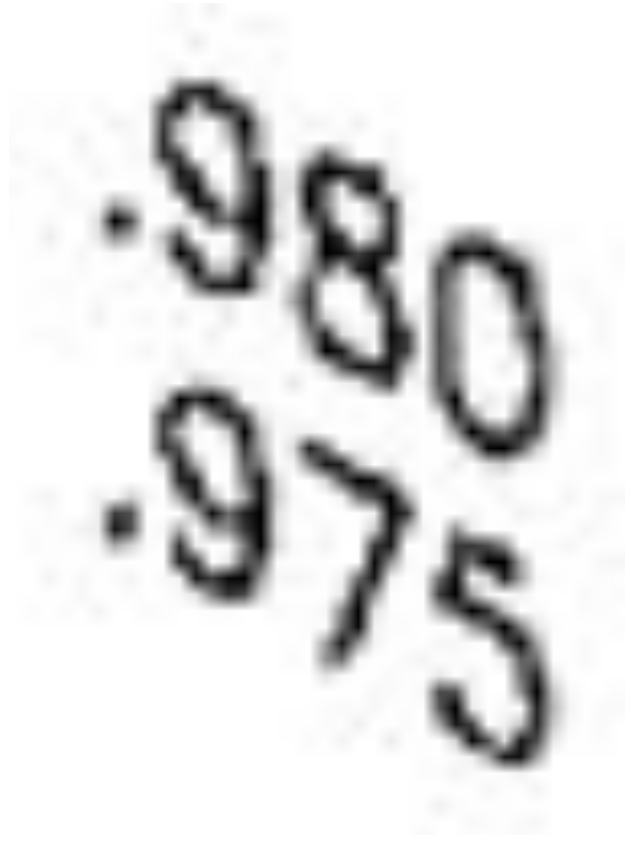
Tolerances

Three basic tolerances that occur most often on working drawings are: *limit dimensions*, *unilateral*, and *bilateral tolerances*.

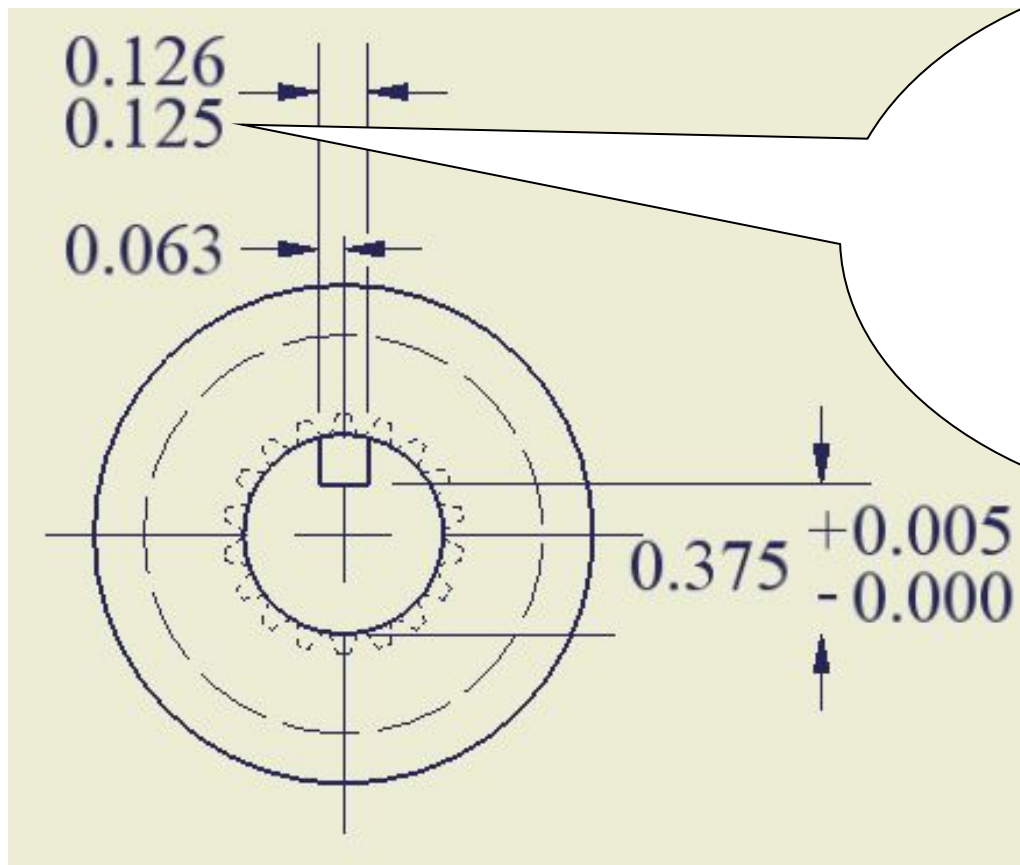


Limit Dimensions

Limit dimensions are two dimensional values stacked on top of each other. The dimensions show the largest and smallest values allowed. Anything in between these values is acceptable.



Limit Dimensions



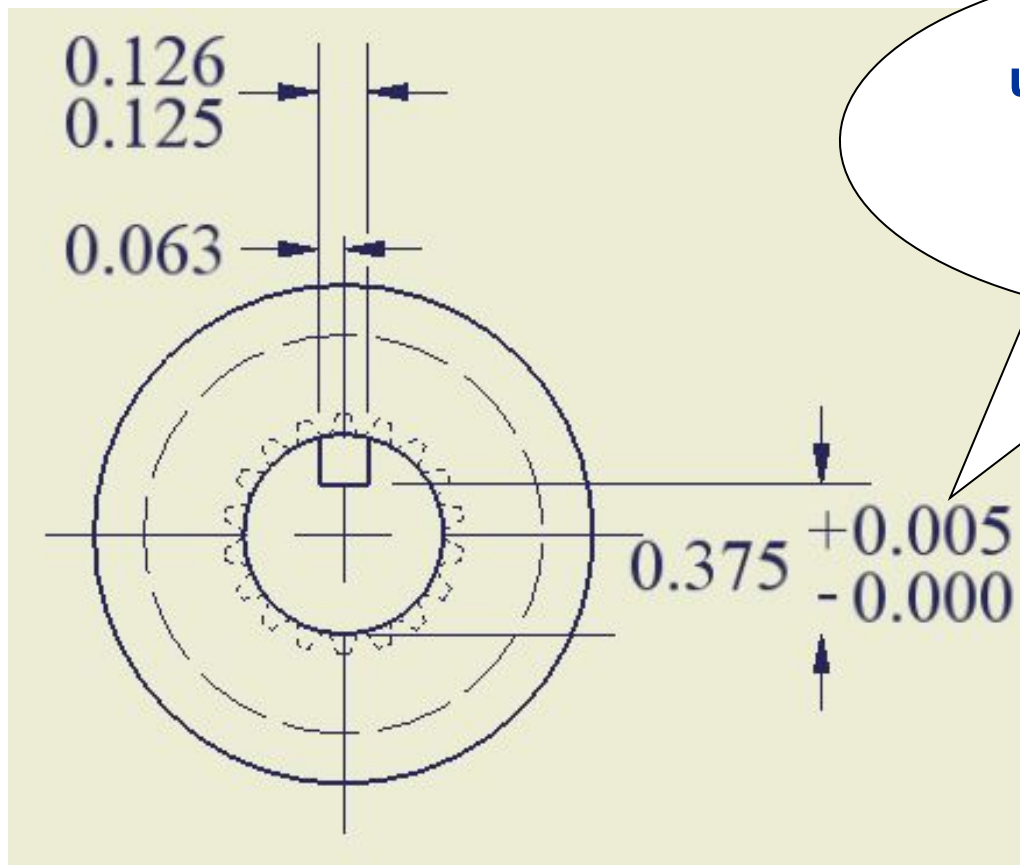
These are limit dimensions, because the upper and lower dimensional sizes are stacked on top of each other.

Unilateral Tolerance

A *unilateral tolerance* exists when a target dimension is given along with a tolerance that allows variation to occur in only one direction.

$.500 \pm .005$
 $.500$

Unilateral Tolerance



This tolerance is unilateral, because the size may only deviate in one direction.

Bilateral Tolerance

A *bilateral tolerance* exists if the variation from a target dimension is shown occurring in both the positive and negative directions.

Ø.500
└┐ Ø1.170 ± .002
↕.776

General Tolerances

If no tolerances are specified at the dimension level, then general tolerances may be applied by deliberately controlling the number of values past the decimal point on each dimension.

Angles = $\pm .5^\circ$

Linear Dimensions

X.X = $\pm .020$

X.XX = $\pm .010$

X.XXX = $\pm .005$

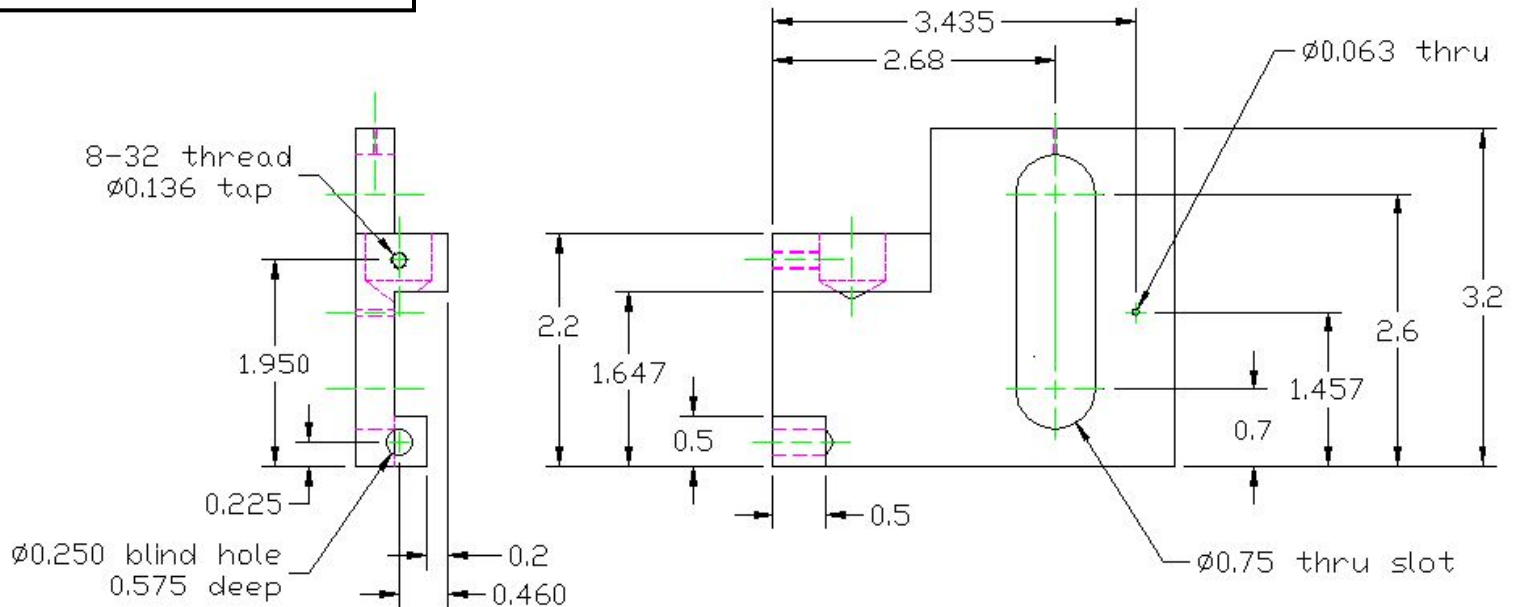
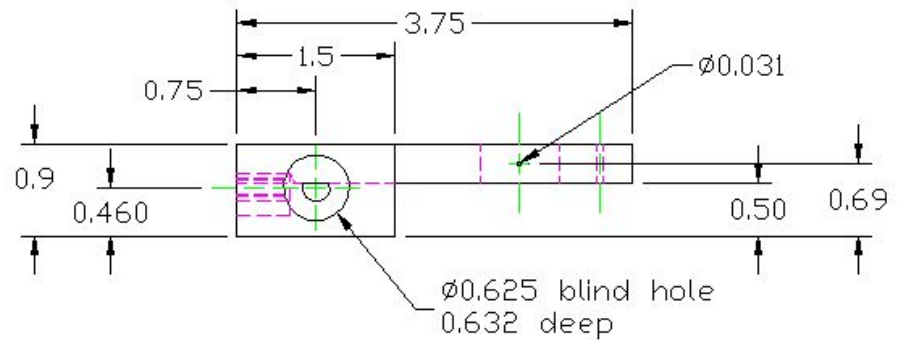
General Tolerances

Tolerances

$$X.X = \pm .020$$

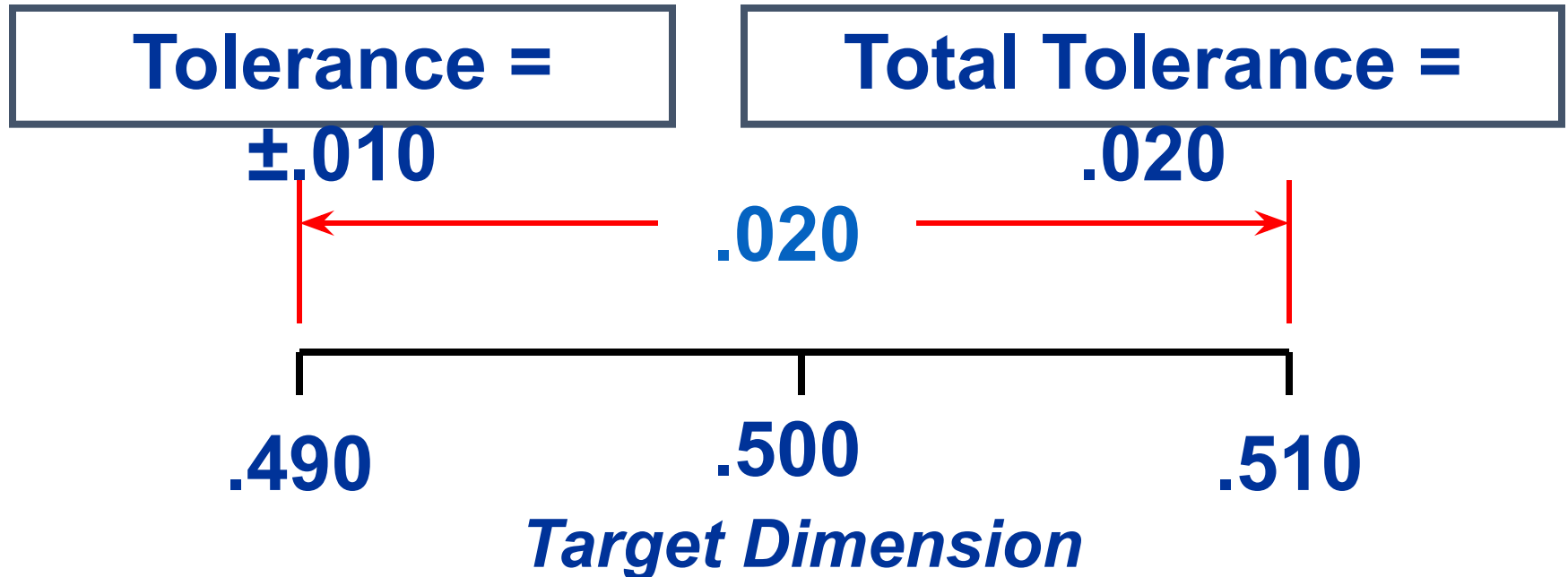
X.XX = ± .010

X.XXX = ± .005



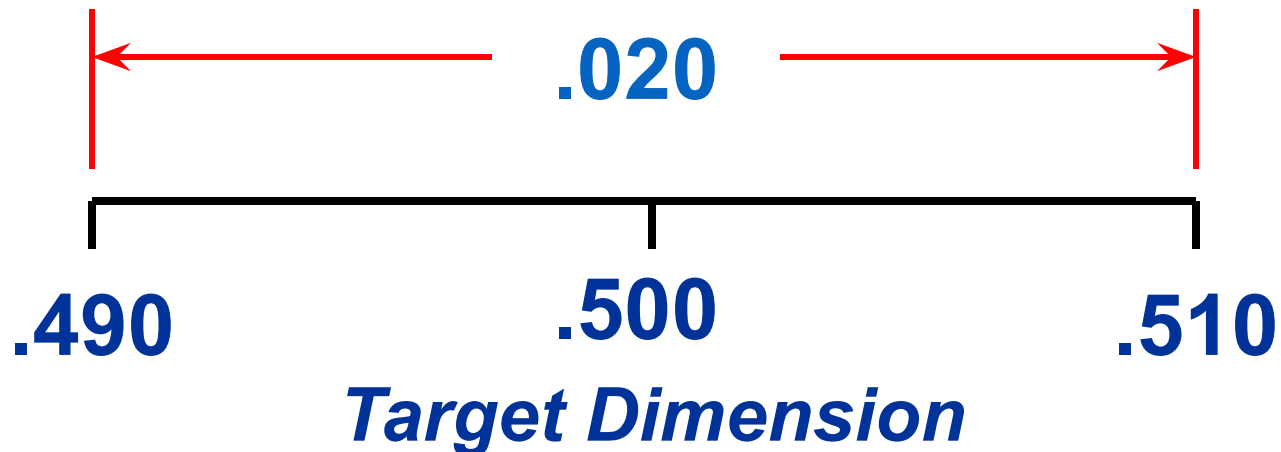
Total Tolerance

The total tolerance is a value that describes the maximum amount of variation.



Total Tolerance

A measuring device should be able to accurately measure within $1/10^{\text{th}}$ of the total blueprint tolerance identified.



Tolerances and Measuring

In this case, a measuring device should be able to take accurate measurements to within two thousandths of an inch.

Total tolerance = .020

$$\frac{1}{10} = \frac{x}{.020}$$

$$10x = .020$$

$$x = .002$$

**X = the minimum accuracy
of the measuring device**

Three Types of Fit

There are three types of fit that should be considered when working with tolerances.

- **Clearance Fit**- have limits of size so prescribed that a clearance always results when mating parts are assembled.
- **Interference Fit**- have limits of size so prescribed that an interference always results when mating parts are assembled.
- **Transition Fit**- have limits of size indicating that either a clearance or an interference may result when mating parts are assembled.

Discussion